Mathematical introduction

Branching, quasi-stationarity, and absorption models

Adam Aldridge

University of Leeds

Contents

Chapter 1

Mathematical introduction

1.1 Overview

The processes studied in this thesis describe populations small enough that their fate is decided by chance rather than by their mean behaviour — a founding cohort of a few individuals, or the contents of a single compartment that may later rupture. At those counts a deterministic rate equation answers the wrong question. It returns a single trajectory, growing or decaying, when what one wants to know is whether the population establishes itself at all, and how large it is if it does. Both are questions about a distribution, and neither is visible in a mean.

This chapter assembles the mathematical framework in which those questions are posed, together with the methods that answer them. The two are kept apart on purpose. Naming the processes first, and only then developing each technique once, on whichever process displays it most clearly, avoids the usual awkwardness of a background chapter, where the same method is introduced three times over because three models happen to need it.

Section 1.2 sets out the *objects*: the classes of Markov process that appear in this thesis. Discrete-time chains come first, the simple random walk as the elementary example and the Galton–Watson branching process as the one that matters here. Continuous-time chains follow, built from exponential clocks: the Poisson process, the linear birth–death process, and the birth–death–catastrophe process in which an abrupt event can remove an entire population at a stroke. Two extensions are then described that leave the time-homogeneous setting: processes whose rates vary with time, worked through on a logistic model of diversification in which the birth rate declines as a niche fills; and coupled systems in which a deterministic differential equation and a Markov chain each drive the other, whose schematic instance here is a compartment that ruptures into a shared medium.

Section 1.3 sets out the *methods*. Discrete time is treated briefly, because the technique that the branching problem of this thesis actually requires — the Koenigs linearisation of a functional iteration — is developed in the following chapter rather than here. Continuous time is treated at length, with the birth–death process as the running example, and covers backward equations, means, variances and higher moments, hitting probabilities, extinction probabilities, and the limiting conditional means that arise when one conditions on a process not yet having been absorbed. Section 1.3.3 then develops the method of characteristics, which converts the master equation for an infinite family of state probabilities into a single first-order partial differential equation for a generating function and solves it; a worked example is carried through in full.

Three threads recur, and it is worth naming them at the outset.

The first is *conditioning on survival*. A subcritical branching process dies out with probability one, so its unconditional mean decays geometrically and records little beyond the certainty of extinction. Restrict attention to those realisations still alive at generation n, however, and the picture changes entirely: the conditional population settles to a fixed distribution, and the mean of that distribution is a constant. For the binary Galton–Watson process with division probability $p < \frac{1}{2}$ that constant is $1/A(p)$, where $A(p) = \lim_n S_n/(2p)^n$ and S_n is the survival probability to generation n. This chapter defines $A(p)$ and shows how the whole limiting conditional law is expressed through it, in section 1.3.2.6. It does not analyse the constant: the exact representations, the bounds, the near-critical asymptotics, and the dynamical obstruction that explains why no elementary formula for it has been found are the subject of the following chapter.

The second is the *near-critical regime*. As p rises towards $\frac{1}{2}$ the process takes ever longer to die, the quasi-stationary mean diverges, and any numerical scheme that works by iterating to stationarity becomes impractical. Just above $\frac{1}{2}$ the size of the founding cohort begins to weigh heavily on whether the process survives at all. The behaviour on either side of $p = \frac{1}{2}$ is what makes small founding populations interesting, and it is also what makes them awkward to compute with. The phenomenology is recorded in section 1.A; the theory of $A(p)$ that eventually tames it belongs to the following chapter.

The third is *absorption*. Extinction and rupture are both absorbing states, and much of this thesis concerns a chain conditioned on not yet having reached one. The generating-function methods of section 1.3 are what make such conditioning computable.

Two remarks on how the chapter is written. First, not all of it is background: the closed form obtained in section 1.B.2 for the absorption–birth–death generating function is a result of this thesis, as is the integral identity of section 1.C on which an alternative derivation of it rests. Everything else is standard, and is presented as such. Second, the argument is in places heuristic — continuum approximations to difference equations, mean-field substitutions, asymptotic matching. Every such step is marked where it occurs, and where a claim can be checked numerically it has been.

1.2 Markov chains

A stochastic process is *Markov* if its future depends on its past only through its present. That single property is what makes every model in this thesis tractable, and this section sets out the classes of Markov process the thesis uses. Standard treatments are Norris [1], Karlin and Taylor [2], and, closest to the applications here, Allen [3].

1.2.1 Discrete-time Markov chains

Let $\{X_n : n = 0, 1, 2, \dots\}$ take values in a countable state space $\mathcal{S}$. The process is a *discrete-time Markov chain* if

$$\mathbb{P}(X_{n+1} = j \mid X_n = i,\ X_{n-1}, \dots, X_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i) =: P_{ij}, \tag{1.1}$$

the right-hand side being independent of n for a *time-homogeneous* chain. The matrix $P = (P_{ij})$ is stochastic: its entries are non-negative and each row sums to one. Multi-step laws follow by decomposing a journey at an intermediate state, which gives the Chapman–Kolmogorov identity

$$\mathbb{P}(X_{n+m} = j \mid X_0 = i) = \sum_{k \in \mathcal{S}} \mathbb{P}(X_n = k \mid X_0 = i)\, \mathbb{P}(X_m = j \mid X_0 = k)\,, \tag{1.2}$$

and hence $\mathbb{P}(X_n = j \mid X_0 = i) = (P^n)_{ij}$: composing steps is multiplying matrices. In principle everything about the chain is therefore contained in P; in practice the interest lies in extracting it, and the rest of this chapter is about how. A state i with $P_{ii} = 1$ is *absorbing*: a chain that reaches it never leaves.

1.2.1.1 Simple random walks

The simplest non-trivial chain moves one step at a time on the integers. Fix $q \in (0, 1)$ and let

$$P_{i,i+1} = q, \qquad P_{i,i-1} = 1 - q, \tag{1.3}$$

so that $X_n = X_0 + \sum_{k=1}^{n} \xi_k$ with the ξ_k independent and equal to $+1$ with probability q and -1 otherwise. The walk is *symmetric* when $q = \frac{1}{2}$. Its mean and variance are immediate from the independence of the increments:

$$\mathbb{E}(X_n) = X_0 + n(2q - 1), \qquad \mathrm{Var}(X_n) = 4nq(1 - q). \tag{1.4}$$

The drift $2q - 1$ decides the long-run behaviour, and the diffusive spread $\sqrt{n}$ is what the mean cannot see — the same tension, in its simplest form, that motivates everything else in this chapter.

Here the whole law is available too, and it is worth writing down once, since it is the distribution that (1.4) summarises. If R of the n increments are $+1$ then $X_n = X_0 + 2R - n$, and R is binomial with parameters n and q, so

$$\mathbb{P}(X_n = X_0 + 2r - n) = \binom{n}{r} q^{\,r} (1 - q)^{\,n-r}, \qquad r = 0, 1, \ldots, n, \tag{1.5}$$

supported on those lattice points with the parity of n (figure 1.1). Almost no other chain in this chapter surrenders its n-step law so cheaply, and the reason is that the increments of the walk are independent of the state it has reached. Once the step distribution depends on the current count — as it does for every branching process below — the n-step law is no longer a convolution and must be reached by the generating functions of section 1.3.1.2.

The random walk is not a peripheral example here. Every continuous-time chain in section 1.2.2 decomposes into a holding time and a jump, and the sequence of jumps is itself a discrete-time chain — the *embedded jump chain*. For the linear birth–death process of section 1.2.2.3, whose rates out of state i are λi and μi, the state-dependence cancels in the ratio and the embedded chain is exactly (1.3) with

$$q = \frac{\lambda}{\lambda + \mu}. \tag{1.6}$$

Absorption at 0 is then the walk's ruin problem, and hitting probabilities for the continuous-time process can be computed on the walk alone; section 1.3.1.1 does exactly that. (1.6) is also the bridge between the discrete-generation and continuous-time models compared in section 1.3.2.6.

Two absorbing barriers are usually imposed. With absorbing states at 0 and M, writing h_i for the probability of reaching M before 0 from state i, the one-step decomposition $h_i = q h_{i+1} + (1 - q) h_{i-1}$ with $h_0 = 0$, $h_M = 1$ has the classical gambler's-ruin solution

$$h_i = \begin{cases} \dfrac{1 - \theta^{\,i}}{1 - \theta^{\,M}}, & \theta := \dfrac{1 - q}{q} \neq 1, \\[2ex] \dfrac{i}{M}, & q = \frac{1}{2}. \end{cases} \tag{1.7}$$

This is the smallest instance of the first-step principle of section 1.3.1.1, and it is worth keeping in view: the entire method of section 1.3 consists of conditioning on one step and solving the resulting self-consistent equation, on state spaces where the equation is no longer a two-term recursion. Figure 1.2 shows both halves of the picture.

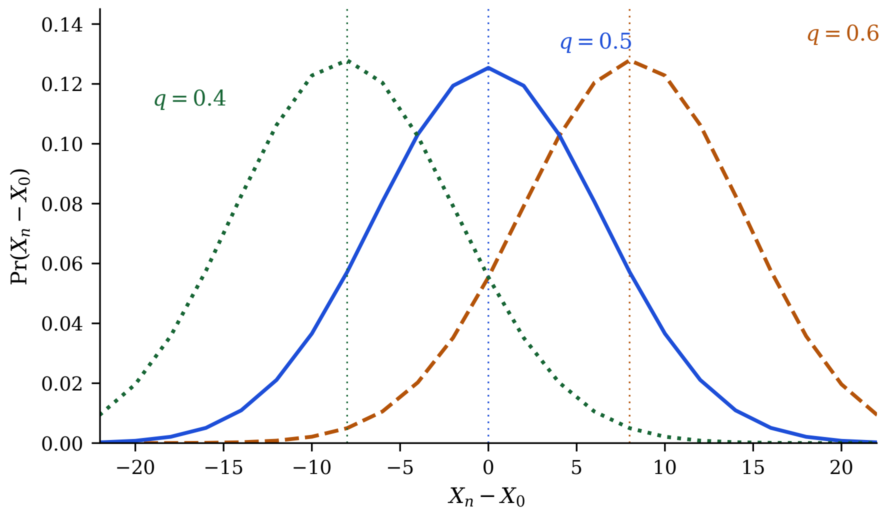


Figure 1.1: The exact displacement law (1.5) at $n = 40$, for $q = 0.4$, 0.5, 0.6; the dotted vertical lines mark the means $(2q-1)n$ of (1.4). Changing the drift slides the whole distribution along the lattice and barely deforms it, because the width $\sqrt{4nq(1-q)}$ varies little over this range of q. Figure 1.2 shows the same process as trajectories; this is the distribution those trajectories are drawn from, and the gap between the two pictures — a smooth law, and paths that wander far from its centre — is the reason a mean is not an answer.

1.2.1.2 Galton–Watson branching processes

The Galton–Watson process is a discrete-time chain on the non-negative integers in which every member of a population independently leaves offspring, or does not, at each generational increment. It is the oldest model of stochastic branching and the central object of this thesis; the standard references are Harris [4] and Athreya and Ney [5].

Suppose each individual produces, independently of every other, a number of offspring distributed as the random variable L, with $\mathbb{P}(L = k) = p_k$ and mean $\mu = \mathbb{E}(L)$. Let Z_n be the size of generation n, started from a single progenitor, $Z_0 = 1$, so that

$$Z_{n+1} = \sum_{i=1}^{Z_n} L_i, \tag{1.8}$$

the L_i being independent copies of L. Conditioning on the previous generation and using the tower property gives $\mathbb{E}(Z_{n+1}) = \mu\mathbb{E}(Z_n)$, hence

$$\mathbb{E}(Z_n) = \mu^n. \tag{1.9}$$

When $\mu < 1$ the expected population decays geometrically, and since Z_n is integer-valued Markov's inequality gives $\mathbb{P}(Z_n \geq 1) \leq \mu^n \to 0$: extinction is certain. The three regimes $\mu < 1$, $\mu = 1$, $\mu > 1$ are called *subcritical*, *critical* and *supercritical*.

Because Z_{n+1} is a random sum of independent copies of L, the composition rule for generating functions (section 1.3.1.2) turns (1.8) into a functional iteration. Writing $\phi(z) = \mathbb{E}\left(z^L\right)$ for the offspring generating function and ϕ_n for that of Z_n,

$$\phi_{n+1} = \phi_n \circ \phi, \tag{1.10}$$

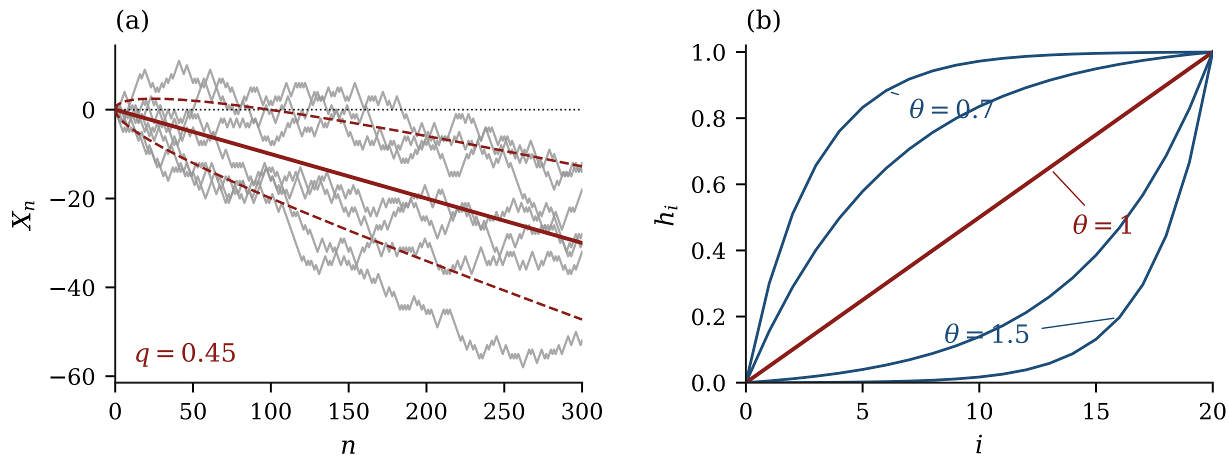


Figure 1.2: The simple random walk of (1.3). (a) Seven realisations at $q = 0.45$ (grey), against the drift line $n(2q-1)$ and the envelope $\pm\sqrt{4nq(1-q)}$ of one standard deviation (red), from (1.4). The drift is linear in n and the spread only $O(\sqrt{n})$, so the drift eventually dominates; over the first hundred steps it does not, and individual paths spend long intervals on the wrong side of the mean. (b) The gambler's-ruin probability (1.7) of reaching $M = 20$ before 0, against the starting state i, for $\theta = (1-q)/q \in \{0.7, 0.85, 1, 1.2, 1.5\}$. The symmetric case $\theta = 1$ is the straight line i/M; $\theta < 1$ bows the curve up and $\theta > 1$ bows it down, and the effect is strongly nonlinear in θ. Only the ratio θ enters, which is why (1.65) can read this figure off the embedded jump chain of a birth–death process without reference to its rates.

so ϕ_n is the n-fold composition of ϕ with itself. That single fact is the source of both the tractability and the difficulty of the model: everything about Z_n is encoded in an iterated map, and iterating a map is easy to do and hard to solve in closed form.

The binary case. Throughout this thesis we specialise to the binary, or *simple*, case, in which an individual either divides into two or dies:

$$p_2 = p \quad \text{(division)}, \qquad p_0 = 1-p \quad \text{(death)}, \qquad p_i = 0 \;\; \text{otherwise}, \tag{1.11}$$

so that $\mu = 2p$, $\mathbb{E}(Z_n) = (2p)^n$, and the offspring generating function is

$$\phi(z) = pz^2 + (1-p). \tag{1.12}$$

Criticality is at $p = \frac{1}{2}$. Writing $u_n = \mathbb{P}(Z_n = 0)$ for the probability of extinction by generation n and setting $z = 0$ in (1.10) gives $u_{n+1} = \phi(u_n)$, and it is more convenient to work with the survival probability $S_n = 1 - u_n = \mathbb{P}(Z_n > 0)$, which satisfies

$$S_{n+1} = 2pS_n - pS_n^2, \qquad S_0 = 1. \tag{1.13}$$

Figure 1.3 shows realisations either side of criticality.

That is as far as the definition needs to go here. (1.13) is a quadratic recursion, and almost everything of interest about the binary process turns out to be a statement about its orbit: whether that orbit reaches zero (proposition 1.6), how fast it does so (section 1.A), and what constant survives in front of the decay (section 1.3.1.4, and then the following chapter at length). The critical case, in which extinction is certain and yet the expected lifetime is infinite, is developed in full in section 1.A.1.

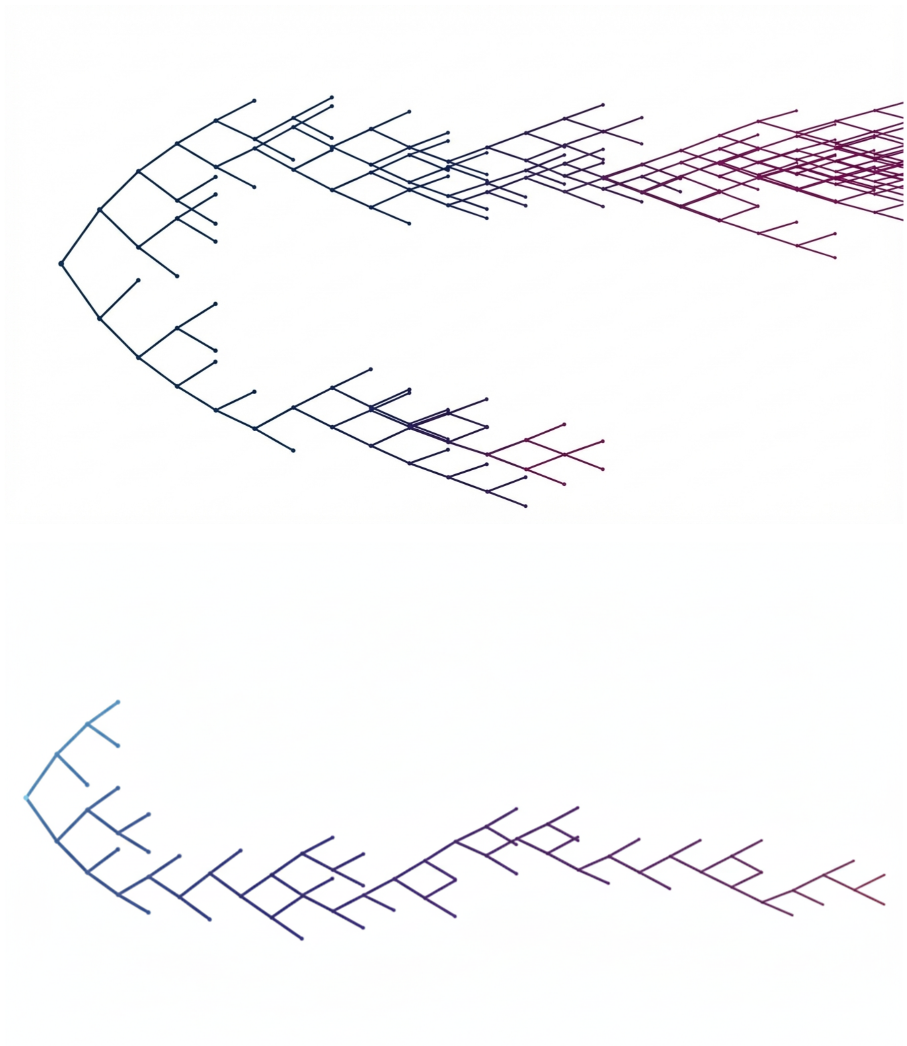


Figure 1.3: Realisations of the simple Galton–Watson process, drawn as genealogies with generation increasing to the right. Top: supercritical, $p = 0.58$, where the number of lines grows without bound. Bottom: subcritical, $p = 0.47$; the same construction, but the tree terminates. Colour encodes generation number.

1.2.2 Continuous-time Markov chains

Let $\{X_t : t \geq 0\}$ take values in a countable state space $\mathcal{S}$, which throughout will be a set of population counts or of tuples of counts. The process is a *continuous-time Markov chain* if, for all $s, t \geq 0$,

$$\mathbb{P}(X_{t+s} = j \mid X_t = i,\ \{X_u\}_{u<t}) = \mathbb{P}(X_{t+s} = j \mid X_t = i)\,, \tag{1.14}$$

and *time homogeneous* if the right-hand side depends on s but not t, in which case one writes $p_{ij}(s)$. Such a chain is specified by its *generator* $Q = (q_{ij})$, whose off-diagonal entries are transition rates,

$$q_{ij} = \lim_{\Delta t \to 0} \frac{p_{ij}(\Delta t)}{\Delta t} \quad (j \neq i), \qquad q_{ii} = -\sum_{j \neq i} q_{ij} =: -q_i, \tag{1.15}$$

so that every row of Q sums to zero. Equivalently, and this is the form used to define models below,

$$\mathbb{P}(X_{t+\Delta t} = j \mid X_t = i) = q_{ij}\,\Delta t + o(\Delta t) \qquad (j \neq i), \tag{1.16}$$

with complementary probability $1 - q_i\Delta t + o(\Delta t)$ of no transition at all. A state with $q_i = 0$ is absorbing.

1.2.2.1 Exponential clocks

Everything in continuous time rests on one distribution, and it is worth seeing why it is forced rather than chosen. A random variable T is exponentially distributed with rate $\theta > 0$, written $T \sim \mathrm{Exp}(\theta)$, if

$$f_T(t) = \theta\mathrm{e}^{-\theta t}, \qquad F_T(t) = 1 - \mathrm{e}^{-\theta t} \qquad (t \geq 0), \tag{1.17}$$

with mean $1/\theta$ and variance $1/\theta^2$. The property that matters is memorylessness: for $t_1, t_2 > 0$,

$$\mathbb{P}(T > t_1 + t_2 \mid T > t_1) = \frac{\mathrm{e}^{-\theta(t_1+t_2)}}{\mathrm{e}^{-\theta t_1}} = \mathrm{e}^{-\theta t_2} = \mathbb{P}(T > t_2)\,, \tag{1.18}$$

so a clock that has already run for time t_1 is statistically indistinguishable from a fresh one. The exponential law is the only continuous distribution with this property, which is exactly why a process assembled from constant-rate events satisfies (1.14): with any other waiting-time law the elapsed time would be informative and the process would not be Markov.

When several events compete, two facts organise everything that follows.

Proposition 1.1 (Competing clocks)**.** *Let $T_1, \dots, T_m$ be independent with $T_i \sim \mathrm{Exp}(\theta_i)$, let $T = \min_i T_i$, and write $\Theta = \sum_i \theta_i$. Then*

$$T \sim \mathrm{Exp}(\Theta), \qquad \mathbb{P}(T_k = T) = \frac{\theta_k}{\Theta}, \tag{1.19}$$

and the identity of the winning clock is independent of T.

Proof. For the first claim, $\mathbb{P}(T > t) = \prod_i \mathbb{P}(T_i > t) = \prod_i \mathrm{e}^{-\theta_i t} = \mathrm{e}^{-\Theta t}$. For the second, condition on the value of T_k and use independence:

$$\mathbb{P}(T_k = T) = \int_0^\infty \theta_k \mathrm{e}^{-\theta_k u} \prod_{i \neq k} \mathrm{e}^{-\theta_i u}\,\mathrm{d}u = \theta_k \int_0^\infty \mathrm{e}^{-\Theta u}\,\mathrm{d}u = \frac{\theta_k}{\Theta}.$$

Repeating the calculation with the integral truncated at t gives $\mathbb{P}(T_k = T,\ T > t) = (\theta_k/\Theta)\mathrm{e}^{-\Theta t} = \mathbb{P}(T_k = T)\,\mathbb{P}(T > t)$, which is the asserted independence. □

Proposition 1.1 is what connects (1.16) to a trajectory: the chain leaves state i after a holding time $\mathrm{Exp}(q_i)$ and then jumps to j with probability q_{ij}/q_i, independently of how long it waited. It also says that a rate becomes a probability as soon as one asks *which* of several things happened rather than when, which is the identification (1.6) of the embedded jump chain. And it is the whole content of the direct simulation method.

Remark 1.2 (Direct simulation)*.* Proposition 1.1 is an algorithm as well as a theorem. Given the current state, compute the rate θ_i of every possible event and their sum Θ; draw $T \sim \mathrm{Exp}(\Theta)$ by inverting (1.17) on a uniform variate; draw the event k with probability θ_k/Θ; advance the clock by T, apply event k, and repeat. Because the two draws are independent and the exponential law is memoryless, the resulting trajectory has exactly the law of the process. This is Gillespie's direct method [6]; it is used in this thesis to check analytical results, never to replace them.

1.2.2.2 The Poisson process

The simplest continuous-time chain counts events that arrive at a constant rate and never depart. Let N_t take values in $\{0, 1, 2, \dots\}$ with the single non-zero rate

$$q_{n,n+1} = \alpha \qquad (n \geq 0), \tag{1.20}$$

so $q_n = \alpha$ for every n. By proposition 1.1 the holding times are independent $\mathrm{Exp}(\alpha)$ variables, and N_t counts how many have elapsed by time t: the process is the renewal count of a sequence of independent exponential waits. Its law follows from the master equation (1.56) below, or directly by induction on n:

$$\mathbb{P}(N_t = n) = \mathrm{e}^{-\alpha t}\frac{(\alpha t)^n}{n!}, \qquad \mathbb{E}(N_t) = \mathrm{Var}(N_t) = \alpha t. \tag{1.21}$$

The equality of mean and variance is characteristic, and is the baseline against which the over-dispersion of branching processes should be read. Figure 1.4 shows a realisation and the law it is drawn from.

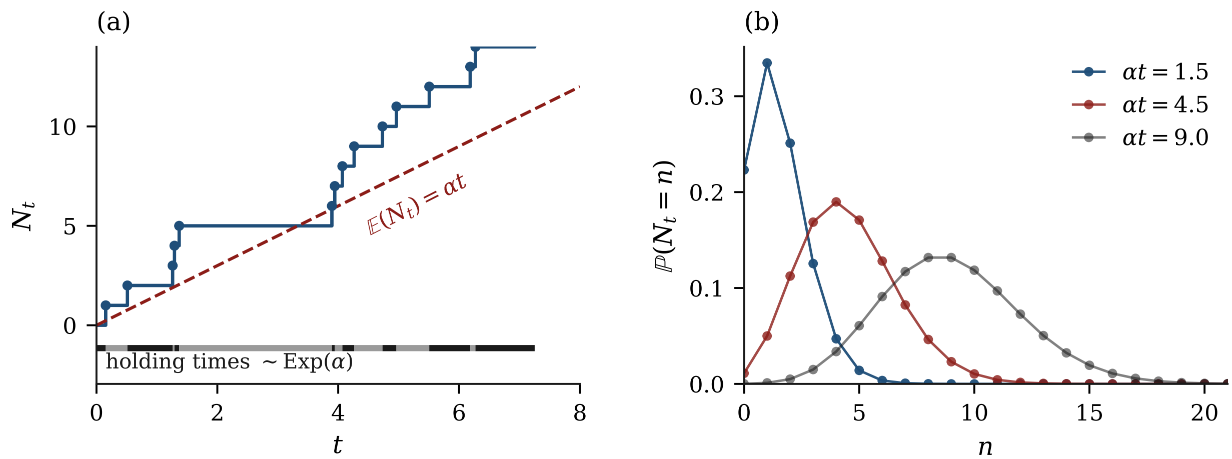


Figure 1.4: The Poisson process of (1.20), at rate $\alpha = 1.5$. (a) One realisation of the counting process N_t, with the independent $\mathrm{Exp}(\alpha)$ holding times of proposition 1.1 marked in alternating shades along the axis, and the mean αt dashed. The path crosses and recrosses its own mean; nothing prevents a long wait followed by three arrivals in quick succession, because each holding time is drawn afresh. (b) The law (1.21) at $t = 1, 3, 6$. Mean and variance are both αt, so the distribution broadens as it advances and its relative width $1/\sqrt{\alpha t}$ shrinks.

Two properties are used later. Counts over disjoint time intervals are independent, and the count over an interval depends only on its length — the process has stationary independent increments, which is the continuous-time statement of memorylessness. And if each arrival is independently retained with probability ϖ, the retained arrivals form a Poisson process of rate $\varpi\alpha$: *thinning* preserves the class. Thinning is what makes the absorption models of section 1.3.3 tractable, where a cohort of exterior particles is taken up one by one at a constant per-particle rate.

The Poisson process is also the $\mu = 0$, state-independent corner of the family treated next. Making the arrival rate proportional to the current count, αn in place of α, gives the pure-birth (Yule) process; adding a departure rate gives the birth–death process.

1.2.2.3 Birth–death processes

Let X_t count individuals, each of which independently gives birth at rate λ and dies at rate μ. The non-zero rates out of state n are

$$q_{n,n+1} = \lambda n, \qquad q_{n,n-1} = \mu n, \tag{1.22}$$

so $q_n = (\lambda + \mu)n$ and $n = 0$ is absorbing. This is the *linear* birth–death process, the continuous-time counterpart of the Galton–Watson process of section 1.2.1.2, and it is the running example of section 1.3.2: nearly every method in this thesis is first demonstrated on it. Its embedded jump chain is the random walk (1.3) with $q = \lambda/(\lambda + \mu)$, so questions about *which* states are visited reduce to section 1.2.1.1 while questions about *when* require the rates themselves.

Figure 1.5 shows the three regimes it can occupy.

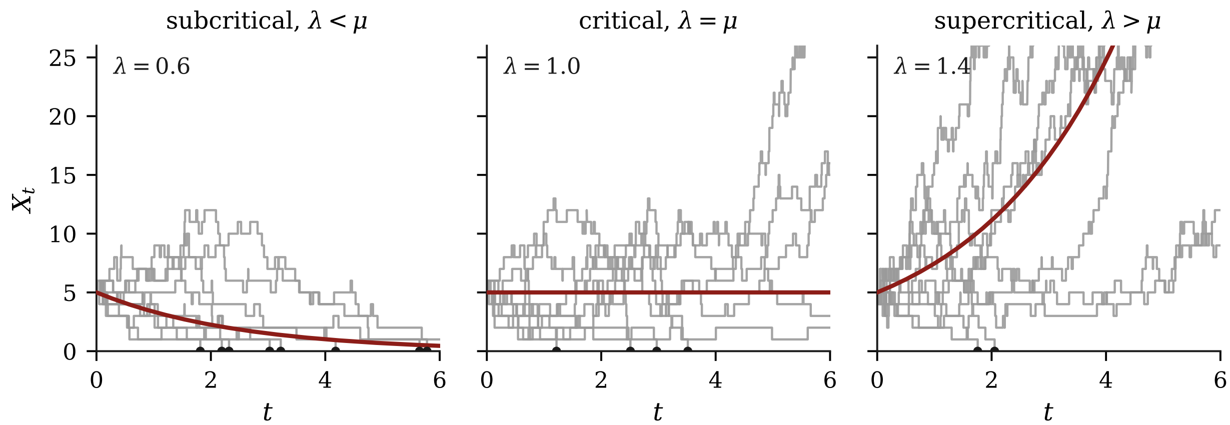


Figure 1.5: Nine realisations of the linear birth–death process (1.22) from $X_0 = 5$, with $\mu = 1$ throughout and the mean $N\mathrm{e}^{(\lambda-\mu)t}$ of (1.61) in red; dots on the axis mark trajectories that reached the absorbing state. The middle panel is the one to dwell on. At criticality the mean is exactly constant, and yet several realisations are already extinct while others have grown fourfold — the mean is the same number in both cases, and describes neither. That gap between a mean and a distribution is what the conditioning of section 1.3.2.6 exists to close. Note also that extinction is not confined to the subcritical panel: a supercritical process still dies out with probability $(\mu/\lambda)^N$, visible here as the paths that fail early on the right.

The general birth–death process allows state-dependent rates λ_n and μ_n; the linear case $\lambda_n = \lambda n$, $\mu_n = \mu n$ is the one in which individuals act independently, and hence the one for which generating functions close. Everything derived in section 1.3.2 — mean, variance, extinction probability, conditional mean — uses that independence.

1.2.2.4 Birth–death–catastrophe processes

The processes at the centre of this thesis add a third channel. Alongside the gradual loss of individuals to death, a compartment may suffer a *catastrophe*: an abrupt event removing the entire population at once.

Definition 1.3 (Birth–death–catastrophe process)**.** Let X_t count individuals in a compartment. Each individual independently gives birth at rate λ and dies at rate μ, and the compartment additionally undergoes a catastrophe — a rupture that removes the entire population at once —

at rate ρ per individual, so that the total catastrophe rate in state n is ρn. Writing † for the post-catastrophe state, the non-zero transition rates out of $n \geq 1$ are

$$q_{n,n+1} = \lambda n, \qquad q_{n,n-1} = \mu n, \qquad q_{n,\dagger} = \rho n, \tag{1.23}$$

and both $n = 0$ and † are absorbing.

Two features distinguish definition 1.3 from the birth–death process, and both matter later. The process has *two* absorbing states, reached by mechanisms that are distinguishable and not interchangeable: extinction empties the compartment gradually, catastrophe empties it in one event and releases its contents. And the catastrophe rate is proportional to the population, so the compartment becomes more likely to rupture the fuller it is — the hazard is state-dependent, which is a genuine complication rather than a parameter choice, as remark 1.4 shows. Figure 1.6 sets the two processes side by side.

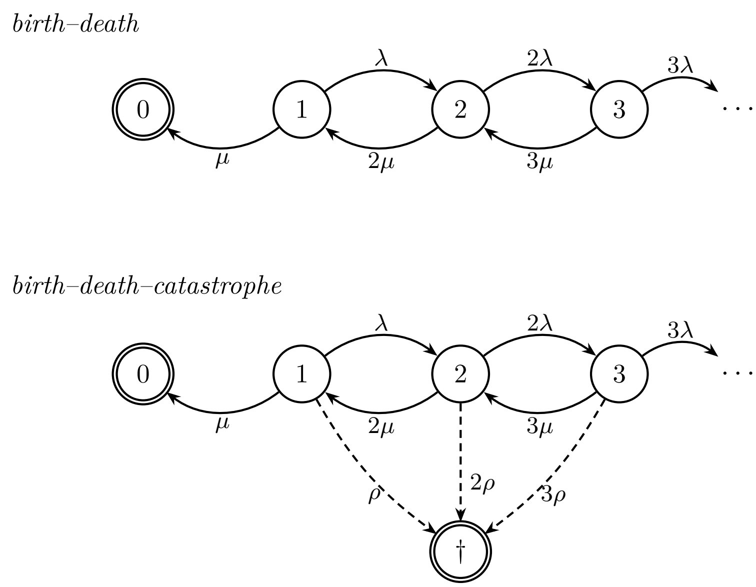


Figure 1.6: Transition-rate diagrams for the linear birth–death process (1.22) and for the birth–death–catastrophe process of definition 1.3. Double circles mark absorbing states. Every rate is proportional to the current count, because individuals act independently. The two differ in one structural respect: the birth–death chain can only be absorbed by stepping down through state 1, whereas catastrophe (dashed) removes the population from *any* state in a single transition, and does so faster the fuller the compartment is. That is why survival mass drains from state 1 alone in the first process but from every positive state in the second, as (1.26) records, and it is what makes † and 0 genuinely different endings rather than two labels for extinction.

It is useful to see the whole family at once. Let the positive state i carry birth rate λ_i, ordinary death rate μ_i and catastrophe rate κ_i, a catastrophe sending the process directly to 0. On bounded test functions the generator is

$$(\mathcal{L}f)(i) = \lambda_i\{f(i+1) - f(i)\} + \mu_i\{f(i-1) - f(i)\} + \kappa_i\{f(0) - f(i)\}, \qquad i \geq 1, \tag{1.24}$$

with $(\mathcal{L}f)(0) = 0$. Restricting to the positive states gives the *killed* subgenerator K, with $K_{i,i+1} = \lambda_i$, $K_{i,i-1} = \mu_i$ for $i \geq 2$, and

$$K_{i,i} = -(\lambda_i + \mu_i + \kappa_i). \tag{1.25}$$

The row mass missing from K is the instantaneous absorption rate: κ_i for $i \geq 2$, and $\mu_1 + \kappa_1$ for $i = 1$. Writing $T_0 = \inf\{t \geq 0 : X_t = 0\}$ and summing the positive-state forward equations,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{P}(T_0 > t) = -\mu_1 \mathbb{P}(X_t = 1) - \sum_{i\geq 1} \kappa_i \mathbb{P}(X_t = i)\,, \tag{1.26}$$

all probabilities on the right being understood before absorption. Ordinary death drains survival mass only from state 1; catastrophe drains it from every positive state, and that asymmetry is the whole difference between the two channels. The consequences for conditioning are taken up in section 1.3.2.6, and the process itself is developed in the later chapters on birth–death–catastrophe processes, with the two-type extension treated in the chapter on multi-type processes and the release of a ruptured compartment's contents into a shared medium in the chapter on compartment rupture.

Remark 1.4 (A state-dependent catastrophe is not a mean-field hazard)*.* It is tempting to replace a state-dependent catastrophe rate by a deterministic hazard evaluated at the mean population, and it does not work. The point is clearest in discrete time. Let κ be the probability that a given individual initiates catastrophe at a step, let $\widetilde{Z}_i$ be the branching population with catastrophe switched off, and let $H_n = \sum_{i=0}^{n-1} \widetilde{Z}_i$ be the cumulative exposure. Conditioning on the entire catastrophe-free genealogy, every individual present independently declines to trigger, so

$$\mathbb{P}(\text{no catastrophe before } n) = \mathbb{E}\Big((1-\kappa)^{H_n}\Big)\,. \tag{1.27}$$

This identity is exact. Replacing H_n by $\mathbb{E}(H_n)$ is not: the map $h \mapsto (1-\kappa)^h$ is convex, so Jensen's inequality gives $\mathbb{E}\Big((1-\kappa)^{H_n}\Big) \geq (1-\kappa)^{\mathbb{E}(H_n)}$, and the substitution systematically understates the chance of getting through. The failure is structural, not a matter of accuracy — the expectation in (1.27) is a nonlinear functional of the whole path, and no deterministic hazard closes it. Two further cautions belong with it. Non-catastrophe is not the same event as population survival: the probability of being both unruptured and non-zero at generation n is $\mathbb{E}\Big(\mathbf{1}\{\widetilde{Z}_n > 0\}(1-\kappa)^{H_n}\Big)$, for which Jensen supplies no bound at all. And once a trigger occurs every surviving lineage is stopped together, so the killed lineages are coupled and cannot be treated as independent after the fact — although their family trees and trigger variables do factorise *before* any trigger, which is what makes (1.27) correct. Closing the model properly therefore requires a marked or multivariate generating function, or the killed-generator viewpoint of (1.25); the elementary calculation this remark displaces, and the size of the gap it leaves, are recorded in section 1.A.5.

1.2.3 Time-inhomogeneous processes

All the chains above are time homogeneous: the rates in (1.16) depend on the state but not on the clock. Dropping that assumption gives a *time-inhomogeneous* chain, with generator $Q(t)$ and transition probabilities depending on both endpoints,

$$p_{ij}(s,t) = \mathbb{P}(X_t = j \mid X_s = i) \qquad (s \leq t), \tag{1.28}$$

in place of the single-argument $p_{ij}(t-s)$. The Chapman–Kolmogorov identity survives in the form $P(s,t) = P(s,u)P(u,t)$ for $s \leq u \leq t$, and differentiating it gives forward and backward equations

$$\frac{\partial}{\partial t}P(s,t) = P(s,t)\,Q(t), \qquad \frac{\partial}{\partial s}P(s,t) = -Q(s)\,P(s,t), \tag{1.29}$$

but the solution is no longer the matrix exponential $\mathrm{e}^{(t-s)Q}$ unless the generators at different times commute. In general one is left with a time-ordered product, and the methods of section 1.3.2 apply only after that complication has been handled.

Much survives nonetheless. The competing-clocks decomposition of proposition 1.1 still holds instantaneously, so direct simulation still applies with a time-dependent total rate — at the cost of thinning or of integrating the rate to find the next event time. And a linear birth–death process with time-varying rates $\lambda(t)$, $\mu(t)$ still has a generating function satisfying a first-order partial differential equation, so the method of characteristics of section 1.3.3 remains available; the characteristics simply acquire time-dependent coefficients.

1.2.3.1 The time-inhomogeneous linear birth–death process

The workhorse of the inhomogeneous setting is the process of section 1.2.2.3 with rates that are prescribed functions of the clock: each individual gives birth at rate $\lambda(t)$ and dies at rate $\mu(t)$, both taken locally integrable. The master equation is (1.57) with time-dependent coefficients, and the generating function accordingly satisfies

$$\frac{\partial G}{\partial t} = \left(\lambda(t)z^2 - (\lambda(t) + \mu(t))z + \mu(t)\right)\frac{\partial G}{\partial z}, \qquad G(z, 0) = z^N, \tag{1.30}$$

which is (1.58) with its quadratic coefficient set in motion. The equation is still first order, so section 1.3.3 still applies; what has been lost is that the coefficient no longer factorises once and for all, and with it the closed solution (1.59).

Two quantities survive intact whatever the rates do. The mean obeys the same closed equation as before, $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}(X_t) = (\lambda(t) - \mu(t))\mathbb{E}(X_t)$, because the rates are still linear in the count, so

$$\mathbb{E}(X_t) = N \exp\left(\int_0^t (\lambda(s) - \mu(s))\mathrm{d}s\right), \tag{1.31}$$

and only the integrated net growth enters, not how it was distributed over the interval. And the extinction probability $u(t) = \mathbb{P}(X_t = 0)$ from a single founder satisfies the Riccati equation

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \mu(t) - (\lambda(t) + \mu(t))u + \lambda(t)u^2, \qquad u(0) = 0, \tag{1.32}$$

which is the backward equation of section 1.3.2.1 with time-dependent coefficients: over the first small interval either the founder dies, contributing $\mu(t)\Delta t$, or it divides, contributing $\lambda(t)\Delta t\, u(t)^2$ since both daughter lineages must then die out in turn, or nothing happens. This is a single scalar equation, integrated numerically without difficulty when it does not solve in closed form, and the full generating function retains the fractional-linear shape of (1.59) with its coefficients replaced by solutions of (1.32).

The same Riccati structure recurs twice more below, and the three occurrences are worth keeping apart. It appears as the continuum approximation (1.100) to the discrete survival recursion, and as the characteristic equation (1.121) of the absorption–birth–death model. Only the middle one is an approximation; (1.32) and (1.121) are exact consequences of their models, and the difference is exactly what section 1.A.2 turns on.

1.2.3.2 The logistic speciation model

The standard application of that framework is a population whose per-capita birth rate falls as the population fills the space available to it. In the macroevolutionary reading of that idea the counted individuals are species and the births are speciation events [3]: let X_t be the standing diversity, let each species go extinct at constant rate μ, and let it speciate at the diversity-dependent rate

$$\lambda(n) = \lambda_0\left(1 - \frac{n}{K}\right), \qquad \lambda_0 > \mu, \tag{1.33}$$

when the standing diversity is n, with K the niche capacity.

(1.33) is not yet of the form treated above, and the distinction is the whole point of the subsection: the rate depends on the *state*, not on the clock. The chain it defines is nonlinear, its moment equations do not close, and none of the machinery of section 1.3.2 reaches it directly. The tractable surrogate replaces the random diversity inside (1.33) by a deterministic mean-field trajectory $\bar{N}(t)$, which turns the rate into a prescribed function of time and the chain into a time-inhomogeneous *linear* process, to which section 1.2.3.1 does apply. That replacement is a modelling step and not an identity. It is the same substitution of a mean for a random census that remark 1.4 shows to fail outright for a state-dependent catastrophe, and what separates the two cases is only what is being asked of it: here it defines a tractable comparison process, there it was asked to return an exact survival probability and did not. Everything that follows is accordingly a statement about the surrogate, not about the chain (1.33) defines.

The mean-field diversity solves the logistic equation

$$\frac{\mathrm{d}\bar{N}}{\mathrm{d}t} = (\lambda(\bar{N}) - \mu)\bar{N} = \gamma\,\bar{N}\left(1 - \frac{\bar{N}}{N_*}\right), \qquad \gamma = \lambda_0 - \mu, \quad N_* = K\left(1 - \frac{\mu}{\lambda_0}\right), \tag{1.34}$$

whose equilibrium N_* is the diversity at which speciation balances extinction. From $\bar{N}(0) = N_0$,

$$\bar{N}(t) = \frac{N_*}{1 + B\mathrm{e}^{-\gamma t}}, \qquad B = \frac{N_*}{N_0} - 1, \tag{1.35}$$

and the surrogate speciation rate $\lambda(t) = \lambda_0\big(1 - \bar{N}(t)/K\big)$ is

$$\lambda(t) = \mu + \frac{\gamma B\mathrm{e}^{-\gamma t}}{1 + B\mathrm{e}^{-\gamma t}}, \qquad \lambda(0) = \lambda_0\left(1 - \frac{N_0}{K}\right), \tag{1.36}$$

relaxing monotonically from $\lambda(0)$ down to μ. The integrated net growth that (1.31) calls for is then available in closed form,

$$\int_0^t (\lambda(s) - \mu)\mathrm{d}s = \ln\left(\frac{1 + B}{1 + B\mathrm{e}^{-\gamma t}}\right), \tag{1.37}$$

so that the mean diversity of the surrogate process is

$$\mathbb{E}(X_t) = N_0\,\frac{1 + B}{1 + B\mathrm{e}^{-\gamma t}} \xrightarrow[t\to\infty]{} N_0(1 + B) = N_*, \tag{1.38}$$

recovering the deterministic equilibrium, as a linear surrogate built on that equilibrium must. Its extinction probability follows by integrating (1.32) with the rate (1.36), and no longer in closed form. Figure 1.7 plots (1.38) for three founding diversities.

Two limits are worth reading off (1.36). At early times $\lambda(t) \approx \lambda(0) > \mu$ and the surrogate is indistinguishable from a homogeneous supercritical process; at late times $\lambda(t) \downarrow \mu$ and it approaches criticality from above. The model therefore drifts, over its own lifetime, into the regime whose discrete-generation counterpart causes the difficulties documented in section 1.A — a fixed carrying capacity manufactures a near-critical process out of a comfortably supercritical one, and does so without any parameter being changed by hand.

1.2.4 Coupled ODE–CTMC systems

The final class leaves the purely stochastic setting. In a *coupled ODE–CTMC system* a continuous, deterministic state $\mathbf{y}(t) \in \mathbb{R}^d$ evolves alongside a discrete, stochastic state $X_t \in \mathcal{S}$, and the two influence one another. Three couplings are possible, and they are genuinely different.

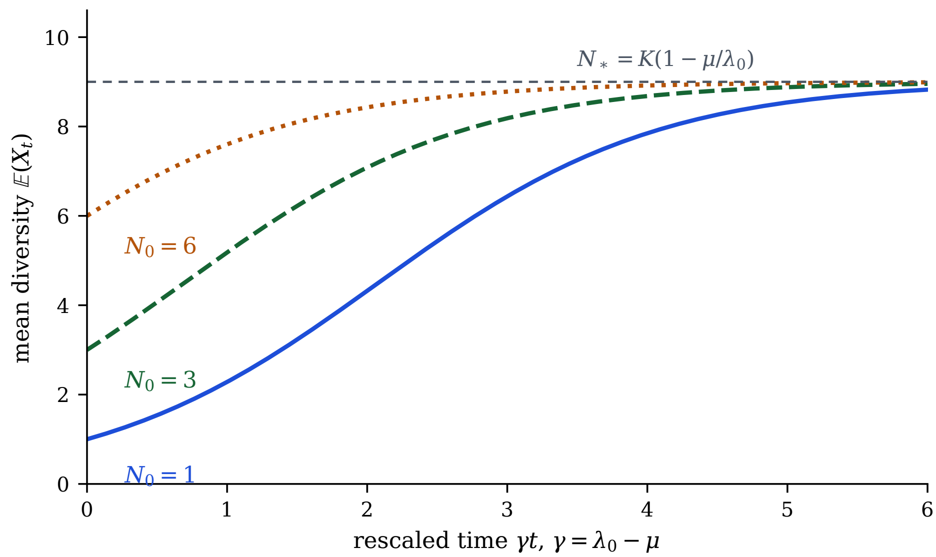


Figure 1.7: Mean diversity (1.38) of the logistic speciation surrogate for founding diversities $N_0 = 1, 3, 6$, with $\mu/\lambda_0 = 0.1$ and $K = 10$, so that $N_* = 9$; the abscissa is the rescaled time γt. Every founding diversity relaxes monotonically to N_*, and the mean of the surrogate coincides with the deterministic solution (1.35) exactly, because the surrogate is linear and its mean therefore closes. The smallest cohort spends a long early interval in near-exponential growth before the diversity dependence binds — and it is over precisely that interval that the surrogate is least trustworthy, since it is there that a single realisation is least like its mean.

(i) *The differential equation depends on the chain.* The vector field is indexed by the discrete state,

$$\frac{\mathrm{d}\mathbf{y}}{\mathrm{d}t} = F(\mathbf{y}(t),\, X_t), \tag{1.39}$$

with X_t an autonomous Markov chain. Between jumps the trajectory follows a fixed flow; at each jump the flow switches. The pair $(\mathbf{y}(t), X_t)$ is a Markov process on $\mathbb{R}^d \times \mathcal{S}$ even though neither component is one alone, and it is a piecewise-deterministic Markov process in the standard classification [7].

(ii) *The chain depends on the differential equation.* The transition rates are functions of the continuous state,

$$q_{ij} = q_{ij}(\mathbf{y}(t)), \tag{1.40}$$

with $\mathbf{y}$ following a prescribed trajectory. This is a time-inhomogeneous chain in the sense of section 1.2.3, with the inhomogeneity generated by the flow rather than imposed directly, and the remarks there apply.

(iii) *Both directions at once.* The vector field depends on the discrete state and the rates depend on the continuous one:

$$\frac{\mathrm{d}\mathbf{y}}{\mathrm{d}t} = F(\mathbf{y}(t), X_t), \qquad q_{ij} = q_{ij}(\mathbf{y}(t)). \tag{1.41}$$

Neither component can now be solved first and substituted into the other, and the joint process must be treated as a whole.

Case (i) alone is already enough to break the methods of section 1.3.2 as stated, because the state space is no longer countable and the generator acquires a transport term: for suitable test functions $f(\mathbf{y}, i)$,

$$(\mathcal{L}f)(\mathbf{y}, i) = F(\mathbf{y}, i) \cdot \nabla_{\mathbf{y}} f(\mathbf{y}, i) + \sum_{j \neq i} q_{ij}(\mathbf{y})\{f(\mathbf{y}, j) - f(\mathbf{y}, i)\}. \tag{1.42}$$

The first term is the deterministic drift along the flow and the second is the familiar jump part; (1.42) covers all three cases above, with the $\mathbf{y}$-dependence of the rates dropped in case (i) and the i-dependence of F dropped in case (ii). Written this way the coupled system is a recognisable object, and the corresponding forward equation is a system of transport equations indexed by $\mathcal{S}$ rather than a system of ordinary differential equations.

Figure 1.8 shows what the two-way case (1.41) looks like in a trajectory. The system drawn there is generic — it is chosen only to make the mechanism visible, and is not a model used elsewhere in this thesis — but it displays the feature that distinguishes coupling (iii) from the two one-way cases. Because the switching rate rises with y, the intervals spent in the second discrete state are not scattered uniformly in time: they cluster where the continuous variable has climbed, and each such interval then pulls y back down. Neither component can be solved first and substituted into the other, because each is continually re-shaping the other's behaviour.

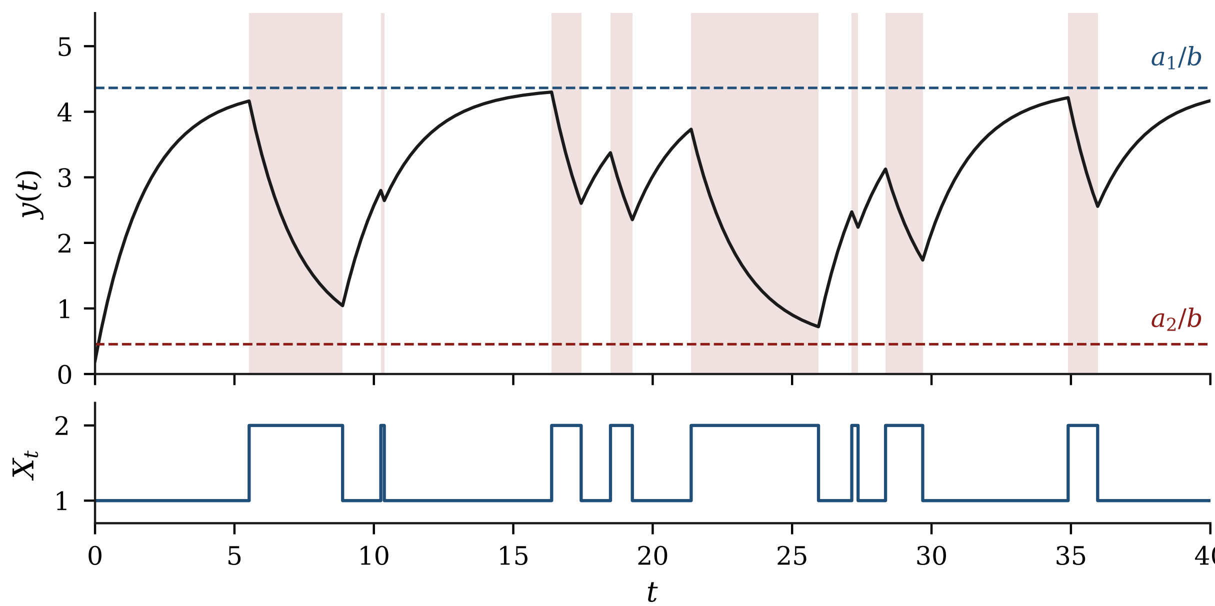


Figure 1.8: A trajectory of a coupled ODE–CTMC system of the two-way form (1.41). The continuous variable follows $\mathrm{d}y/\mathrm{d}t = a(X_t) - by$, so it relaxes towards a_1/b while $X_t = 1$ and towards a_2/b while $X_t = 2$ (dashed lines); the chain switches $1 \to 2$ at rate κy, which depends on the continuous state, and back at a constant rate. Upper panel: $y(t)$, with the intervals spent in state 2 shaded. Lower panel: the discrete component X_t. The visits to state 2 concentrate where y is large, which is the y-dependence of the rate made visible; the resulting descents then remove the very condition that made them likely. The joint process $(y(t), X_t)$ is Markov although neither component is, and its generator is (1.42). The parameters are illustrative only.

1.2.4.1 Simulating a coupled system

The competing-clocks construction of proposition 1.1 extends to (1.41) with one qualification: the rates are no longer constant between events, because $\mathbf{y}$ moves. Given the current state $(\mathbf{y}, i)$, integrate the flow with the discrete component held at i to obtain $\mathbf{y}_s$ for $s \geq t$; the total jump rate $q_i(\mathbf{y}_s) = \sum_{j \neq i} q_{ij}(\mathbf{y}_s)$ is then a known function of s, and the next jump time T has survival function

$$\mathbb{P}(T > s) = \exp\left(-\int_t^s q_i(\mathbf{y}_u)\,\mathrm{d}u\right), \tag{1.43}$$

exactly as in section 1.2.3; the destination is then drawn with probability $q_{ij}(\mathbf{y}_T)/q_i(\mathbf{y}_T)$, the state is updated, and the procedure repeats. Where q_i admits a constant majorant the integral can be inverted by rejection, and otherwise by quadrature. Analytically the position is less comfortable. If the coupling is weak one closes the system by replacing $\mathbf{y}$ by its mean-field trajectory — which returns a time-inhomogeneous chain in the sense of section 1.2.3, and the logistic surrogate of section 1.2.3.2 is exactly that closure — or by replacing X_t by its mean, which returns a closed differential equation. When the coupling is strong neither closure is exact, and the working objects become joint expectations satisfying hybrid moment equations that mix the jump generator with the drift.

1.2.4.2 A schematic instance: rupture into a shared medium

It is worth seeing the framework carry one concrete mechanism, and the natural one here is suggested by definition 1.3. Let X_t be the contents of a compartment, evolving as the birth–death–catastrophe process of that definition, and let $y(t) \geq 0$ be the concentration of material in a medium the compartment sits in. Between catastrophes the medium clears at a constant per-unit rate δ,

$$\frac{\mathrm{d}y}{\mathrm{d}t} = -\delta y, \tag{1.44}$$

and at a catastrophe time τ the entire contents of the compartment are released into the medium at once,

$$y(\tau+) = y(\tau-) + c\,X_{\tau-}, \tag{1.45}$$

with $c > 0$ the material carried per individual. That is the first arrow of the coupling: the discrete component drives the flow. The second is equally present, since the catastrophe rate ρ, and any rate at which a surviving compartment takes material back up, may themselves be functions of the current concentration, so that a fuller medium hastens the next rupture. Both dependences are then active and the system is of the two-way form (1.41).

This is a schematic: it fixes the mechanism and the shape of the state, and the models actually built on it are developed in the chapter on compartment rupture. Two of its features are worth isolating even so, because they survive into every instance and neither is visible in (1.42) as written. The continuous variable is *impulsive*: (1.45) is a jump map, not a vector field, so the natural description is a flow punctuated by resets rather than a single differential equation, and the generator acquires a corresponding jump term in y. And the quantity released, $X_{\tau-}$, is the state of the chain at a stopping time rather than a parameter of the model: what the rupture finds is whatever the conditioned process happens to hold, so its law is the quasi-stationary problem of section 1.3.2.6 evaluated at a random time. That is why systems of this kind are approached with the generating-function and first-step machinery of section 1.3 rather than closed by simulation alone. Figure 1.9 shows one realisation.

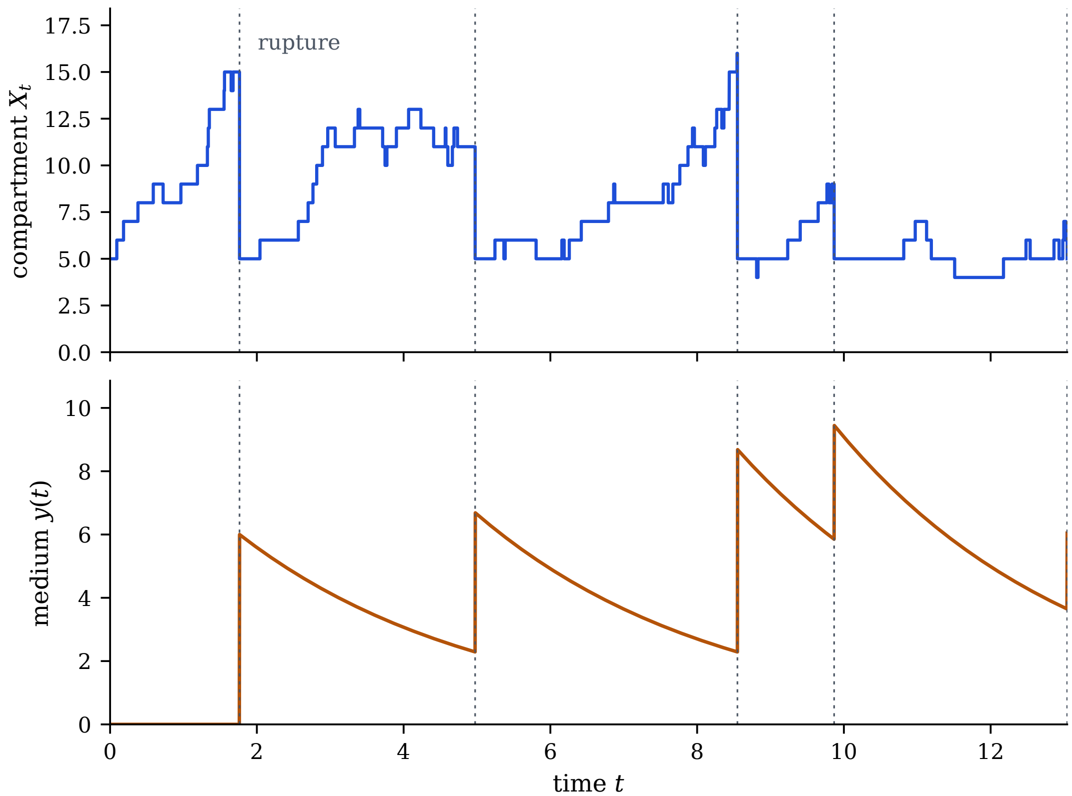


Figure 1.9: One exact event-driven realisation of the schematic (1.44) and (1.45), with $\lambda = 0.55$, $\mu = 0.25$, $\rho = 0.08$, $X_0 = 5$, $\delta = 0.30$ and $c = 0.4$; so that the picture continues past the first rupture, the compartment is re-seeded with X_0 individuals after each one. Upper panel: the compartment contents X_t, growing between ruptures and collapsing at them. Lower panel: the medium $y(t)$, decaying exponentially between ruptures and receiving the impulsive release $c\,X_{\tau-}$ at each. Dotted verticals mark the rupture times, aligned across the panels. The releases differ in size because each is the compartment's state at the moment its catastrophe clock happened to fire, and a rate proportional to the contents biases that state upwards, a fuller compartment rupturing sooner. That is why (1.45) cannot be replaced by a fixed increment.

1.3 Methods for Markov chains

Having set out the processes, we turn to the techniques for extracting information from them. The organising idea is the same throughout and is worth stating once before it is applied: *condition on one step, and solve the self-consistent equation that results.* In discrete time the step is a generation and the equation is a functional iteration; in continuous time the step is the first jump and the equation is a differential one; for generating functions the step is an infinitesimal interval and the equation is a partial differential equation. The three settings look different and are the same gesture.

1.3.1 Discrete-time methods

Discrete time is treated briefly here, for a specific reason. The techniques below reduce every question about the Galton–Watson process of section 1.2.1.2 to a question about the orbit of the quadratic map (1.13), and the method that actually resolves that question — the Koenigs linearisation, which conjugates the map to a multiplication and turns the orbit into a geometric sequence — is the subject of the following chapter. What follows is the machinery needed to reach that point.

1.3.1.1 First-step analysis

When the question is a hitting probability or a mean absorption time rather than a full time course, conditioning on the first step replaces the evolution problem by an algebraic one, because after that step the chain starts afresh from wherever it landed.

Proposition 1.5 (First-step principle)**.** *Let $\{X_n\}$ be a discrete-time Markov chain with transition matrix P and absorbing set $\mathcal{A}$, and let h_i be the probability of ever reaching $\mathcal{A}$ from state i. Then $h_i = 1$ for $i \in \mathcal{A}$, and for every transient i,*

$$h_i = \sum_j P_{ij}\, h_j. \tag{1.46}$$

The same conditioning applied to a continuous-time chain with generator Q gives

$$h_i = \sum_{j\neq i} \frac{q_{ij}}{q_i}\, h_j, \tag{1.47}$$

and the expected time to absorption satisfies $k_i = 1/q_i + \sum_{j\neq i}(q_{ij}/q_i)k_j$.

Proof. Absorption occurs eventually if and only if it occurs from the state reached at the first step; conditioning on that state and using the Markov property gives (1.46). In continuous time the chain waits an $\mathrm{Exp}(q_i)$ time and then, by proposition 1.1, jumps to j with probability q_{ij}/q_i independently of the waiting time, which gives (1.47); adding the expected holding time $1/q_i$ to the same decomposition gives the equation for k_i. □

Note what (1.47) does not contain: the holding times have vanished. Hitting *probabilities* for a continuous-time chain depend only on the embedded jump chain of section 1.2.1.1, not on the rates themselves. Mean hitting times do depend on the rates, which is why $1/q_i$ appears in the companion system. The gambler's-ruin formula (1.7) is the two-term instance of (1.46), and section 1.3.2.4 applies it to the birth–death process.

For a branching process the first-step decomposition produces something better than a linear system. Because the offspring of the founder generate independent subpopulations, conditioning on the first generation gives a *fixed-point equation*: the ultimate extinction probability s from a single individual satisfies

$$s = \phi(s), \tag{1.48}$$

ϕ being the offspring generating function. That is the whole content of proposition 1.6.

1.3.1.2 Generating functions

For a random variable X taking values in $\{0, 1, 2, \dots\}$ with $p_k = \mathbb{P}(X = k)$, the probability generating function is

$$\phi(z) = \mathbb{E}\Big(z^X\Big) = \sum_{k=0}^{\infty} p_k z^k, \tag{1.49}$$

convergent at least for $|z| \le 1$. The series determines the distribution, since $p_k = \phi^{(k)}(0)/k!$, and moments are read off at $z = 1$:

$$\phi(1) = 1, \qquad \phi'(1) = \mathbb{E}(X)\,, \qquad \phi''(1) = \mathbb{E}(X(X-1))\,. \tag{1.50}$$

Two structural properties do most of the work. If X and Y are independent then $\phi_{X+Y} = \phi_X \phi_Y$, so sums become products. And if N is an independent count with probability generating function ψ, the random sum $X_1 + \cdots + X_N$ of independent copies of X has probability generating function $\psi \circ \phi$: *composition* of generating functions corresponds to compounding, which is exactly the structure of (1.8) and hence the source of (1.10).

For a pair of counts the definition extends to

$$G(x, y) = \mathbb{E}\Big(x^X y^Y\Big) = \sum_i \sum_j \mathbb{P}(X = i,\ Y = j)\ x^i y^j, \tag{1.51}$$

with marginals recovered by setting the other variable to one; the absorption models of section 1.3.3 need this form, and section 1.D records how to invert it. The same construction with n arguments handles vector-valued states and is the prepared ground for the multi-type models of later chapters; it is not developed here.

1.3.1.3 Functional iteration

Applying (1.10) repeatedly, the generating function of Z_n is the n-fold composite $\phi_n = \phi^{\circ n}$, and every question about the Galton–Watson process becomes a question about iterating a single map. For the binary law (1.12) this is the quadratic recursion (1.13) for the survival probability, obtained by evaluating at $z = 0$ and taking complements.

Proposition 1.6 (Certainty of extinction)**.** *For the binary Galton–Watson process, S_n decreases to a limit S_∞ which is a fixed point of $S \mapsto 2pS - pS^2$. Consequently*

$$S_\infty = \begin{cases} 0, & p \le \frac{1}{2}, \\ \dfrac{2p-1}{p}, & p > \frac{1}{2}. \end{cases} \tag{1.52}$$

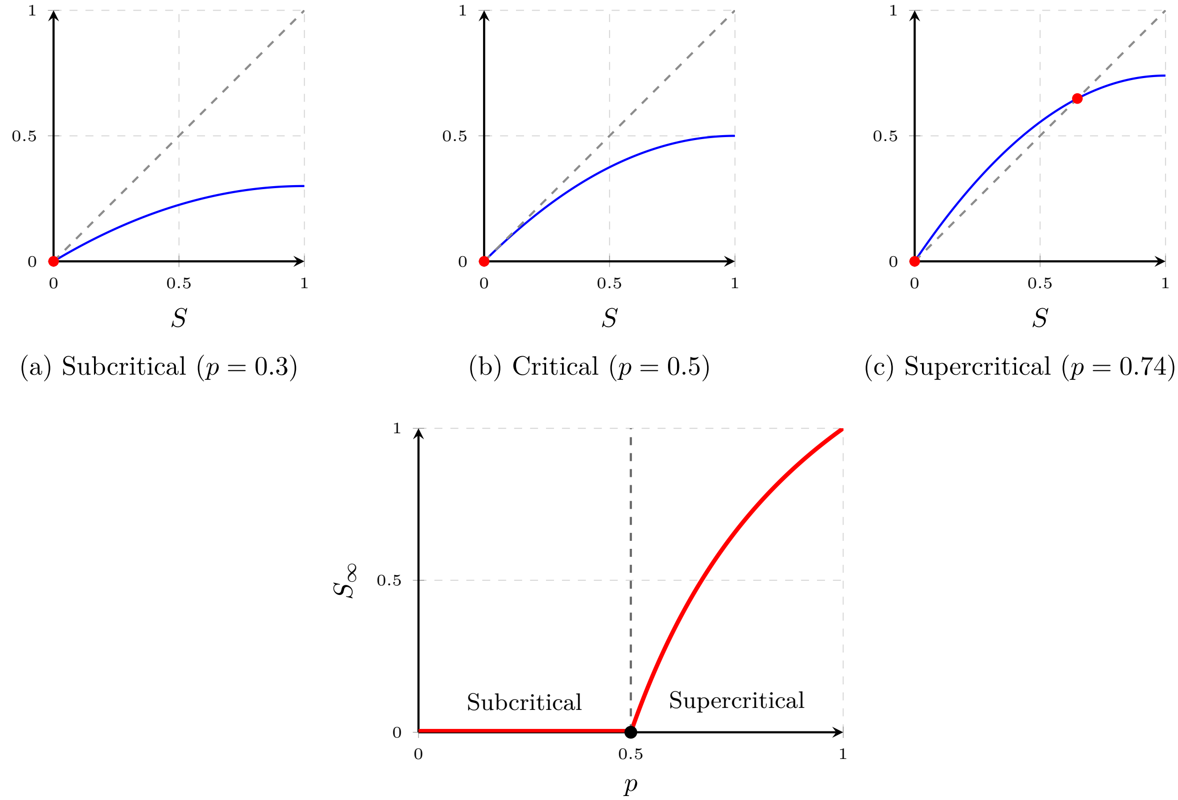
Proof. A population cannot return from extinction, so u_n is non-decreasing and bounded by 1; equivalently S_n is non-increasing and bounded below by 0, so it converges, and its limit must be a fixed point. Setting $S = 2pS - pS^2$ gives $pS\big(S + \frac{1}{p} - 2\big) = 0$, with roots $S = 0$ and $S = 2 - 1/p = (2p-1)/p$. For $p < \frac{1}{2}$ the second root is negative and carries no probabilistic meaning, so the only admissible limit is 0; at $p = \frac{1}{2}$ the two roots collide there. For $p > \frac{1}{2}$ the origin turns repelling and the positive root is attained. □

Figure 1.10 shows the transition together with the change of stability that produces it. The qualitative picture is the standard one for a transcritical bifurcation, and it is worth noticing that the whole of proposition 1.6 is a statement about the fixed points of a quadratic map: the probabilistic content has been fully converted into dynamics.

1.3.1.4 Where the discrete-time methods run out

Fixed points are the easy part. The hard question is not *whether* $S_n \to 0$ in the subcritical case but *how*, and there the elementary methods stop. Since the quadratic term in (1.13) becomes negligible once S_n is small, the decay is asymptotically geometric with ratio $2p$. The ratio is immediate; the constant standing in front of it is not,

$$A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n}, \qquad \text{so that} \qquad S_n \sim A(p)\,(2p)^n. \tag{1.53}$$



(a) Subcritical ($p = 0.3$)

(b) Critical ($p = 0.5$)

(c) Supercritical ($p = 0.74$)

(d) The transition in S_∞

Figure 1.10: Fixed points of the survival recursion (1.13). Panels (a)–(c) show the map $S \mapsto 2pS - pS^2$ (blue) against the diagonal (dashed), with the fixed points marked in red. Below and at criticality the origin is the only fixed point in $[0, 1]$ and it is attracting, so extinction is certain; at $p = \frac{1}{2}$ the map is tangent to the diagonal there. For $p > \frac{1}{2}$ the origin turns repelling and a stable positive fixed point $S_\infty = (2p - 1)/p$ separates from it, traced in panel (d). The survival probability is identically zero below the critical point and rises continuously above it.

We call $A(p)$ the *amplitude* of the decay, and use it under that name throughout. The amplitude accumulates every early nonlinear correction made before the recursion attenuates to $S_{n+1} \approx 2pS_n$, which is precisely what the late-time behaviour has already forgotten. No argument confined to late generations can therefore produce it, and no elementary formula for it is known.

The technique that addresses this is a change of coordinate. Substituting $S_n = 2w_n$ and $r = 2p$ turns (1.13) into the logistic map $w_{n+1} = rw_n(1 - w_n)$, and one asks for a coordinate in which that map becomes multiplication by r — a solution ψ_r of the Schröder equation $\psi_r(f_r(z)) = r\psi_r(z)$, the Koenigs linearising function. In that coordinate the orbit is exactly geometric and its amplitude can be read off. The following chapter carries this out, identifies $A(p)$ with twice a value of ψ_{2p}, and proves that ψ_r is hypertranscendental for every r in the subcritical window — which is why the elementary methods of this subsection could not have succeeded. Nothing further about $A(p)$ is developed here; it is used below as a named object.

1.3.2 Continuous-time methods

The continuous-time methods carry the technical weight of this chapter. They are presented on the linear birth–death process of section 1.2.2.3 throughout, both because it is the process the thesis actually uses and because it is simple enough that each method can be seen working rather than merely stated.

1.3.2.1 Forward and backward equations

Differentiating the Chapman–Kolmogorov identity $p_{ij}(t+\Delta t) = \sum_k p_{ik}(t)p_{kj}(\Delta t)$ at $\Delta t = 0$ gives the *forward* equations,

$$\frac{\mathrm{d}p_{ij}(t)}{\mathrm{d}t} = \sum_k p_{ik}(t)\, q_{kj}, \qquad \text{that is} \qquad \frac{\mathrm{d}P}{\mathrm{d}t} = PQ, \tag{1.54}$$

which condition on the last jump before time t. Differentiating in the other slot gives the *backward* equations,

$$\frac{\mathrm{d}P}{\mathrm{d}t} = QP, \tag{1.55}$$

which condition on the first jump after time 0 — first-step analysis in differential form. The two carry the same information but suit different questions: forward equations track the distribution as it spreads, backward equations track a functional of the whole future and are the natural setting for absorption probabilities and for the generating function of a branching process.

Written out for a fixed initial state, with $p_j(t)$ the probability of occupying state j, (1.54) becomes the *master equation*

$$\frac{\mathrm{d}p_j(t)}{\mathrm{d}t} = \sum_{k\neq j} q_{kj}\, p_k(t) - q_j\, p_j(t) : \tag{1.56}$$

probability flows into j from every state that can reach it and out of j at total rate q_j. Every model in this thesis is specified by writing down (1.16) and reading off (1.56).

For the birth–death rates (1.22) the master equation is

$$\frac{\mathrm{d}p_n(t)}{\mathrm{d}t} = \lambda(n-1)p_{n-1}(t) + \mu(n+1)p_{n+1}(t) - (\lambda+\mu)n\, p_n(t), \tag{1.57}$$

an infinite coupled system, and on its own it is not much use. The step that makes it tractable is to collapse it into a single equation for a generating function. Multiply (1.57) by z^n, sum over $n \geq 0$, and shift indices so that every sum is again of the form (1.49); the factors of n produced by the linear rates become derivatives ∂_z, and the three terms become $\lambda z^2 \partial_z G$, $\mu\partial_z G$ and $-(\lambda+\mu)z\partial_z G$. Hence

$$\frac{\partial G}{\partial t} = (\lambda z^2 - (\lambda+\mu)z + \mu)\frac{\partial G}{\partial z} = (\lambda z - \mu)(z-1)\frac{\partial G}{\partial z}, \qquad G(z,0) = z^N, \tag{1.58}$$

for a process started from N individuals. This is the template for every generating-function calculation in this thesis. Note where the linearity of the rates was used: it is what makes the coefficient a polynomial in z and the equation first order, and hence what puts it within reach of section 1.3.3. The quadratic coefficient factorises with roots $z = 1$ and $z = \mu/\lambda$, and the same factorisation reappears in section 1.B.2.

Solving (1.58) along its characteristics gives the classical result of Kendall [8],

$$G(z,t) = \left(\frac{\mu(1-z) - (\mu-\lambda z)\mathrm{e}^{-(\lambda-\mu)t}}{\lambda(1-z) - (\mu-\lambda z)\mathrm{e}^{-(\lambda-\mu)t}}\right)^N, \qquad \lambda \neq \mu. \tag{1.59}$$

Everything in the next four subsections is extracted from this one formula, or obtained more cheaply by a route that (1.59) then confirms.

1.3.2.2 Mean population size

Differentiating (1.59) at $z = 1$ is possible and unpleasant. It is quicker to take the mean of the master equation directly: multiplying (1.57) by n and summing, the terms telescope to

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}(X_t) = (\lambda - \mu)\mathbb{E}(X_t)\,, \qquad \mathbb{E}(X_0) = N, \tag{1.60}$$

whence

$$\mathbb{E}(X_t) = N\mathrm{e}^{(\lambda-\mu)t}. \tag{1.61}$$

The mean obeys a closed equation — exactly the deterministic rate equation one would have written down without any probability at all — and this is a special feature of *linear* rates. It is also the reason the mean is uninformative here: (1.61) is the same whether the process is a near-certain extinction with a rare enormous excursion or a steady moderate growth, and distinguishing those is the entire problem.

1.3.2.3 Variance and higher moments

The second moment is obtained the same way. Multiplying (1.57) by n^2 and summing gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}\left(X_t^2\right) = 2(\lambda - \mu)\mathbb{E}\left(X_t^2\right) + (\lambda + \mu)\mathbb{E}(X_t)\,, \tag{1.62}$$

a linear equation driven by the known mean (1.61). Solving with the integrating factor $\mathrm{e}^{-2(\lambda-\mu)t}$ and subtracting $\mathbb{E}(X_t)^2$ gives, for $\lambda \neq \mu$,

$$\mathrm{Var}(X_t) = N\,\frac{\lambda+\mu}{\lambda-\mu}\,\mathrm{e}^{(\lambda-\mu)t}\left(\mathrm{e}^{(\lambda-\mu)t} - 1\right), \tag{1.63}$$

and in the critical case $\lambda = \mu$ the limit is $\mathrm{Var}(X_t) = 2N\lambda t$. Two readings are worth recording. In the subcritical case $\lambda < \mu$ both mean and variance decay, but the variance decays more slowly — the coefficient of variation grows, and the process becomes relatively *more* random as it dies. At criticality the mean is constant and the variance grows without bound, which is the analytic signature of the heavy tail examined in section 1.A.

The pattern generalises. Multiplying the master equation by n^{ϱ} produces an equation for the ϱth moment driven by the lower ones, so the moments can be found recursively without ever solving (1.58). Equivalently, differentiating (1.58) repeatedly at $z = 1$ and using (1.50) gives the same hierarchy; the hierarchy closes at each level precisely because the rates are linear. For a process with nonlinear rates the moment equations do not close, and the generating-function route of section 1.3.3 becomes not merely convenient but necessary.

1.3.2.4 Hitting probabilities

Hitting probabilities are where first-step analysis earns its place, because they require no time-dependent solution at all. By proposition 1.5 the probability h_i of ever reaching a target set from state i satisfies (1.47), which involves only the ratios q_{ij}/q_i — the embedded jump chain. For the birth–death rates (1.22) those ratios are independent of the state,

$$\frac{q_{n,n+1}}{q_n} = \frac{\lambda n}{(\lambda+\mu)n} = \frac{\lambda}{\lambda+\mu}, \qquad \frac{q_{n,n-1}}{q_n} = \frac{\mu}{\lambda+\mu}, \tag{1.64}$$

so the embedded chain is the simple random walk of section 1.2.1.1 with $q = \lambda/(\lambda+\mu)$, and (1.7) answers the question immediately. With absorbing barriers at 0 and M, the probability of reaching M before extinction, starting from i individuals, is

$$h_i = \frac{1-(\mu/\lambda)^i}{1-(\mu/\lambda)^M} \qquad (\lambda \neq \mu), \qquad h_i = \frac{i}{M} \qquad (\lambda = \mu), \tag{1.65}$$

since $\theta = (1-q)/q = \mu/\lambda$. Letting $M \to \infty$ gives the probability of unbounded growth, $1 - (\mu/\lambda)^i$ when $\lambda > \mu$ and 0 otherwise, which is the complement of the extinction probability found independently in section 1.3.2.5. The two routes agreeing is a useful check on both.

Mean hitting times behave differently and the difference is instructive. They depend on the rates and not merely on their ratios, and at criticality $\lambda = \mu$ the mean time to absorption is infinite even though absorption is certain — the continuous-time counterpart of the phenomenon derived in full in section 1.A.

1.3.2.5 Extinction probabilities

Extinction by time t is the event $X_t = 0$, so its probability is the generating function evaluated at the origin: $p_0(t) = G(0,t)$. From (1.59),

$$p_0(t) = \begin{cases} \left(\dfrac{\mu - \mu e^{(\mu-\lambda)t}}{\lambda - \mu e^{(\mu-\lambda)t}}\right)^N, & \lambda \neq \mu, \\[2ex] \left(\dfrac{\lambda t}{1+\lambda t}\right)^N, & \lambda = \mu, \end{cases} \tag{1.66}$$

the critical case following by taking $\mu \to \lambda$ in the first line. Letting $t \to \infty$ recovers the threshold: extinction is certain when $\lambda \leq \mu$ and occurs with probability $(\mu/\lambda)^N$ when $\lambda > \mu$, which is the complement of (1.65) with $M \to \infty$, as it must be.

Figure 1.11 plots (1.66), alongside the law it leaves behind.

The survival probability is $S^{(N)}(t) = 1 - p_0(t)$, and for a single founder, $N = 1$, (1.66) rearranges to

$$S(t) = \frac{\lambda-\mu}{\lambda - \mu e^{(\mu-\lambda)t}} = \frac{\mu-\lambda}{\mu}\, e^{-(\mu-\lambda)t} \left(1 - \frac{\lambda}{\mu} e^{-(\mu-\lambda)t}\right)^{-1} \sim \frac{\mu-\lambda}{\mu}\, e^{-(\mu-\lambda)t} \tag{1.67}$$

as $t \to \infty$, in the subcritical case $\mu > \lambda$. The last form is the one that matters below: the survival probability decays exponentially with a definite *amplitude* $(\mu-\lambda)/\mu$, and because (1.67) is elementary at every t, that amplitude can simply be read off. Contrast (1.53), where the analogous amplitude is available only as a limit.

1.3.2.6 Conditional means

Absorption is often inevitable — for $\lambda < \mu$, and for the whole subcritical branching range — and yet a process on its way to absorption need not be featureless. In many cases it settles, long before

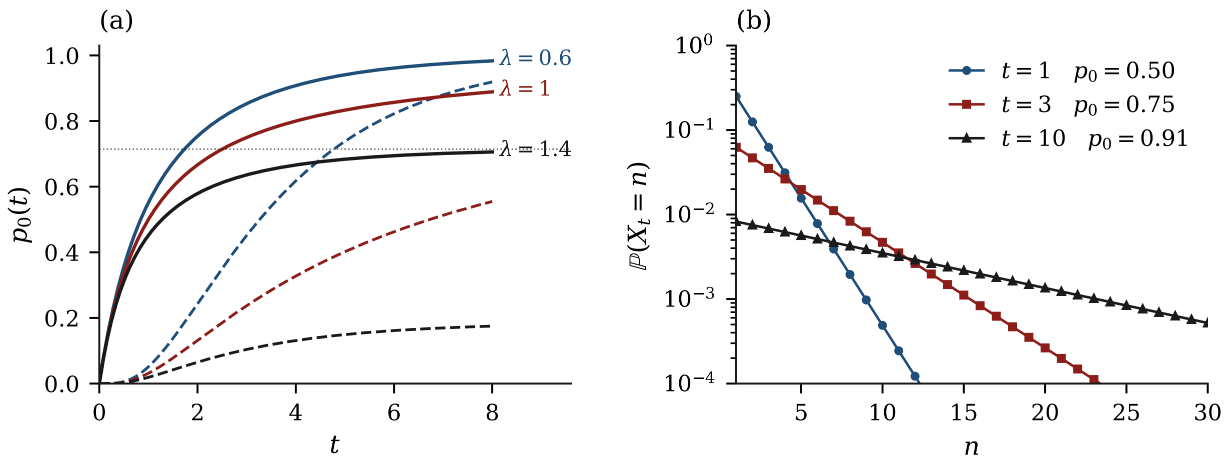


Figure 1.11: (a) The extinction probability (1.66) against time, with $\mu = 1$ and $\lambda = 0.6,\, 1,\, 1.4$; solid curves start from $N = 1$, dashed from $N = 5$. Below and at criticality every curve climbs to one, however large the founding cohort; above it the curves level off at $(\mu/\lambda)^N$, marked dotted for $N = 1$, and a larger cohort buys a permanently smaller extinction probability rather than merely a delay. (b) The critical case in detail, from one founder: the law of X_t on the positive states at $t = 1, 3, 10$, on a logarithmic scale. Absorbing mass accumulates at zero — p_0 rises from 0.50 to 0.91 — while the surviving tail flattens and stretches. The mean is exactly 1 at all three times. Neither the growing chance of extinction nor the lengthening tail is visible in that number, which is the sense in which the mean of (1.61) answers the wrong question.

it is absorbed, into a distribution that no longer changes with time: a distribution describing the population conditional on its not yet having been absorbed. The phenomenon is *quasi-stationarity* and the limit is the Yaglom limit [9]; modern accounts are Méléard and Villemonais [10] and Collet, Martínez and San Martín [11].

Definition 1.7 (Quasi-stationary and Yaglom laws)**.** Let $\{X_t\}$ be a Markov process with absorbing state 0. A probability distribution ν on the non-absorbing states is *quasi-stationary* if $\mathbb{P}(X_t = j \mid X_t \neq 0) = \nu_j$ for all $t \geq 0$ and $j \neq 0$ whenever $X_0 \sim \nu$. If, from a specified initial state, $\mathbb{P}(X_t = j \mid X_t \neq 0) \to \nu_j$, then ν is the corresponding *Yaglom limit*.

Existence of the Yaglom limit for the processes used here is classical, from Yaglom [9] with modern treatments in [4, 5, 10], and is taken from that literature rather than re-proved. What is computed below is the limiting conditional mean, explicitly, in both time scales. The distinction matters: convergence of moments is consistent with the Yaglom theorem but does not on its own establish convergence of the full conditional law.

The continuous-time constant. Since $X_t = 0$ contributes nothing to its own expectation,

$$\mathbb{E}(X_t \mid X_t > 0) = \frac{\mathbb{E}(X_t)}{S^{(N)}(t)}, \tag{1.68}$$

and for $N = 1$ substituting (1.61) and (1.67) gives, exactly,

$$\mathbb{E}(X_t \mid X_t > 0) = \frac{\mathrm{e}^{(\lambda-\mu)t}\left(\lambda - \mu\mathrm{e}^{(\mu-\lambda)t}\right)}{\lambda - \mu} = \frac{\lambda\mathrm{e}^{(\lambda-\mu)t} - \mu}{\lambda - \mu}. \tag{1.69}$$

For $\mu > \lambda$ the exponential decays and the limit is a constant,

$$\lim_{t\to\infty} \mathbb{E}(X_t \mid X_t > 0) = \frac{\mu}{\mu - \lambda} = \frac{1}{1 - \lambda/\mu} =: \frac{1}{A_\mathrm{c}}, \tag{1.70}$$

the reciprocal of the amplitude identified in (1.67). To compare this with the discrete-generation model the two parametrisations must be matched, and proposition 1.1 is what matches them: the probability that the birth clock wins is $p = \lambda/(\lambda + \mu)$, so $\lambda/\mu = p/(1-p)$, and

$$A_\mathrm{c}(p) = \frac{1-2p}{1-p} = \frac{2\varepsilon}{1+\varepsilon}, \qquad \frac{1}{A_\mathrm{c}(p)} = \frac{1-p}{1-2p}, \qquad \varepsilon := 1-2p. \tag{1.71}$$

The symbol ε is used for the distance from criticality throughout this thesis. (1.71) is elementary and complete: A_c is a rational function of p, decreasing from $A_\mathrm{c}(0) = 1$ to $A_\mathrm{c}(\frac{1}{2}^-) = 0$, with $A_\mathrm{c}(p) \to 2\varepsilon$ near criticality.

The reason a closed form was available deserves emphasis, because it is exactly what fails in discrete time. The survival probability (1.67) is elementary at *every* time, not merely asymptotically, because (1.58) is a linear first-order equation with elementary characteristics. The amplitude is a value of that solution; nothing had to accumulate.

The discrete-time constant. For the binary Galton–Watson process the same decomposition gives $\mathbb{E}(Z_n \mid Z_n > 0) = (2p)^n/S_n$, and (1.53) supplies the limit:

$$\mathbb{E}(Z_n \mid Z_n > 0) \xrightarrow[n\to\infty]{} \frac{(2p)^n}{A(p)(2p)^n} = \frac{1}{A(p)}. \tag{1.72}$$

The geometric factors cancel exactly, and the limiting conditional mean is the reciprocal of the amplitude $A(p)$. Carrying the same argument to the second moment gives a limiting conditional variance

$$\mathrm{Var}(Z_n \mid Z_n > 0) \longrightarrow \frac{1}{A(p)}\left(1 + \frac{1}{1-2p}\right) - \frac{1}{A(p)^2}, \tag{1.73}$$

again a function of p and $A(p)$ alone; the second moment behind it is computed in section 1.A.4. Figure 1.12 shows the convergence of (1.72), and shows at the same time the difficulty: the closer p lies to $\frac{1}{2}$, the higher the plateau and the longer the process takes to reach it.

The two constants compared, and the handover. Setting the two side by side,

$$A_\mathrm{c}(p) = \frac{1-2p}{1-p} \quad \text{(a closed form)}, \qquad A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n} \quad \text{(a limit, and no more)}, \tag{1.74}$$

figure 1.13 raises the obvious question: why does the continuous curve start at 1 where the discrete one starts at $\frac{1}{2}$, given that the two agree at the critical end? A factor of two at one endpoint and none at the other means that no constant rescaling of A_c can reproduce A.

That is where this chapter stops. The following chapter proves the strict bound $A(p) > \frac{1}{2}A_\mathrm{c}(p)$, shows that the ratio A/A_c increases monotonically from $\frac{1}{2}$ at $p = 0$ to 1 at $p = \frac{1}{2}^-$, accounts for both endpoints, derives exact product and series representations for $A(p)$ together with two-sided bounds and the near-critical asymptotic $A(p) \sim 2\varepsilon$, identifies $A(p)$ with a value of the Koenigs linearising function of section 1.3.1.4, and proves that this function is hypertranscendental — which is why no elementary expression for $A(p)$ has been found. For the purposes of the present chapter, two facts suffice and both are established there: the limit (1.53) exists with $0 < A(p) \le \frac{1}{2}$, so the quasi-stationary mean is at least 2; and $A(p)$ is computable to arbitrary accuracy by iterating (1.13), which is how the values quoted in figure 1.12 were obtained.

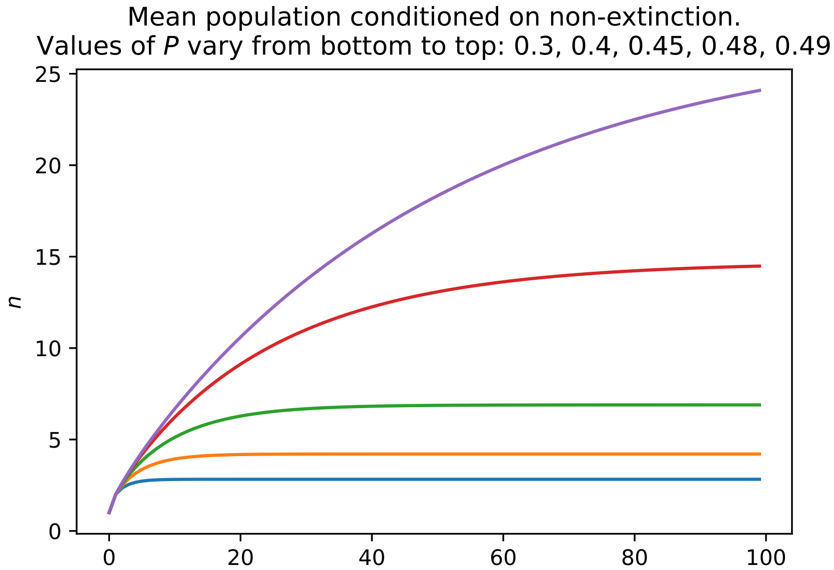


Figure 1.12: The conditional mean $(2p)^n/S_n$ against generation number n, with S_n obtained by iterating (1.13). From bottom to top, $p = 0.3$, 0.4, 0.45, 0.48, 0.49. Each curve approaches the constant $1/A(p)$; the limiting values are 2.825, 4.208, 6.893, 14.703 and 27.476. The lowest three have essentially converged by $n = 100$, but the $p = 0.49$ curve has reached only 24.1 of its limiting 27.5 — an early sign of a near-critical slowdown that the following chapter analyses and works around.

Conditioning under two absorbing mechanisms. One structural point about the birth–death–catastrophe process of definition 1.3 belongs here rather than in the following chapter, because it concerns quasi-stationarity as a method. If a distribution ν on the positive states is quasi-stationary in the sense of definition 1.7 and the relevant sums are finite, then for some $\theta > 0$

$$\nu K = -\theta\nu, \qquad \mathbb{P}_\nu(T_0 > t) = \mathrm{e}^{-\theta t}, \tag{1.75}$$

with K the killed subgenerator (1.25). Indeed quasi-stationarity writes the unnormalised positive-state law at time t as $a(t)\nu$; the semigroup property forces $a(s+t) = a(s)a(t)$, and right-continuity gives $a(t) = \mathrm{e}^{-\theta t}$. Differentiating at zero yields the left-eigenvector relation, and summing its coordinates against (1.26) identifies the decay rate as

$$\theta = \mu_1\nu_1 + \sum_{i\geq 1} \kappa_i\nu_i. \tag{1.76}$$

A quasi-stationary law is thus a left eigenvector of the killed generator and its eigenvalue is the survival decay rate — the continuous-time statement of what $A(p)$ and $(2p)^n$ do together in (1.53). One limiting case is worth isolating: if $\kappa_i \equiv \kappa$ is constant, the catastrophe clock is independent of the birth–death process, the killed semigroup factorises as $P_t^K = \mathrm{e}^{-\kappa t} P_t^{\mathrm{BD},K}$, and the extra clock shifts the decay rate by κ while leaving the conditional law unchanged. A state-dependent sequence such as $\kappa_i = i\kappa$ weights paths by their accumulated exposure and generally changes the quasi-stationary law itself. Whether a quasi-stationary distribution survives when the process is conditioned on *neither* extinction nor rupture is not settled by the argument above — the two mechanisms drain survival mass from different parts of the state space, by (1.26) — and is best approached by simulation in the first instance.

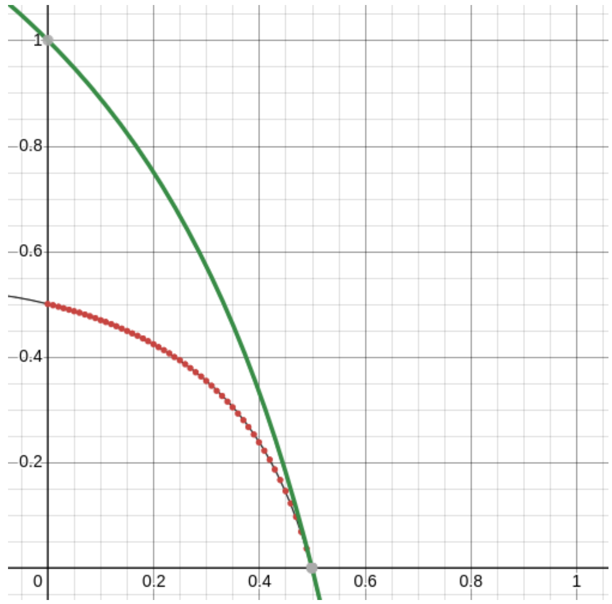


Figure 1.13: The continuous-time constant $A_c(p) = (1-2p)/(1-p)$ (green) against high-precision numerical values of the discrete-time constant $A(p)$ (red points). Both vanish at $p = \frac{1}{2}$, but at $p = 0$ the continuous curve starts at 1 and the discrete one at $\frac{1}{2}$, and the two are not related by a constant factor in between. The black line is a formula obtained by symbolic regression; it tracks the points closely but is not exact, and the following chapter both accounts for the discrepancy between the two curves exactly and explains why the fitted curve cannot be right.

1.3.3 The method of characteristics

Every generating-function equation met above is first order: (1.58) for the birth–death process, and the bivariate equations of section 1.B for the absorption models. That is not a coincidence but a consequence of linear rates, and it is worth exploiting, because first-order equations have a general solution method. This subsection presents it.

1.3.3.1 The idea

A first-order partial differential equation says that a certain directional derivative of the unknown function is prescribed. If one can find the curves along which that direction points — the *characteristics* — then along each such curve the equation collapses to an ordinary differential equation, which can be solved. The solution is then transported along the curve back to a surface where the initial data are known.

Concretely, suppose G satisfies

$$\frac{\partial G}{\partial t} = a(x,y)\frac{\partial G}{\partial x} + b(x,y)\frac{\partial G}{\partial y}, \qquad G(x,y,0) = G_0(x,y). \tag{1.77}$$

Reparametrise the independent variables as $(x,y,t) \to (\sigma,\eta,\tau)$, where σ and η label a characteristic curve and τ runs along it. Along such a curve G is a function of τ alone, so the chain rule gives

$$\frac{\mathrm{d}G}{\mathrm{d}\tau} = \frac{\partial G}{\partial t}\frac{\mathrm{d}t}{\mathrm{d}\tau} + \frac{\partial G}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}\tau} + \frac{\partial G}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}\tau}. \tag{1.78}$$

Comparing this with (1.77) rewritten as $0 = -\partial_t G + a\,\partial_x G + b\,\partial_y G$ and matching the coefficient of each partial derivative gives the *characteristic system*

$$\frac{\mathrm{d}G}{\mathrm{d}\tau} = 0, \qquad \frac{\mathrm{d}t}{\mathrm{d}\tau} = -1, \qquad \frac{\mathrm{d}x}{\mathrm{d}\tau} = a(x,y), \qquad \frac{\mathrm{d}y}{\mathrm{d}\tau} = b(x,y). \tag{1.79}$$

Four ordinary differential equations have replaced one partial differential equation. The first says that G is constant along characteristics, so G depends only on the labels (σ,η); the second says the parameter runs backwards in time, which is what allows the initial surface to be reached from any (x,y,t); the last two are the curves themselves.

The parametrisation is fixed by anchoring the labels to the initial surface,

$$x(\sigma,\eta,0) = \sigma, \qquad y(\sigma,\eta,0) = \eta, \qquad t(\sigma,\eta,0) = 0, \tag{1.80}$$

so that $G(\sigma,\eta,\tau) = G_0(\sigma,\eta)$. The whole problem then reduces to a change of variables: express σ and η in terms of (x,y,t) by integrating the characteristic equations and substitute. That final substitution is where the work lies, and it is the step that repays a worked example.

Figure 1.14 shows the geometry for the x-equation of the example that follows. The picture is worth holding onto, because it is the same in every model below: a family of curves foliating the domain, each carrying one value of G up from the initial surface, so that evaluating G anywhere means identifying which curve one is standing on.

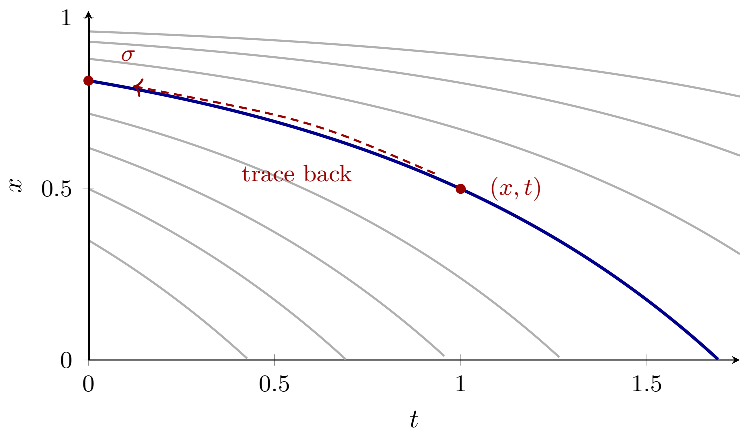


Figure 1.14: Characteristics of the x-equation (1.87), $\mathrm{d}x/\mathrm{d}\tau = \mu(1-x)$ with $\tau = -t$, drawn for $\mu = 1$. Each grey curve is one characteristic, $x(t) = 1 - (1-\sigma)\mathrm{e}^{\mu t}$, labelled by the point σ at which it meets the initial surface $t = 0$, drawn as the heavy vertical line; together they foliate the domain, and exactly one passes through any given (x,t). Since G is constant along characteristics, its value at a point is its value at the foot of that point's curve — so solving the equation amounts to running the curve backwards from (x,t) to $t = 0$ and reading off σ, here $\sigma = 1 - (1-x)\mathrm{e}^{-\mu t}$. The full problem is three-variate; this is its projection onto the (x,t) plane, which is where the cascade starts, because (1.87) does not involve y.

1.3.3.2 A worked example: absorption–death

The example is chosen so that the answer can be checked. Consider a single compartment that takes in particles from a surrounding medium, in which a resident particle may then die; figure 1.15

sets out the geometry, which is common to every absorption model in this thesis. Let X_t count particles inside the compartment and Y_t those outside, and let the events be

$$
\begin{aligned}
\mathbb{P}(\Delta X = 1,\ \Delta Y = -1 \mid Y_t = j) &= j\alpha\,\Delta t, && \text{(absorption)}\\
\mathbb{P}(\Delta X = -1,\ \Delta Y = 0 \mid X_t = i) &= i\mu\,\Delta t, && \text{(death)}\\
\mathbb{P}(\Delta X = 0,\ \Delta Y = 0 \mid X_t = i,\ Y_t = j) &= 1 - (j\alpha + i\mu)\Delta t, && \text{(no event)}
\end{aligned}
$$

with terms of $o(\Delta t)$ suppressed. Each exterior particle is independently taken up at rate α and each interior one dies at rate μ. The initial condition is $X_0 = 0$, $Y_0 = N$.

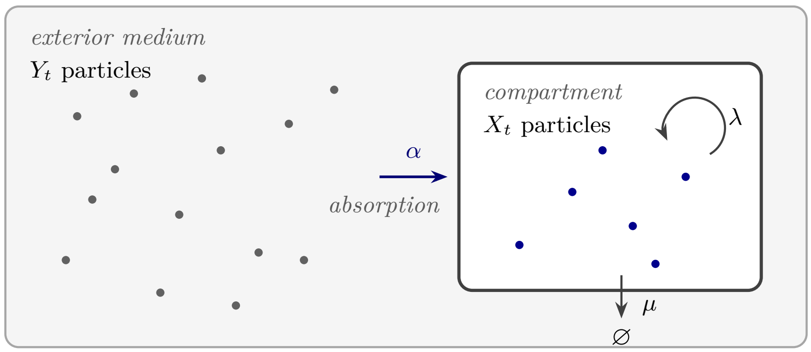


Figure 1.15: The single-compartment absorption models. Particles begin in the exterior medium and are taken up one at a time, each independently at rate α, so the absorption channel carries total rate $Y_t\alpha$; once inside, a particle dies at rate μ and, in the fullest version of the model, divides at rate λ. Only the three transitions drawn here occur, and each is proportional to the count in the compartment it acts on — which is what makes the generating-function equation first order and puts it within reach of section 1.3.3.1. Switching channels off gives the family in order of difficulty: with $\lambda = \mu = 0$ the particles are merely relocated (section 1.B.1); with $\lambda = 0$ this is the worked example below; with all three active the interior state space becomes infinite and section 1.B.2 must solve the equation directly. Note that nothing returns from the compartment to the medium, which is why the exterior count is unaffected by the interior dynamics.

The forward equations (1.56) read

$$\frac{\mathrm{d}p_{i,j}}{\mathrm{d}t} = \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} - (\alpha j + \mu i)p_{i,j}, \tag{1.81}$$

and the construction of section 1.3.2.1, applied to the bivariate generating function (1.51), converts them into

$$\frac{\partial G}{\partial t} = \mu(1-x)\frac{\partial G}{\partial x} + \alpha(x-y)\frac{\partial G}{\partial y}, \qquad G(x,y,0) = y^N. \tag{1.82}$$

Why this example is the right one to work through: because the answer is already available by an independent route. Without births the particles never interact, so each of the N of them is independently in one of three states at time t — outside, inside, or dead — and the generating function is a power of the single-particle one. The single-particle state probabilities satisfy $\mathrm{d}p_{0,1}/\mathrm{d}t = -\alpha p_{0,1}$ and $\mathrm{d}p_{1,0}/\mathrm{d}t = \alpha p_{0,1} - \mu p_{1,0}$, giving

$$p_{0,1}(t) = \mathrm{e}^{-\alpha t}, \qquad p_{1,0}(t) = \frac{\alpha}{\mu - \alpha}\left(\mathrm{e}^{-\alpha t} - \mathrm{e}^{-\mu t}\right), \qquad p_{0,0}(t) = 1 - p_{0,1}(t) - p_{1,0}(t), \tag{1.83}$$

plotted in figure 1.16, and hence

$$G(x,y,t) = \left(1 - \tfrac{\mu}{\mu-\alpha}\mathrm{e}^{-\alpha t} + \tfrac{\alpha}{\mu-\alpha}\mathrm{e}^{-\mu t} + \tfrac{\alpha}{\mu-\alpha}(\mathrm{e}^{-\alpha t} - \mathrm{e}^{-\mu t})x + \mathrm{e}^{-\alpha t}y\right)^N. \tag{1.84}$$

So (1.82) can be solved by characteristics with the answer known in advance, which is exactly what one wants when learning to trust a method.[1]

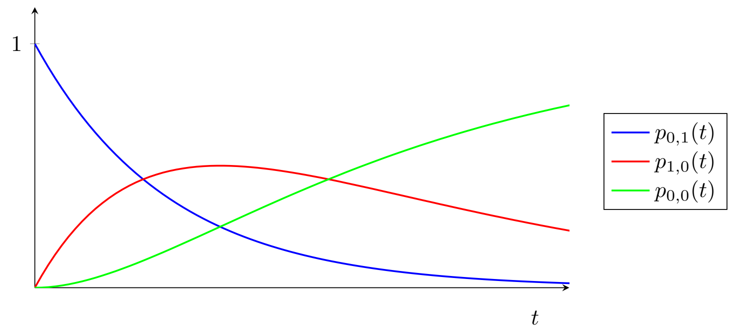


Figure 1.16: State probabilities for the single-particle absorption–death model, (1.83). The particle starts outside ($p_{0,1}$, blue), is absorbed into the compartment ($p_{1,0}$, red, which rises and then decays as death takes over), and finally dies ($p_{0,0}$, green, absorbing).

Comparing (1.82) with (1.77) gives $a = \mu(1-x)$ and $b = \alpha(x-y)$, so the characteristic system (1.79) is

$$\frac{\mathrm{d}G}{\mathrm{d}\tau} = 0, \tag{1.85}$$
$$\frac{\mathrm{d}t}{\mathrm{d}\tau} = -1, \tag{1.86}$$
$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = \mu(1-x), \tag{1.87}$$
$$\frac{\mathrm{d}y}{\mathrm{d}\tau} = \alpha(x-y). \tag{1.88}$$

The four equations are taken in turn, and none of them is hard.

The equation for G, (1.85). G is constant along characteristics, so it is a function of (σ, η) alone. Under (1.80) the initial condition $G(x,y,0) = y^N$ becomes $G(\sigma,\eta,0) = \eta^N$, so

$$G(\sigma,\eta,\tau) = \eta^N. \tag{1.89}$$

The whole problem now reduces to expressing η in terms of (x,y,t).

The equation for t, (1.86). Integrating gives $t = -\tau + c(\sigma,\eta)$, and (1.80) fixes $c = 0$:

$$t = -\tau. \tag{1.90}$$

[1] I am grateful to Benjamin Sharp for the advice on applying the method of characteristics to three-variate problems that unlocked this calculation.

The equation for x, (1.87). Separating variables gives $1 - x = c(\sigma, \eta)\mathrm{e}^{-\mu\tau}$, and (1.80) gives $c = 1 - \sigma$:

$$1 - x = (1 - \sigma)\mathrm{e}^{-\mu\tau}. \tag{1.91}$$

Note that this determines σ in terms of x and τ, which is what will be substituted at the end.

The equation for y, (1.88). This is the only one that needs the previous answer. Substituting (1.91),

$$\frac{\mathrm{d}y}{\mathrm{d}\tau} = \alpha(1 - (1 - \sigma)\mathrm{e}^{-\mu\tau} - y),$$

a linear equation, solved with integrating factor $\mathrm{e}^{\alpha\tau}$:

$$y = 1 - \frac{\alpha}{\alpha - \mu}(1 - \sigma)\mathrm{e}^{-\mu\tau} + c(\sigma, \eta)\mathrm{e}^{-\alpha\tau}.$$

Applying (1.80) gives $c(\sigma, \eta) = \eta + \frac{\alpha}{\alpha - \mu}(1 - \sigma) - 1$, so

$$y = 1 - \frac{\alpha}{\alpha - \mu}(1 - \sigma)\mathrm{e}^{-\mu\tau} + \left(\eta + \frac{\alpha}{\alpha - \mu}(1 - \sigma) - 1\right)\mathrm{e}^{-\alpha\tau}. \tag{1.92}$$

Recovering the solution. Rearranging (1.92) for η, substituting σ from (1.91) and $\tau = -t$ from (1.90), gives

$$\eta = 1 - \frac{\mu}{\mu - \alpha}\mathrm{e}^{-\alpha t} + \frac{\alpha}{\mu - \alpha}\mathrm{e}^{-\mu t} + \frac{\alpha}{\mu - \alpha}(\mathrm{e}^{-\alpha t} - \mathrm{e}^{-\mu t})x + \mathrm{e}^{-\alpha t}y,$$

and (1.89) then reproduces (1.84) exactly. The two routes agree.

1.3.3.3 What the example shows

Three things are worth extracting before the method is used on a problem where no check is available.

The structure is always the same. The equations for G and for t are universal: G is constant on characteristics and $\tau = -t$. Only the equations for the state variables change from model to model, and they are ordinary differential equations in one variable each, solved in cascade — x first, since (1.87) does not involve y, then y using the result. The triangular structure is a consequence of the model, not of the method: absorption moves particles from outside to inside, so the exterior count depends on the interior one and not conversely.

The difficulty is concentrated in one integral. In the worked example (1.87) was linear and its solution elementary, so the substitution into (1.88) produced another linear equation. When the interior dynamics include births, the coefficient of $\partial_x G$ acquires a quadratic term and (1.87) becomes a Riccati equation; it still solves in closed form, but the subsequent integral $\int \mathrm{e}^{\alpha\tau}x(\tau)\mathrm{d}\tau$ no longer does, and the solution acquires a special function. That is the absorption–birth–death model, carried out in full in section 1.B.2, where the closed form obtained is a result of this thesis.

The method needs no independence. This is the point of using it at all. In the worked example the independence of the particles gave the answer without any partial differential equation, and characteristics merely confirmed it. Once replication is allowed inside the compartment, a single particle can generate unboundedly many, the state space becomes infinite, and the independence shortcut yields an infinite sum with nothing gained. The characteristic method is indifferent to that: it uses only the first-order structure of the equation, which survives.

The remaining absorption models — absorption alone, and absorption with birth and death — together with the domain, resonance and coefficient-extraction questions they raise, are collected in section 1.B.

1.A Critical behaviour, small populations and conditional moments

Section 1.2.1.2 defined the binary Galton–Watson process and proposition 1.6 settled whether it dies out. This appendix settles how, what happens when it starts from more than one individual, and what the second moment of the conditioned population does. The material is used in later chapters and referred to from sections 1.1 and 1.3.2.6, but it is not needed to follow the methods of section 1.3, so it is collected here.

1.A.1 The critical case: a power law and an infinite expected lifetime

At $p = \frac{1}{2}$ the expectation is constant, $\mathbb{E}(Z_n) = 1$ for every n, and yet extinction is certain by proposition 1.6. The two facts are reconciled by the tail: rare realisations survive for enormously long times and grow enormously large, and they do so at exactly the rate needed to hold the mean at one.

Let $T = \inf\{n : Z_n = 0\}$ be the extinction time, so that the process passes through generations $0, 1, \ldots, T-1$ and is dead at T. Writing T as a sum of indicators and taking expectations term by term — legitimate because every term is non-negative —

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} \mathbb{P}(T > n) = \sum_{n=0}^{\infty} \mathbb{P}(Z_n > 0) = \sum_{n=0}^{\infty} S_n. \tag{1.93}$$

Everything therefore turns on how fast S_n decays at criticality, and that can be established directly from the exact recursion.

Proposition 1.8 (Critical survival probability)**.** *For the critical binary process, $p = \frac{1}{2}$,*

$$S_n \sim \frac{2}{n} \qquad (n \to \infty). \tag{1.94}$$

Proof. At $p = \frac{1}{2}$ the recursion (1.13) reads

$$S_{n+1} = S_n - \tfrac{1}{2} S_n^2 = S_n \left(1 - \frac{S_n}{2}\right). \tag{1.95}$$

By proposition 1.6, $S_n \downarrow 0$, so the reciprocal sequence tends to infinity. Its increments satisfy

$$\frac{1}{S_{n+1}} - \frac{1}{S_n} = \frac{1}{S_n\left(1 - \frac{S_n}{2}\right)} - \frac{1}{S_n} = \frac{1}{2\left(1 - \frac{S_n}{2}\right)} \longrightarrow \frac{1}{2}. \tag{1.96}$$

Summation of the convergent increments — equivalently, the Stolz–Cesàro theorem — gives $(1/S_n)/n \to \frac{1}{2}$, which is (1.94). $\square$

Substituting into (1.93), the tail of the sum behaves like twice the harmonic series,

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} S_n \sim \sum_{n=1}^{\infty} \frac{2}{n},$$

which diverges (figure 1.17). Since every S_n is positive, $\mathbb{E}(T) = \infty$: the expected lifetime of the critical Galton–Watson process is infinite even though extinction occurs with probability one. The point probabilities follow from the exact decrement in (1.95),

$$\mathbb{P}(T = n+1) = S_n - S_{n+1} = \tfrac{1}{2} S_n^2 \sim \frac{2}{n^2}. \tag{1.97}$$

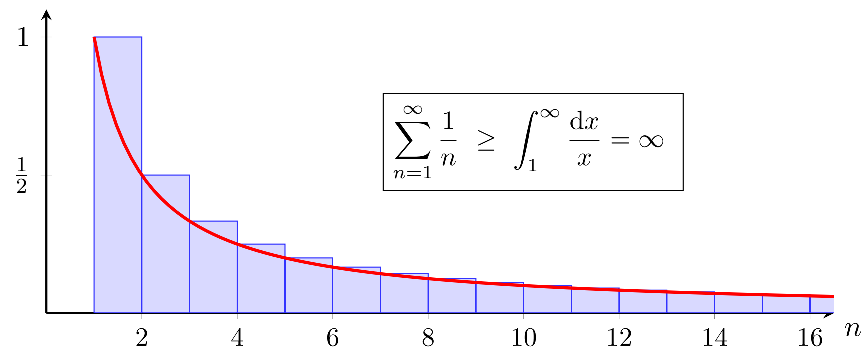


Figure 1.17: Divergence of the harmonic series. Each blue rectangle has area $1/n$ and contains the region under the red curve $1/x$ on $[n, n+1]$, so the sum of the areas exceeds $\int_1^\infty x^{-1}\mathrm{d}x = \infty$. Because $S_n \sim 2/n$ at criticality, the same divergence carries over to $\mathbb{E}(T) = \sum_n S_n$.

The total progeny obeys a different exponent, and it can be computed exactly. Let $\mathcal{T} = \sum_{n\geq 0} Z_n$. A finite binary genealogy with k division vertices carries $k+1$ terminal vertices, hence $2k+1$ individuals in all, and the number of ordered full binary trees with k internal vertices is the Catalan number $C_k = (k+1)^{-1}\binom{2k}{k}$. Each such tree is realised with probability $p^k(1-p)^{k+1}$, so

$$\mathbb{P}(\mathcal{T} = 2k+1) = C_k\, p^k (1-p)^{k+1}, \tag{1.98}$$

and $\mathbb{P}(\mathcal{T} = 2k) = 0$. At $p = \frac{1}{2}$, Stirling's formula applied to C_k gives

$$\mathbb{P}(\mathcal{T} = 2k+1) \sim \frac{1}{2\sqrt{\pi}}\, k^{-3/2}, \tag{1.99}$$

the classical 3/2 law of critical branching, holding along the odd support.

The heavy tails responsible are visible directly in simulation. A power-law distributed quantity carries a modest but non-negligible chance of an extraordinarily large realisation, and a sample mean computed from such a quantity does not settle: the larger the sample, the larger the mean, because the occasional enormous value keeps pushing the normalised sum upward.

1.A.2 A continuum view of the survival recursion

The critical asymptotic (1.94) is also reachable by treating n as continuous, and the comparison is instructive because the same substitution is made elsewhere in this thesis where it cannot be repaired so easily.

Once n is large the unit generational increment is small compared with the scale on which S_n varies, so the difference on the left of (1.95) may be replaced by a derivative:

$$\frac{\mathrm{d}S}{\mathrm{d}n} = -\tfrac{1}{2}S^2, \qquad \text{with general solution} \qquad S(n) = \frac{2}{n + \mathrm{const}}.$$

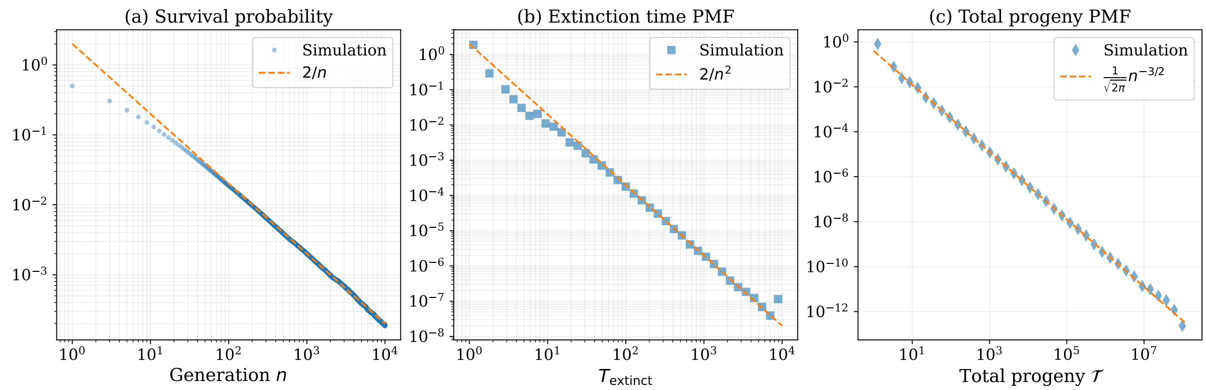


Figure 1.18: Power-law behaviour in the critical Galton–Watson process ($p = 0.5$), from 10^6 realisations. (a) Survival probability $S_n = \mathbb{P}(Z_n > 0)$, showing the late-generation scaling $S_n \sim 2/n$ of (1.94). (b) Probability mass function of the extinction time $T = \inf\{n : Z_n = 0\}$, with heavy tail $\mathbb{P}(T = n) \sim 2/n^2$. (c) Probability mass function of the total progeny $\mathcal{T} = \sum_{k\geq 0} Z_k$, displaying the $n^{-3/2}$ asymptotic (1.99) on its odd support.

This replaces a difference equation by a differential one, and nothing in the step justifies the replacement beyond the smallness of the increment relative to the solution. Proposition 1.8 shows that the conclusion is nevertheless correct.

Away from criticality the same substitution is less benign. With $\varepsilon = 1 - 2p$ the recursion (1.13) rearranges to $S_{n+1} - S_n = -\varepsilon S_n - pS_n^2$, and the continuum replacement gives the Riccati equation

$$\frac{\mathrm{d}S}{\mathrm{d}n} = -\varepsilon S - pS^2, \tag{1.100}$$

which the substitution $u = 1/S$ linearises to $u' = \varepsilon u + p$, with solution $u(n) = K\mathrm{e}^{\varepsilon n} - p/\varepsilon$ and hence

$$S(n) = \frac{\varepsilon}{K\varepsilon\,\mathrm{e}^{\varepsilon n} - p}. \tag{1.101}$$

Setting $\varepsilon = 0$ recovers (1.94); keeping $\varepsilon > 0$ recovers the geometric decay $S_n \sim A(p)(2p)^n$ of (1.53), since $(2p)^n = \mathrm{e}^{n\ln(1-\varepsilon)}$ agrees with $\mathrm{e}^{-\varepsilon n}$ to leading order. But the constant K is fixed by the initial condition, that is by the early generations, where the approximation is invalid; the continuum equation cannot supply it. Since K is precisely $A(p)$, this is the analytic statement of the obstruction described in section 1.3.1.4: no argument of this kind will produce the amplitude, and a different method is needed. The following chapter supplies it.

Note that the same Riccati structure reappears in section 1.B.2 as the characteristic equation of a partial differential equation, where it *can* be solved outright — the difference being that there it is an exact consequence of the model rather than a continuum approximation to a difference equation.

1.A.3 Survival of small populations

Near criticality, with p and $1 - p$ each close to a half, a lineage dies at its first round almost half the time. Should it survive one generation, both its offspring fail at their own first attempt with probability around 0.25, so the lineage reaches the second generation only with probability about 0.375:

$$0.375 \approx \underbrace{(1 - 0.5)}_{\text{no extinction at the first step}} \times \underbrace{(1 - 0.5 \times 0.5)}_{\text{nor at the second}}.$$

More generally, survival to a given generation is obtained by iterating (1.13). At exactly critical branching the extinction probabilities for $n = 1, 2, 3, 4$ are $u_n = 0.500,\ 0.625,\ 0.695,\ 0.741$ to three significant figures, so the probability of survival decays quickly over the first few rounds, and does so almost as fast just above criticality as just below. Figure 1.19 makes the point: whatever advantage supercriticality confers, it is not an early-generation advantage.

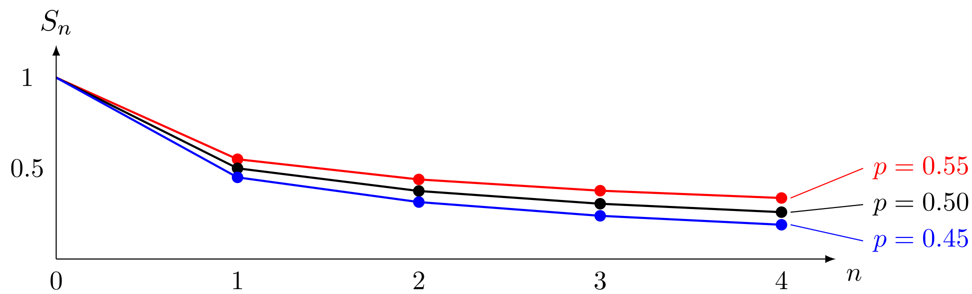


Figure 1.19: Survival probability over the first four generations from a single progenitor. Even in the supercritical case ($p = 0.55$) the early decay is steep, and the three curves have barely separated by $n = 4$.

An initial cohort of size k

Suppose now $Z_0 = k$: the process begins with a group rather than an individual. Under supercritical branching, growth becomes more predictably exponential as the count rises, because a large number of individuals lets the expected dynamics show through the noise. The founding cohort is precisely the regime in which that noise dominates instead.

The lineages are independent and each is extinct by generation n with probability u_n, so all k have gone with probability u_n^k and

$$S_n^{(k)} = 1 - u_n^k. \tag{1.102}$$

Letting $n \to \infty$ and using (1.52), the ultimate establishment probability is 0 for $p \leq \frac{1}{2}$ and $1 - \big((1-p)/p\big)^k$ for $p > \frac{1}{2}$: a larger cohort helps decisively above criticality and, in the long run, not at all below it. Figures 1.20 and 1.21 show $S_n^{(k)}$ through 500 generations for $k = 1, 10, 100, 1000$. The improvement in longevity that a larger first cohort buys is acutely sensitive to criticality, and the two panels of figure 1.21 differ only in the third decimal place of p.

Conditioning and apparent early growth

The conditioning of section 1.3.2.6 has a mirror image worth recording, because it shows that the effect is not confined to the subcritical case.

Consider an ensemble of independent homogeneous birth–death processes, all with the same constant rates, and retain only those still alive at some large time. The retained subpopulation is automatically enriched for early rapid growth: a realisation that grew quickly at the outset is far more likely to have escaped the early extinction risk documented in figure 1.19, and so far more likely to be in the retained set. No change in the underlying rates is required to produce the enrichment, and the apparent slowdown that follows is not a slowdown at all but a reversion to the mean. The effect is known as the *push of the past*, and was identified in a setting far from the present one, as an explanation for a systematic pattern in longevity and diversity data [12].

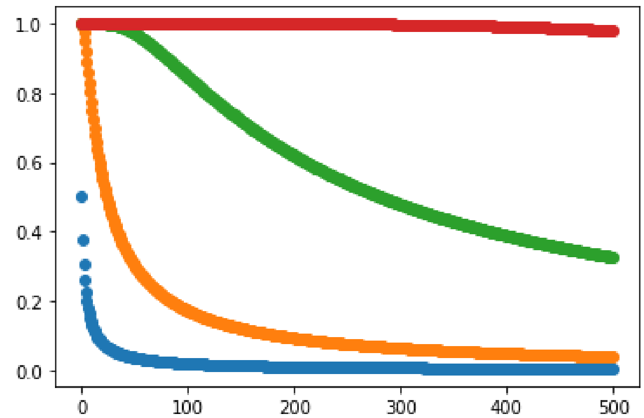


Figure 1.20: Survival probability $S_n^{(k)}$ from (1.102) out to 500 generations at exact criticality, $p = 0.5$, for $k = 1, 10, 100, 1000$ (blue through to red). A larger founding cohort buys longevity, but at criticality every curve is still headed for zero.

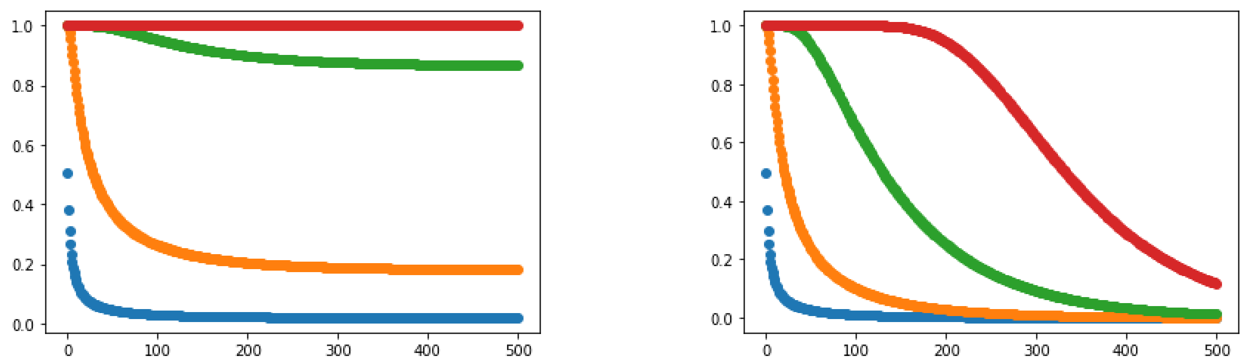


Figure 1.21: As figure 1.20, but with p displaced slightly from criticality: $p = 0.505$ on the left (supercritical), $p = 0.495$ on the right (subcritical). A displacement of half a per cent separates curves that were nearly coincident in figure 1.20.

That is the conditioning of section 1.3.2.6 seen from the other side. There a subcritical process was conditioned on survival and its mean settled to $1/A(p)$ rather than decaying; here a critical or supercritical process is conditioned on survival to a late time and its early growth appears inflated. In both cases the conditioning, not the mechanism, does the work: the selection is performed on trajectories after they have been generated, and the steep early decline of figure 1.19 is the source of the resulting bias.

1.A.4 The limiting conditional variance

Section 1.3.2.6 quoted the limiting conditional variance (1.73) without deriving it. The derivation is short, it uses only the composition rule (1.10), and it is worth having, because the mean of a distribution says nothing about whether the distribution itself is settling down: a second moment that also stabilises is the least one should ask of a population claimed to be quasi-stationary.

Write ϕ_n for the generating function of Z_n, so that $\phi_n = \phi \circ \phi_{n-1}$, and recall from (1.50) that $\phi_n'(1) = \mathbb{E}(Z_n)$ and $\phi_n''(1) = \mathbb{E}(Z_n(Z_n - 1))$. For the binary law (1.12) the derivative of the offspring function is $\phi'(z) = 2pz$, and that single fact makes the chain rule collapse. Differentiating the composition once,

$$\phi_n'(z) = \phi'(\phi_{n-1}(z))\,\phi_{n-1}'(z) = 2p\,\phi_{n-1}(z)\,\phi_{n-1}'(z), \tag{1.103}$$

which at $z = 1$ recovers $\mathbb{E}(Z_n) = 2p\mathbb{E}(Z_{n-1}) = (2p)^n$, since $\phi_{n-1}(1) = 1$. Differentiating (1.103) again,

$$\phi_n''(z) = 2p\Big\{\phi_{n-1}'(z)^2 + \phi_{n-1}(z)\,\phi_{n-1}''(z)\Big\}, \tag{1.104}$$

so, writing $v_n = \phi_n''(1)$ and evaluating at $z = 1$,

$$v_n = (2p)^{2n-1} + 2p\,v_{n-1}, \qquad v_0 = 0. \tag{1.105}$$

The recursion is linear and unrolls to a geometric sum, $v_n = \sum_{k=1}^{n}(2p)^{n-k}(2p)^{2k-1} = (2p)^n \sum_{k=1}^{n}(2p)^{k-1}$, that is

$$\mathbb{E}(Z_n(Z_n-1)) = (2p)^n\,\frac{1-(2p)^n}{1-2p}, \qquad \text{whence} \qquad \mathbb{E}\Big(Z_n^2\Big) = (2p)^n\left(1 + \frac{1-(2p)^n}{1-2p}\right). \tag{1.106}$$

The check at $n = 1$ is immediate: the right-hand side is $4p$, and Z_1 equals 2 with probability p and 0 otherwise.

Now condition. Subcriticality means $2p < 1$, so $(2p)^{2n}$ is negligible beside $(2p)^n$ at large n, and dividing (1.106) by $S_n \sim A(p)(2p)^n$ from (1.53) gives

$$\mathbb{E}\Big(Z_n^2 \mid Z_n > 0\Big) = \frac{\mathbb{E}(Z_n^2)}{S_n} \xrightarrow[n\to\infty]{} \frac{1}{A(p)}\left(1 + \frac{1}{1-2p}\right), \tag{1.107}$$

the geometric factors cancelling exactly as they did for the mean in (1.72). Both conditional moments therefore converge, and subtracting the square of the first from the second gives (1.73).

Two remarks are in order. The limit (1.107) is again a function of p and the single constant $A(p)$ alone, so the second moment adds no new unknown — the whole quasi-stationary description of the binary process runs through one number, and that concentration of the difficulty into a single constant is what the following chapter sets out to analyse. And convergence of two moments is not convergence of a law; it is consistent with the Yaglom limit of definition 1.7 and corroborates it, but the limit itself is taken from the literature cited there rather than established here.

1.A.5 The discrete rupture calculation

Remark 1.4 asserted that a state-dependent catastrophe does not reduce to a mean-field hazard. The elementary calculation behind that assertion is recorded here; it is retained as a bridge to the birth–death–catastrophe models of later chapters, not as a theory of them.

Let κ be the probability that a given individual initiates catastrophe at a step. A single individual fails to do so with probability $1-\kappa$, and a cohort of size N with $(1-\kappa)^N$.

Birth and catastrophe alone

With no death, the census is known with certainty: from a single progenitor the population doubles at each step prior to catastrophe. The total exposure over generations $0, \ldots, n-1$ is $1 + 2 + \cdots + 2^{n-1} = 2^n - 1$, so the probability of surviving to generation n is

$$S_n = \prod_{i=0}^{n-1}(1-\kappa)^{2^i} = (1-\kappa)^{\sum_{i=0}^{n-1} 2^i} = (1-\kappa)^{2^n-1}. \tag{1.108}$$

The conditional probability of rupture at step n itself, given that every earlier step was passed, is

$$1-(1-\kappa)^{2^{n-1}}, \tag{1.109}$$

2^{n-1} being the number exposed at that step. (1.109) is a conditional probability and should not be confused with the unconditional probability of the process ending at step n.

Death included

Once death is allowed the census is random, and (1.108) is replaced by the exact but implicit (1.27), in which the cumulative exposure $H_n = \sum_{i=0}^{n-1} \widetilde{Z}_i$ appears inside an expectation. The natural simplification replaces H_n by its mean,

$$S_n \overset{?}{=} (1-\kappa)^{\mathbb{E}(H_n)}, \qquad \mathbb{E}(H_n) = \sum_{i=0}^{n-1} (2p)^i = \begin{cases} \dfrac{1-(2p)^n}{1-2p}, & p \neq \frac{1}{2}, \\ n, & p = \frac{1}{2}, \end{cases} \tag{1.110}$$

and by the Jensen argument of remark 1.4 this is a lower bound rather than an identity. Nor are the steps independent, since passing one is evidence of a smaller population and so raises the chance of passing the next; both effects push the estimate the same way.

The exact object

The bound is crude, but nothing forces one to stop there: the quantity (1.27) has an exact closed recursion, and it is the marked generating function that remark 1.4 called for. Define

$$F_n(s) = \mathbb{E}\left(s^{\widetilde{Z}_n}(1-\kappa)^{H_n}\right), \qquad F_0(s) = s, \tag{1.111}$$

which carries the population and its accumulated exposure jointly, the first in the variable s and the second in the weight. Condition on what the founder does at the first step. It dies with probability $1-p$, contributing nothing further; it divides with probability p, after which the two daughter lineages evolve independently and each contributes a factor F_n; and either way the founder itself is exposed for one step, contributing $1-\kappa$. Hence

$$F_{n+1}(s) = (1-\kappa)\,\phi\big(F_n(s)\big) = (1-\kappa)\big(1-p+p\,F_n(s)^2\big), \tag{1.112}$$

a functional iteration of exactly the type of (1.10), differing from it only by the constant factor the exposure contributes at each step.

Two evaluations carry the quantities of interest. $F_n(1)$ is (1.27), the probability of no catastrophe in the first n steps; and $F_n(1) - F_n(0)$ is the probability of being both unruptured and non-extinct at generation n, the joint event for which remark 1.4 observed that Jensen supplies no bound at all. Since (1.112) is exact and costs one composition per generation to iterate, it is the object to use whenever (1.110) is too crude — which is the honest position: the elementary estimate is indicative, and the recursion is what replaces it.

1.B The absorption models in full

Section 1.3.3 developed the method of characteristics on one absorption model. This appendix collects the family that model belongs to: the simplest member, in which particles are only taken up;

and the hardest, in which they may replicate once inside, and for which the closed form obtained is a result of this thesis.

The setting is a single compartment drawing particles from a surrounding medium. Once several compartments share a medium, a question arises that the models of section 1.2 do not address: what becomes of the contents of the first compartment to rupture? The shared medium acquires, at a stroke, a potentially large cohort of particles available for subsequent absorption by the compartments that remain. The analysis elsewhere in this thesis assumes *immediate transfer*, in which the entire contents of a ruptured compartment relocate at once. The alternative is an *absorption period*, during which absorption events redistribute the released cohort among the remaining compartments and only then do birth, death and rupture resume. Working towards models of that kind, this appendix treats the case in which a cohort of particles enters a single compartment.

Throughout, X_t and Y_t count particles inside and outside the compartment, the initial condition is $X_0 = 0$, $Y_0 = N$, and events have exponentially distributed inter-event times, so $\{(X_t, Y_t)\}$ is a continuous-time Markov chain in the sense of section 1.2.2.

1.B.1 Absorption only

The simplest case has no dynamics inside the compartment at all:

$$\begin{aligned} \mathbb{P}(\Delta X = 1,\ \Delta Y = -1 \mid Y_t = j) &= j\alpha\,\Delta t, && \text{(absorption)} \\ \mathbb{P}(\Delta X = 0,\ \Delta Y = 0 \mid Y_t = j) &= 1 - j\alpha\,\Delta t, && \text{(no event)} \end{aligned}$$

the factor j recording that each exterior particle is independently subject to the same probability $\alpha\Delta t$ of being taken up. The forward equations are

$$\frac{\mathrm{d}p_{i,j}}{\mathrm{d}t} = \alpha(j+1)p_{i-1,j+1} - \alpha j\,p_{i,j}, \tag{1.113}$$

and the construction of section 1.3.2.1 gives

$$\frac{\partial G}{\partial t} = \alpha(x-y)\frac{\partial G}{\partial y}, \qquad G(x,y,0) = y^N. \tag{1.114}$$

The sums involved are finite, because $p_{i,j}(t) = 0$ whenever $i + j \neq N$: particles in this model are neither created nor destroyed.

Because the particles never interact, the generating function factorises. Writing G_n for the function with n initial exterior particles and decomposing $X_t = \sum_k X_t^{(k)}$, $Y_t = \sum_k Y_t^{(k)}$ into single-particle contributions, independence gives

$$G_n(x,y,t) = \mathbb{E}\left(\prod_{k=1}^{n} x^{X_t^{(k)}} y^{Y_t^{(k)}}\right) = \prod_{k=1}^{n}\mathbb{E}\left(x^{X_t^{(k)}} y^{Y_t^{(k)}}\right) = \big(G_1(x,y,t)\big)^n. \tag{1.115}$$

For a single particle only two states are reachable, so $\mathrm{d}p_{0,1}/\mathrm{d}t = -\alpha p_{0,1}$ and $\mathrm{d}p_{1,0}/\mathrm{d}t = \alpha p_{0,1}$, giving $p_{0,1}(t) = \mathrm{e}^{-\alpha t}$ and $p_{1,0}(t) = 1 - \mathrm{e}^{-\alpha t}$ (figure 1.22); hence

$$G_n(x,y,t) = \left(\left(1 - \mathrm{e}^{-\alpha t}\right)x + \mathrm{e}^{-\alpha t}y\right)^n. \tag{1.116}$$

This satisfies (1.114) with the required initial condition, and the form is recognisable: $X_t \sim \mathrm{Binomial}(n, 1 - \mathrm{e}^{-\alpha t})$, as it must be, since the n particles are absorbed independently and each has been taken up by time t with probability $1 - \mathrm{e}^{-\alpha t}$. This is the thinning property of section 1.2.2.2 in another guise.

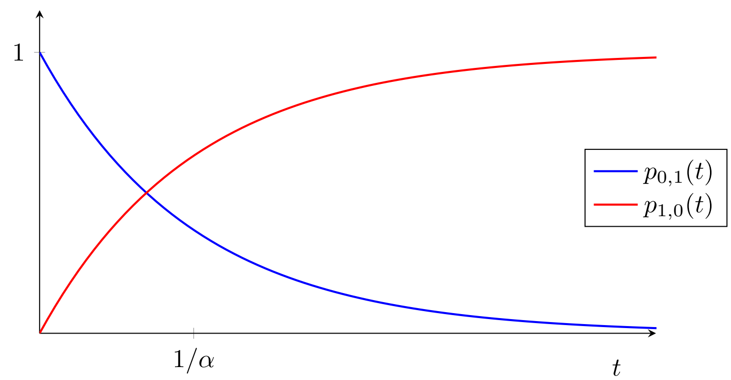


Figure 1.22: State probabilities for the single-particle absorption-only model. The particle begins outside the compartment ($p_{0,1}$, decaying exponentially at rate α) and ends inside it ($p_{1,0}$). There is nowhere else for it to go, so the two curves sum to one at every time.

1.B.2 Absorption–birth–death

Introducing replication for particles inside the compartment, the independence property is still present but no longer serves as a shortcut: from a single initial particle, birth events make infinitely many states reachable, so G_1 is an infinite sum and there is nothing to be gained by factorising. The partial differential equation must be solved directly — which is what section 1.3.3 was developed for.

Birth at per-capita rate λ is added to the absorption–death model of section 1.3.3.2:

$$\begin{aligned}
\mathbb{P}(\Delta X = 1,\ \Delta Y = -1 \mid Y_t = j) &= j\alpha\,\Delta t, && \text{(absorption)}\\
\mathbb{P}(\Delta X = 1,\ \Delta Y = 0 \mid X_t = i) &= i\lambda\,\Delta t, && \text{(birth)}\\
\mathbb{P}(\Delta X = -1,\ \Delta Y = 0 \mid X_t = i) &= i\mu\,\Delta t, && \text{(death)}\\
\mathbb{P}(\Delta X = 0,\ \Delta Y = 0 \mid X_t = i,\ Y_t = j) &= 1 - (j\alpha + i\mu + i\lambda)\Delta t, && \text{(no event)}
\end{aligned}$$

with forward equation

$$\frac{\mathrm{d}p_{i,j}}{\mathrm{d}t} = \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} + \lambda(i-1)p_{i-1,j} - (\alpha j + \mu i + \lambda i)p_{i,j}, \tag{1.117}$$

and generating-function equation

$$\frac{\partial G}{\partial t} = \left(\mu - (\mu+\lambda)x + \lambda x^2\right)\frac{\partial G}{\partial x} + \alpha(x-y)\frac{\partial G}{\partial y}, \qquad G(x,y,0) = y^N, \tag{1.118}$$

whose x-coefficient is exactly the quadratic met in (1.58) for the linear birth–death process. The characteristic system (1.79) is

$$\frac{\mathrm{d}G}{\mathrm{d}\tau} = 0, \tag{1.119}$$

$$\frac{\mathrm{d}t}{\mathrm{d}\tau} = -1, \tag{1.120}$$

$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = \mu - (\mu+\lambda)x + \lambda x^2, \tag{1.121}$$

$$\frac{\mathrm{d}y}{\mathrm{d}\tau} = \alpha(x-y), \tag{1.122}$$

with the initial conditions (1.80).

The equations for G and t. As in section 1.3.3.2, and for the same universal reasons,

$$G(\sigma,\eta,\tau) = \eta^N, \qquad t = -\tau. \tag{1.123}$$

The equation for x, (1.121). This is the Riccati equation anticipated in section 1.A.2. Its right-hand side factorises, and the roots are elementary rather than the surd expression that mechanical use of the quadratic formula suggests:

$$\mu - (\mu+\lambda)x + \lambda x^2 = \lambda(x-x_+)(x-x_-), \qquad x_+ = 1, \quad x_- = \frac{\mu}{\lambda}. \tag{1.124}$$

The discriminant is $\left(\frac{\mu+\lambda}{2\lambda}\right)^2 - \frac{\mu}{\lambda} = \left(\frac{\lambda-\mu}{2\lambda}\right)^2$, a perfect square, so the surds cancel. Name the combination that appears throughout,

$$b_1 = \lambda(x_+ - x_-) = \lambda - \mu, \tag{1.125}$$

the net interior growth rate. Separating variables and applying (1.80) gives the characteristic in the compact form

$$\frac{x-x_+}{x-x_-} = \left(\frac{\sigma-x_+}{\sigma-x_-}\right)\mathrm{e}^{b_1\tau}, \tag{1.126}$$

or, solved for x,

$$x(\tau) = \frac{x_+(\sigma-x_-) - x_-(\sigma-x_+)\mathrm{e}^{b_1\tau}}{(\sigma-x_-) - (\sigma-x_+)\mathrm{e}^{b_1\tau}}. \tag{1.127}$$

The equation for y, (1.122). With an integrating factor this is

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\left(\mathrm{e}^{\alpha\tau}y\right) = \alpha\,\mathrm{e}^{\alpha\tau}x(\tau), \tag{1.128}$$

and the integral on the right is where the difficulty lies. Rather than substitute (1.127) and split the resulting quotient — which leads to two hypergeometric integrals and a good deal of bookkeeping — it is cleaner to expand $x(\tau)$ first. Writing $x - x_- = (x-x_+) + (x_+ - x_-)$ in (1.126) and solving for x gives

$$x(\tau) = x_+ + (x_+ - x_-)\frac{\zeta(\tau)}{1-\zeta(\tau)}, \qquad \zeta(\tau) = \left(\frac{\sigma-x_+}{\sigma-x_-}\right)\mathrm{e}^{b_1\tau}, \tag{1.129}$$

so that for $|\zeta|<1$, $x(\tau) = x_+ + (x_+ - x_-)\sum_{n\geq 1}\zeta(\tau)^n$. Every term is now a pure exponential in τ and integrates immediately. Writing $Z = (\sigma-x_+)/(\sigma-x_-)$, so that $\zeta(\tau) = Z\mathrm{e}^{b_1\tau}$,

$$\begin{aligned}
\alpha\int \mathrm{e}^{\alpha\tau}x(\tau)\,\mathrm{d}\tau &= \alpha x_+\int \mathrm{e}^{\alpha\tau}\mathrm{d}\tau + \alpha(x_+ - x_-)\sum_{n=1}^{\infty} Z^n \int \mathrm{e}^{(\alpha+nb_1)\tau}\mathrm{d}\tau \\
&= \mathrm{e}^{\alpha\tau}\left[x_+ + \alpha(x_+ - x_-)\sum_{n=1}^{\infty}\frac{\zeta^n}{\alpha+nb_1}\right] + \text{const.}
\end{aligned}$$

Setting $\delta = \alpha/b_1$ and using $\alpha(x_+ - x_-)/b_1 = \alpha/\lambda$ by (1.125), this defines the function that carries the whole solution:

$$\Psi(\zeta) = x_+ + \frac{\alpha}{\lambda}\sum_{n=1}^{\infty}\frac{\zeta^n}{n+\delta}, \qquad \delta = \frac{\alpha}{b_1} = \frac{\alpha}{\lambda-\mu}. \tag{1.130}$$

Integrating (1.128) gives $\mathrm{e}^{\alpha\tau}y = \mathrm{e}^{\alpha\tau}\Psi(\zeta(\tau)) + \text{const}$, and the initial condition $y(\sigma,\eta,0) = \eta$, at which $\zeta(0) = Z$, fixes the constant as $\eta - \Psi(Z)$, so

$$\eta = \mathrm{e}^{\alpha\tau}\Big[y - \Psi\big(\zeta(\tau)\big)\Big] + \Psi(Z). \tag{1.131}$$

Finally, put $\tau = -t$ and write $W = (x-x_+)/(x-x_-)$, so that $\zeta(\tau) = W$ and $Z = W\mathrm{e}^{b_1 t}$ by (1.126). Substituting into (1.131) and applying (1.123) gives the solution.

Proposition 1.9 (Generating function for absorption–birth–death)**.** *With $x_+ = 1$, $x_- = \mu/\lambda$, $b_1 = \lambda - \mu$, $\delta = \alpha/b_1$, $W = (x-x_+)/(x-x_-)$ and Ψ as in (1.130),*

$$\boxed{G(x,y,t) = \Big\{\mathrm{e}^{-\alpha t}\big[y - \Psi(W)\big] + \Psi\big(W\mathrm{e}^{b_1 t}\big)\Big\}^N.} \tag{1.132}$$

Three checks confirm the result. At $t = 0$ the two Ψ terms cancel and $G = y^N$, the required initial condition. At $x = y = 1$ we have $W = 0$ and $\Psi(0) = x_+ = 1$, so $G = 1$ and probability is conserved. And setting $x = 1$ alone gives $G(1,y,t) = (\mathrm{e}^{-\alpha t}y + 1 - \mathrm{e}^{-\alpha t})^N$, so $Y_t \sim \mathrm{Binomial}(N, \mathrm{e}^{-\alpha t})$ — exactly right, since the exterior particles are absorbed independently at rate α and what happens inside the compartment cannot affect them. Beyond these, (1.132) has been verified numerically against direct integration of (1.117), agreeing to around 10^{-13} across sub- and supercritical interior regimes and for $N = 1, 2, 3$.

1.B.3 A parameter-regular representation

The form (1.132) is compact, but its derivation separated an antiderivative into pieces that are individually singular at certain parameter values. The underlying probability calculation supplies a second representation that stays finite everywhere, and it settles the questions of domain and resonance below without any analytic continuation.

Let $F(x,t)$ be the generating function, at time t, of an interior linear birth–death process started from a single particle. Conditioning on the first event gives its backward equation,

$$\frac{\partial F}{\partial t} = \lambda F^2 - (\lambda+\mu)F + \mu, \qquad F(x,0) = x, \tag{1.133}$$

which is (1.58) read along characteristics. For $\lambda \neq \mu$ its solution is the ratio already met in (1.126),

$$\frac{F(x,t) - 1}{F(x,t) - x_-} = \frac{x-1}{x-x_-}\,\mathrm{e}^{b_1 t}, \tag{1.134}$$

while at $\lambda = \mu$ the roots coincide and

$$F(x,t) = 1 - \frac{1-x}{1+\lambda t(1-x)}. \tag{1.135}$$

Now follow a single exterior particle. Either it has not been absorbed by time t, which happens with probability $\mathrm{e}^{-\alpha t}$ and contributes y; or it is absorbed at some time $v \in (0,t)$, with density

$\alpha e^{-\alpha v}$, and thereafter founds an interior birth–death process of duration $t - v$. Conditioning on the absorption time,

$$g(x, y, t) = e^{-\alpha t} y + \alpha e^{-\alpha t} \int_0^t e^{\alpha v} F(x, v)\, dv, \qquad G(x, y, t) = g(x, y, t)^N, \tag{1.136}$$

the second identity because the N exterior particles are independent throughout. (1.136) is real and finite for $x, y \in [0, 1]$ and all positive rates, including $\lambda = \mu$ where (1.132) degenerates. For $\lambda \neq \mu$, substituting (1.134) into the integral and integrating term by term returns (1.132); so (1.136) also specifies unambiguously which continuation of the special-function expression is intended. The three checks above now need almost no algebra: at $t = 0$ the integral vanishes and $g = y$; at $x = y = 1$ we have $F(1, v) = 1$ so $g = 1$; and setting $x = 1$ gives the binomial exterior marginal directly.

1.B.4 Hypergeometric form, domain and resonance

The series in (1.130) is a Gauss hypergeometric function in disguise. With

$${}_2F_1(a, b; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!}, \qquad (q)_n = \begin{cases} 1, & n = 0, \\ q(q+1)\cdots(q+n-1), & n \geq 1, \end{cases} \tag{1.137}$$

where $(q)_n$ is the rising Pochhammer symbol [13], the case $a = 1$, $b = \delta$, $c = \delta + 1$ collapses to a single ratio. Since $(1)_n = n!$ and

$$\frac{(\delta)_n}{(\delta+1)_n} = \frac{\delta(\delta+1)\cdots(\delta+n-1)}{(\delta+1)\cdots(\delta+n-1)(\delta+n)} = \frac{\delta}{\delta+n}, \tag{1.138}$$

we have ${}_2F_1(1, \delta; \delta+1; z) = \sum_{n\geq 0} \frac{\delta}{\delta+n} z^n$. Comparing with (1.130) and using $\alpha/(\lambda\delta) = b_1/\lambda = x_+ - x_-$,

$$\Psi(\zeta) = x_+ + (x_+ - x_-)\Big[{}_2F_1(1, \delta; \delta+1; \zeta) - 1\Big]. \tag{1.139}$$

(1.132) together with (1.139) is the closed form. It is not pretty, but it is a genuine closed form in terms of a standard special function, and the elementary series (1.130) is what one would actually evaluate. An alternative derivation, by way of the integral $\int e^{h\tau}/(P + Qe^{k\tau})\, d\tau$, is given in section 1.C; that route is longer, but it produces a general identity of independent use.

Remark 1.10 (Domain of validity). The series (1.130) converges for $|\zeta| < 1$, and the arguments appearing in (1.132) are W and $We^{b_1 t}$. In the subcritical interior regime $\mu > \lambda$ we have $b_1 < 0$ and $x_- = \mu/\lambda > 1$, so for $x \in [0, 1]$

$$0 \leq W \leq \frac{\lambda}{\mu} < 1, \qquad 0 \leq We^{b_1 t} \leq W :$$

the series representation is valid on the whole of the unit square, which is the case of most interest here. When $\lambda > \mu$ the arguments may leave the unit disc or cross the standard hypergeometric cut; one may then continue analytically, tracking the branch along the characteristic, but the real integral (1.136) is usually safer. A convenient continuation is the Euler integral

$${}_2F_1(1, \delta; \delta+1; \zeta) = \delta \int_0^1 \frac{u^{\delta-1}}{1 - \zeta u}\, du \qquad (\delta > 0,\ \zeta \notin [1, \infty)),$$

which was used for the numerical verification quoted above. The apparent singularity of W at $x = x_-$ is a coordinate singularity only: $F(x_-, t) = x_-$ for every t, and the limit of (1.136) there is perfectly regular.

Remark 1.11 (Resonance). The denominators in (1.130) vanish when $n + \delta = 0$ for some integer $n \geq 1$, that is when

$$\alpha = n(\mu - \lambda), \qquad n = 1, 2, \dots \tag{1.140}$$

This is the classical resonance condition for linearising the y-equation: the decay rate α of the absorption channel coincides with a harmonic of the interior relaxation rate $\mu - \lambda$. The generating function itself is perfectly regular there. The two Ψ terms of (1.132) each acquire a pole, but their combination does not: in the resonant term the difference

$$\frac{(W e^{b_1 t})^n - e^{-\alpha t} W^n}{n + \delta}$$

has a finite limit as $\delta \to -n$, producing the familiar secular factor $t\, e^{-\alpha t}$ in place of a simple exponential. Equivalently (1.136) has no singular denominator at all and supplies the limit directly. Resonance is therefore a defect of the separated antiderivative, not of the generating function; the resonant parameter values form a set of measure zero, but the degeneracy should be kept in mind when α happens to sit near an integer multiple of $\mu - \lambda$.

1.B.5 Recovering state probabilities

A generating function is useful only insofar as its coefficients can be recovered. In every model above,

$$p_{i,j}(t) = [x^i y^j]\, G(x, y, t) = \frac{1}{i!\, j!} \left.\frac{\partial^{i+j} G}{\partial x^i \partial y^j}\right|_{x=y=0}. \tag{1.141}$$

For absorption only and absorption–death the generating function is a power of an affine form and the extraction is immediate, giving trinomial state probabilities; section 1.D carries this out. For absorption–birth–death the parameter-regular representation separates the two extractions cleanly. Writing

$$h(x, t) := \alpha e^{-\alpha t} \int_0^t e^{\alpha v} F(x, v)\, dv, \qquad g(x, y, t) = e^{-\alpha t} y + h(x, t), \tag{1.142}$$

the exterior variable appears only in the finite binomial expansion of g^N, so

$$p_{i,j}(t) = \binom{N}{j} e^{-\alpha t j}\, [x^i]\, h(x, t)^{N-j}, \qquad 0 \leq j \leq N. \tag{1.143}$$

The remaining univariate coefficient can be obtained by differentiation, by convolution of truncated series, or by Cauchy's integral formula on a circle $|x| = \varrho < 1$.

1.C An integral identity for the hypergeometric function

The route to (1.130) taken in section 1.B.2 expands the characteristic $x(\tau)$ in a geometric series and integrates term by term. An alternative is to integrate the quotient (1.127) directly, splitting the numerator into two pieces, each of the form

$$I(\tau) = \int \frac{e^{h\tau}}{P + Q\, e^{k\tau}}\, d\tau \tag{1.144}$$

with h, k, P, Q constant. That route is longer and easier to get wrong — the two pieces carry different prefactors, and dropping one produces a formula that still satisfies the initial condition while violating $G(1, 1, t) = 1$ — but the identity it rests on is useful in its own right, so it is recorded and proved here.

Proposition 1.12. *For $h/k > 0$ and $-\frac{Q}{P}\mathrm{e}^{k\tau} \notin [1, \infty)$,*

$$\int \frac{\mathrm{e}^{h\tau}}{P + Q\mathrm{e}^{k\tau}}\,\mathrm{d}\tau = \frac{\mathrm{e}^{h\tau}}{Ph}\,{}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}\mathrm{e}^{k\tau}\right) + \textit{const.} \tag{1.145}$$

Proof. We use the Euler integral representation of the hypergeometric function [13],

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)}\int_0^1 u^{b-1}(1-u)^{c-b-1}(1-zu)^{-a}\,\mathrm{d}u, \tag{1.146}$$

Γ being the gamma function.

Start from (1.144) and substitute $v = \mathrm{e}^{k\tau}$, so that $\mathrm{d}\tau = k^{-1}v^{-1}\mathrm{d}v$:

$$I = \frac{1}{k}\int \frac{v^{h/k}}{P + Qv}\,v^{-1}\mathrm{d}v = \frac{1}{Pk}\int \frac{v^{h/k-1}}{1 + \frac{Q}{P}v}\,\mathrm{d}v.$$

Fix the antiderivative by writing this as a definite integral from 0 to v in a dummy variable w; the integral converges at the lower limit precisely because $h/k > 0$:

$$I = \frac{1}{Pk}\int_0^v \frac{w^{h/k-1}}{1 + \frac{Q}{P}w}\,\mathrm{d}w.$$

Now rescale, $w = vu$, so that $\mathrm{d}w = v\,\mathrm{d}u$ with v held fixed:

$$I = \frac{1}{Pk}\int_0^1 \frac{(vu)^{h/k-1}}{1 + \frac{Q}{P}vu}\,v\,\mathrm{d}u = \frac{v^{h/k}}{Pk}\int_0^1 \frac{u^{h/k-1}}{1 + \frac{Q}{P}vu}\,\mathrm{d}u. \tag{1.147}$$

The remaining integrand matches that of (1.146) with $b = h/k$, $c = h/k + 1$, $a = 1$ and $z = -\frac{Q}{P}v$, since then $(1-u)^{c-b-1} = 1$ and $(1-zu)^{-a} = (1 + \frac{Q}{P}vu)^{-1}$. Hence, using $\Gamma(z+1) = z\Gamma(z)$,

$$\int_0^1 \frac{u^{h/k-1}}{1 + \frac{Q}{P}vu}\,\mathrm{d}u = \frac{\Gamma\left(\frac{h}{k}\right)\Gamma(1)}{\Gamma\left(\frac{h}{k} + 1\right)}\,{}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}v\right) = \frac{k}{h}\,{}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}v\right).$$

Substituting into (1.147) gives

$$I = \frac{v^{h/k}}{Ph}\,{}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}v\right),$$

and restoring $v = \mathrm{e}^{k\tau}$ gives (1.145). □

Remark 1.13. Applied to the two pieces of (1.127) — the first with $h = \alpha$, the second with $h = \alpha + b_1$, and both with $k = b_1$, $P = \sigma - x_-$ and $Q = -(\sigma - x_+)$ — proposition 1.12 reproduces (1.139) once the prefactors $\alpha x_+(\sigma - x_-)$ and $-\alpha x_-(\sigma - x_+)$ are carried through correctly. The factor $(\sigma - x_+)/(\sigma - x_-)$ attached to the second piece is easily lost, and its loss is invisible at $t = 0$; checking $G(1, 1, t) = 1$ catches it immediately.

1.D Extracting state probabilities from a generating function

Every generating function obtained above arrived as a formula rather than as an explicit series — the absorption-only case, for instance, as

$$G(x,y,t) = \Big((1-\mathrm{e}^{-\alpha t})x + \mathrm{e}^{-\alpha t}y\Big)^n. \tag{1.148}$$

Such a formula can always be converted back to the series form

$$G(x,y,t) = \sum_{i=0}^{\infty}\sum_{j=0}^{\infty} p_{i,j}(t)\, x^i y^j, \tag{1.149}$$

which recovers the time-dependent state probabilities. All that is required is that G be repeatedly partially differentiable in x and y.

The reason is the bivariate Maclaurin expansion,

$$f(x,y) = \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{\varrho=0}^{k} \binom{k}{\varrho} \partial_x^{\varrho}\partial_y^{k-\varrho} f\Big|_{(0,0)}\; x^{\varrho} y^{k-\varrho},$$

whose coefficient of $x^i y^j$, taking $k = i+j$ and $\varrho = i$, is $\frac{1}{(i+j)!}\binom{i+j}{i}\partial_x^i\partial_y^j f\big|_{(0,0)}$. Hence

$$p_{i,j}(t) = \frac{1}{i!\,j!}\,\partial_x^i\partial_y^j G\Big|_{(0,0)}. \tag{1.150}$$

Equivalently, and more stably in floating point when G is a special-function expression whose high derivatives are unwieldy, the same coefficient is the bivariate Cauchy integral

$$p_{i,j}(t) = \frac{1}{(2\pi\mathrm{i})^2} \oint_{|x|=\varrho_x} \oint_{|y|=\varrho_y} \frac{G(x,y,t)}{x^{\,i+1}y^{\,j+1}}\,\mathrm{d}y\,\mathrm{d}x \tag{1.151}$$

over circles inside the domain of analyticity.

The multilinear case

Both (1.116) and (1.84) have the form

$$G(x,y,t) = \big(c(t) + f(t)\,x + g(t)\,y\big)^n, \tag{1.152}$$

for which the derivatives are immediate. Each differentiation lowers the power by one and brings down a factor, so

$$\partial_x^a\partial_y^b G = \frac{n!}{(n-a-b)!}\, f^a g^b \big(c + fx + gy\big)^{n-a-b} \qquad (a+b \le n), \tag{1.153}$$

and zero for $a+b > n$. Evaluating at the origin and applying (1.150),

$$p_{i,j}(t) = \frac{n!}{i!\,j!\,(n-i-j)!}\; f(t)^i g(t)^j c(t)^{\,n-i-j}, \qquad i+j \le n, \tag{1.154}$$

and $p_{i,j} = 0$ otherwise. This is the trinomial distribution, which is exactly what it should be: the n particles are independent, and at time t each is in one of three states, with probabilities c (neither inside nor outside — that is, dead), f (inside) and g (outside). Note in particular that the falling factorial $n!/(n-i-j)!$ in (1.153) is *not* n^{i+j}; the two agree only to leading order in n, and using the latter breaks normalisation.

Applied to the absorption models

For absorption only, (1.148), we have $c = 0$, $f = 1 - \mathrm{e}^{-\alpha t}$ and $g = \mathrm{e}^{-\alpha t}$. The factor c^{n-i-j} then vanishes unless $i + j = n$, and (1.154) gives

$$p_{i,j}(t) = \binom{n}{i}\big(1 - \mathrm{e}^{-\alpha t}\big)^i \big(\mathrm{e}^{-\alpha t}\big)^j, \qquad i + j = n,$$

so $X_t \sim \mathrm{Binomial}\big(n, 1 - \mathrm{e}^{-\alpha t}\big)$ and $Y_t = n - X_t$, as anticipated in section 1.B.1.

For absorption–death, (1.84), the three functions are

$$c(t) = 1 - \frac{\mu}{\mu - \alpha}\mathrm{e}^{-\alpha t} + \frac{\alpha}{\mu - \alpha}\mathrm{e}^{-\mu t}, \qquad f(t) = \frac{\alpha}{\mu - \alpha}\big(\mathrm{e}^{-\alpha t} - \mathrm{e}^{-\mu t}\big), \qquad g(t) = \mathrm{e}^{-\alpha t},$$

which are precisely $p_{0,0}$, $p_{1,0}$ and $p_{0,1}$ of (1.83), so that

$$p_{i,j}(t) = \frac{n!}{i!\, j!\, (n - i - j)!}\, f(t)^i g(t)^j c(t)^{\,n-i-j}.$$

Each particle independently ends up dead, absorbed, or still outside, with those three probabilities, and the joint law is trinomial. That the generating function factorises in this way is (1.115) read backwards.

The absorption–birth–death generating function of proposition 1.9 is not of the form (1.152), and (1.150) must be applied to (1.132) directly, or — better — the exterior variable separated first as in (1.143), leaving a univariate coefficient extraction from $h(x,t)^{N-j}$ that may be done by convolution of truncated series or by (1.151) reduced to one circle.

1.E Figure candidate gallery (temporary)

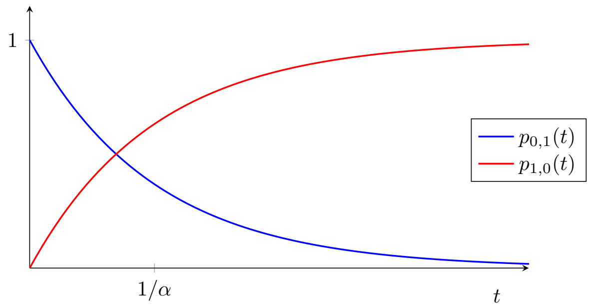
This appendix is a temporary asset-selection aid and forms no part of the chapter's argument. It displays every figure binary collected from the Claude, Qwen, Grok and Codex merges, together with the assets currently used in this project, which carry the `best` tag. Each panel is labelled with its candidate filename: the prefix is the source, and the remainder is the path the file occupied in that source's `figures/` tree, with directory separators written as double underscores. Variants sharing a filename stem are grouped so that they can be compared side by side. Nothing has been dropped as redundant, so byte-identical binaries appear more than once by design. After selection, this appendix and the unused candidates will be removed.

There are 85 candidates here in 32 groups: 24 with more than one variant, and 8 one-offs.

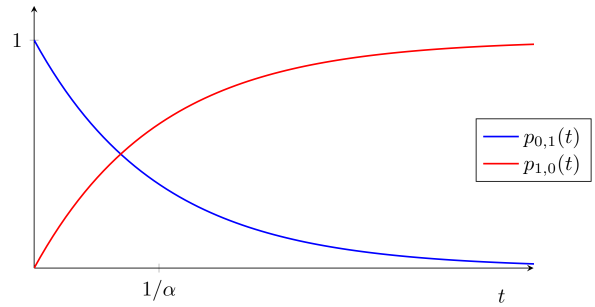
Index of groups

Comparison groups: two or more variants

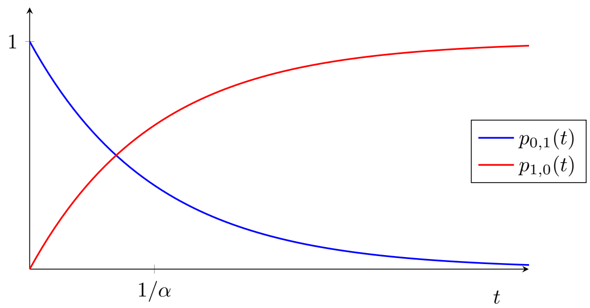
Group: `abs1`



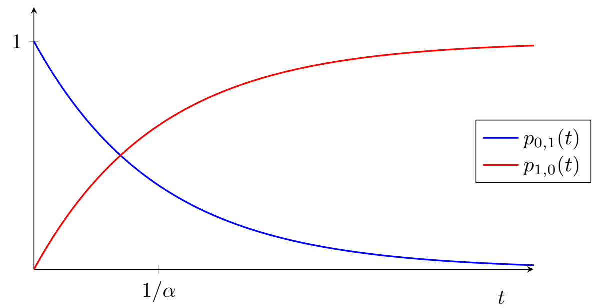
`best__abs1.pdf`



`claude__abs1.pdf`

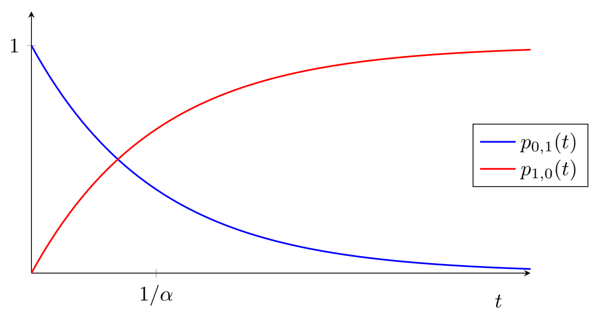


`qwen__abs1.pdf`



`grok__IMG_ch3__abs1.pdf`

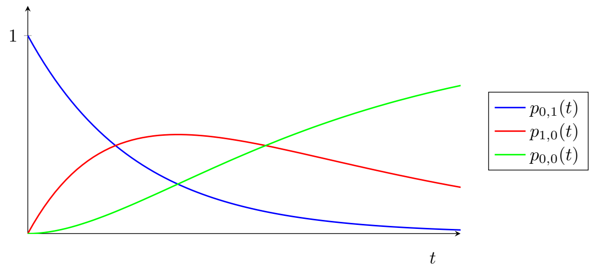
Figure 1.23: Candidates for group `abs1` (5 variants), part 1 of 2. Filenames are printed beneath the panels.



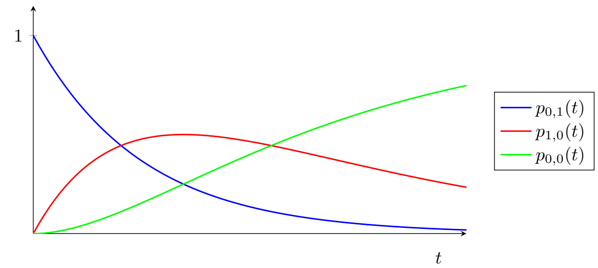
`codex__IMG_ch3__abs1.pdf`

Figure 1.24: Candidates for group `abs1` (5 variants), part 2 of 2. Filenames are printed beneath the panels.

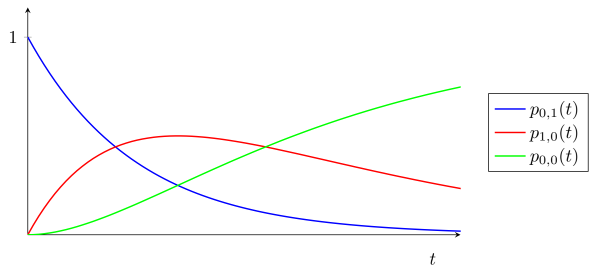
Group: `abs2`



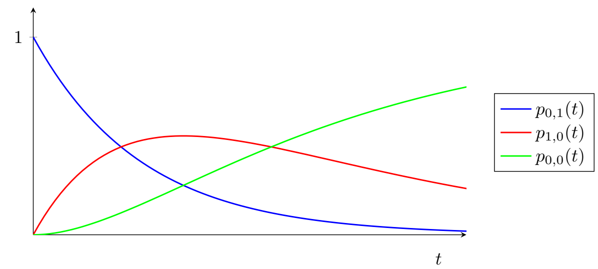
best__abs2.pdf



claude__abs2.pdf

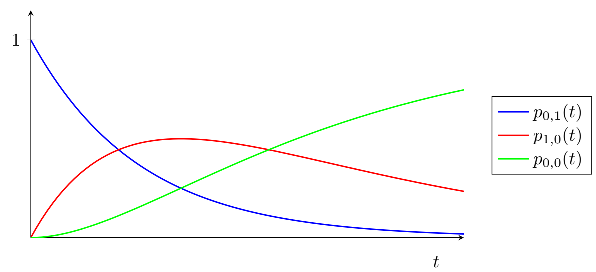


qwen__abs2.pdf



grok__IMG_ch3__abs2.pdf

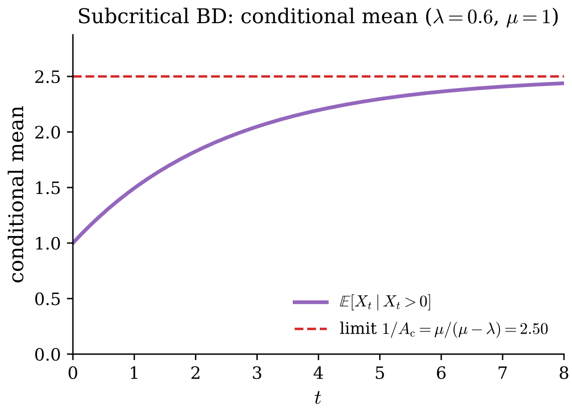
Figure 1.25: Candidates for group `abs2` (5 variants), part 1 of 2. Filenames are printed beneath the panels.



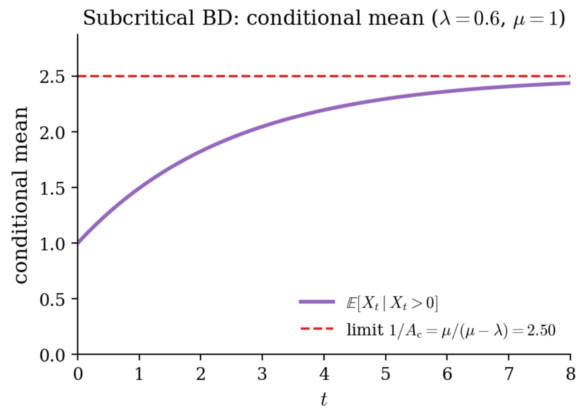
codex__IMG_ch3__abs2.pdf

Figure 1.26: Candidates for group `abs2` (5 variants), part 2 of 2. Filenames are printed beneath the panels.

Group: `bd_conditional_mean`



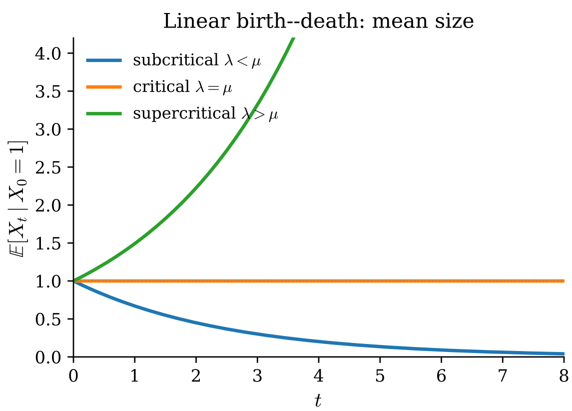
grok__tikz_gen__bd_conditional_mean.pdf



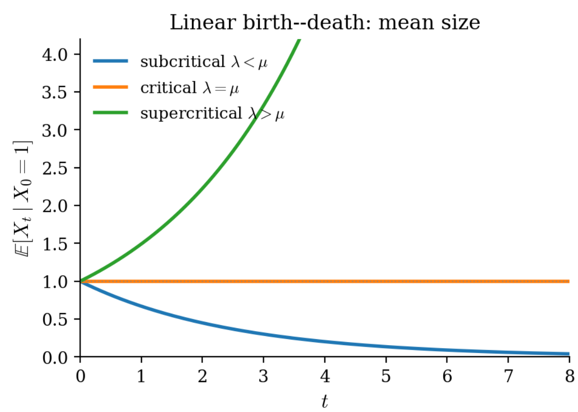
grok__tikz_gen__bd_conditional_mean.png

Figure 1.27: Candidates for group `bd_conditional_mean` (2 variants). Filenames are printed beneath the panels.

Group: `bd_mean_regimes`



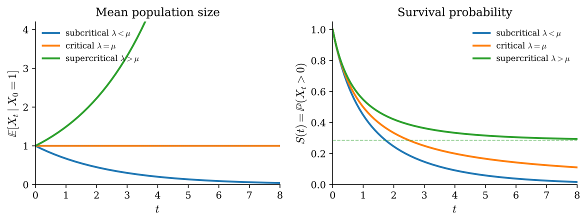
grok__tikz_gen__bd_mean_regimes.pdf



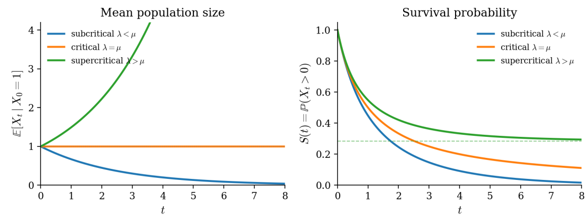
grok__tikz_gen__bd_mean_regimes.png

Figure 1.28: Candidates for group `bd_mean_regimes` (2 variants). Filenames are printed beneath the panels.

Group: `bd_mean_survival_panel`



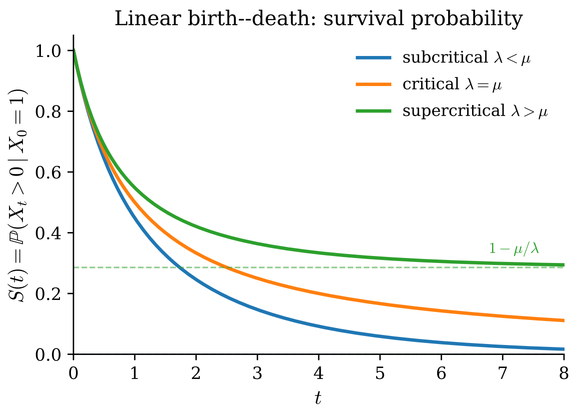
grok__tikz_gen__bd_mean_survival_panel.pdf



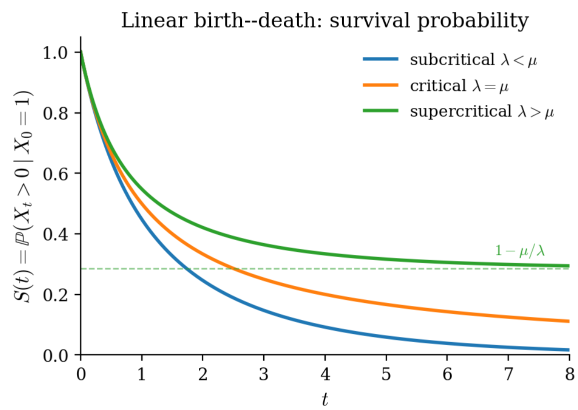
grok__tikz_gen__bd_mean_survival_panel.png

Figure 1.29: Candidates for group `bd_mean_survival_panel` (2 variants). Filenames are printed beneath the panels.

Group: `bd_survival_regimes`



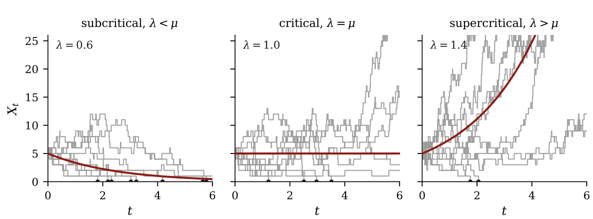
grok__tikz_gen__bd_survival_regimes.pdf



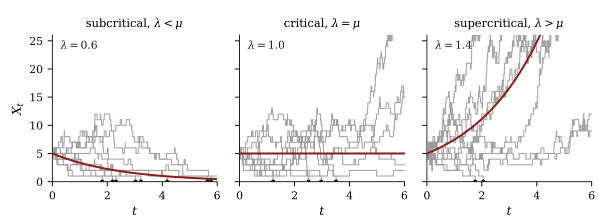
grok__tikz_gen__bd_survival_regimes.png

Figure 1.30: Candidates for group `bd_survival_regimes` (2 variants). Filenames are printed beneath the panels.

Group: `birth_death_paths`



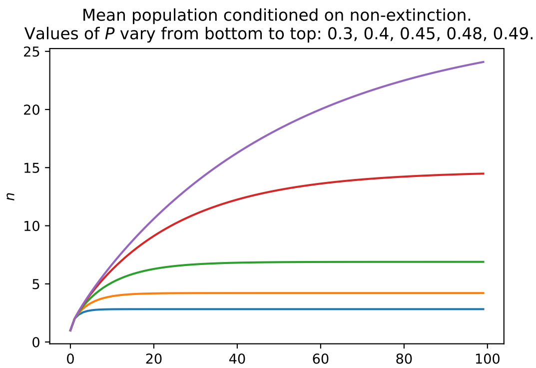
best__birth_death_paths.pdf



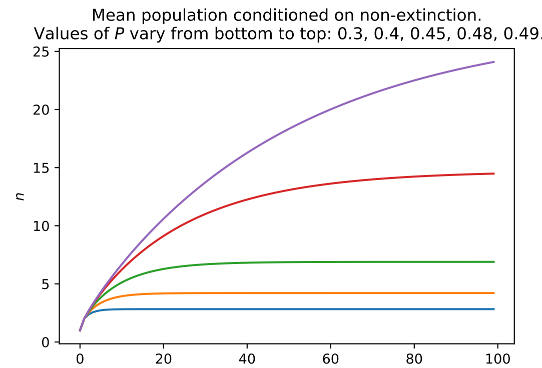
claude__birth_death_paths.pdf

Figure 1.31: Candidates for group `birth_death_paths` (2 variants). Filenames are printed beneath the panels.

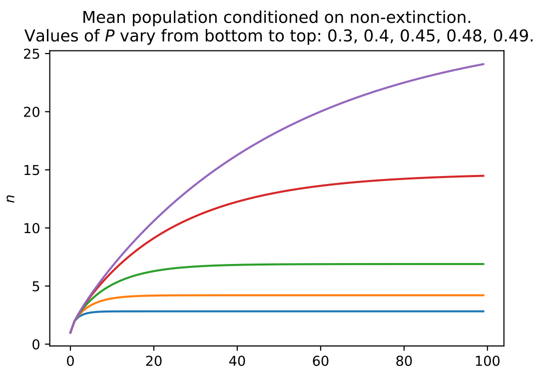
Group: `conditionalmean`



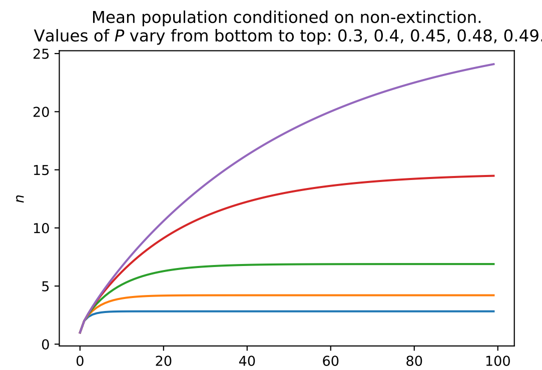
`best__conditionalMean.pdf`



`claude__conditionalMean.pdf`

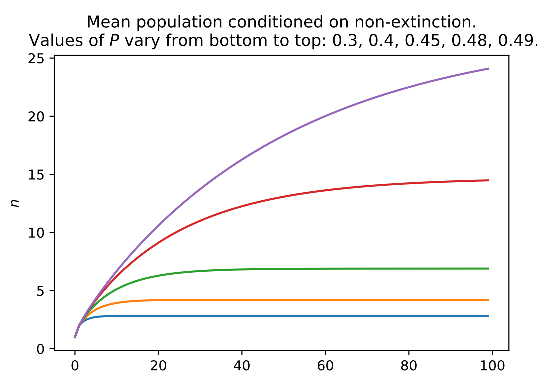


`qwen__conditionalMean.pdf`



`grok__IMG_ch3__conditionalMean.pdf`

Figure 1.32: Candidates for group `conditionalmean` (5 variants), part 1 of 2. Filenames are printed beneath the panels.



`codex__IMG_ch3__conditionalMean.pdf`

Figure 1.33: Candidates for group `conditionalmean` (5 variants), part 2 of 2. Filenames are printed beneath the panels.

Group: coupled_ode_ctmc

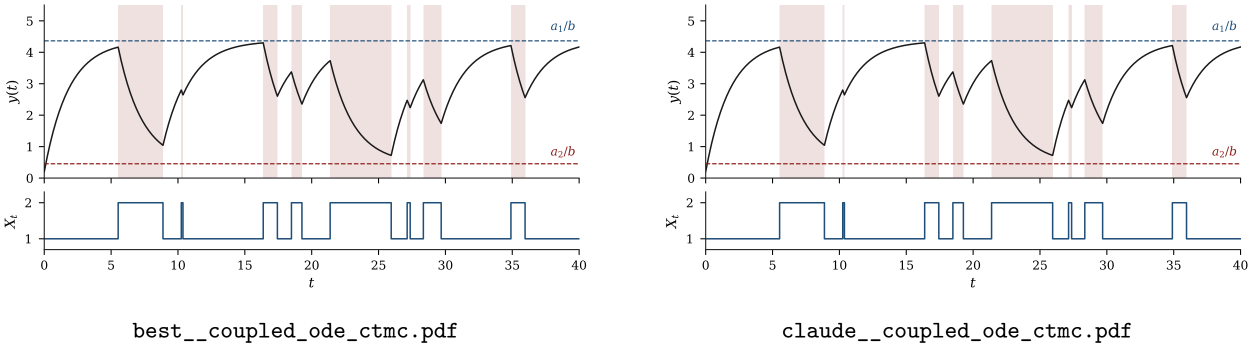
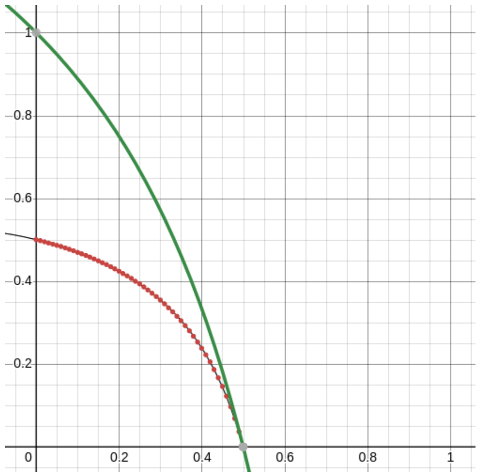
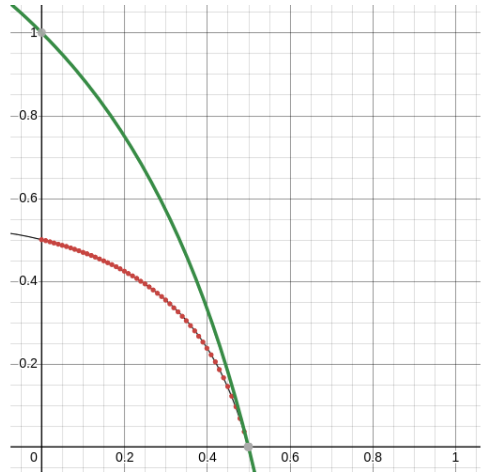


Figure 1.34: Candidates for group coupled_ode_ctmc (2 variants). Filenames are printed beneath the panels.

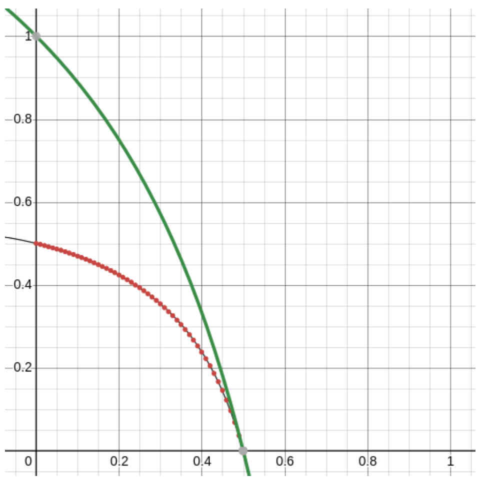
Group: dtcta



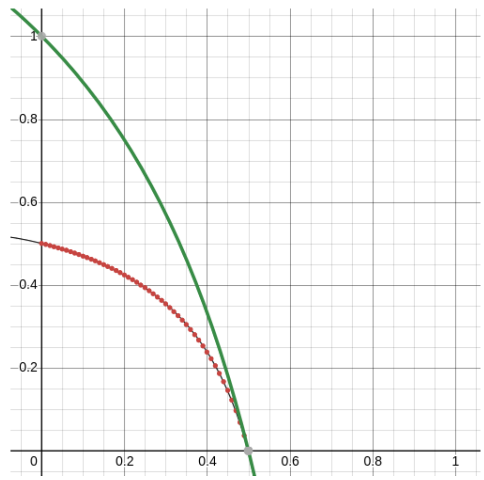
best__dtctA.png



claude__dtctA.png



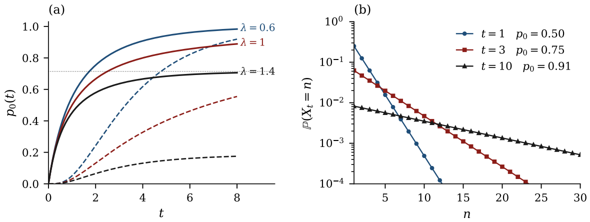
grok__IMG_ch3__dtctA.png



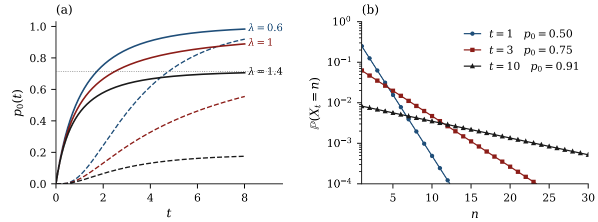
codex__IMG_ch3__dtctA.png

Figure 1.35: Candidates for group `dtcta` (4 variants). Filenames are printed beneath the panels.

Group: `extinction_and_law`



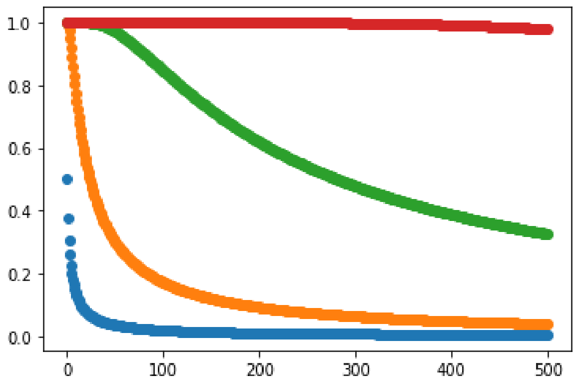
best__extinction_and_law.pdf



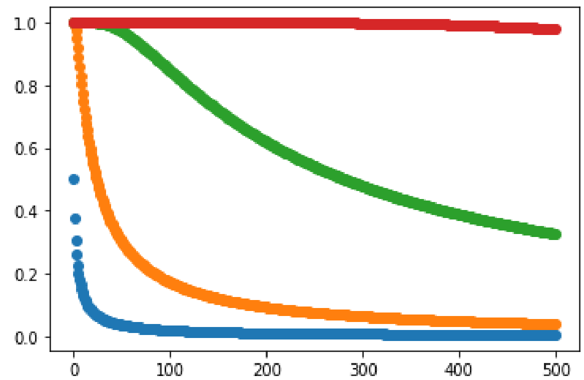
claude__extinction_and_law.pdf

Figure 1.36: Candidates for group `extinction_and_law` (2 variants). Filenames are printed beneath the panels.

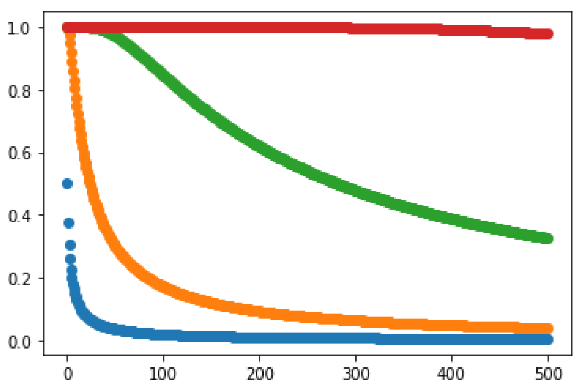
Group: `kvals`



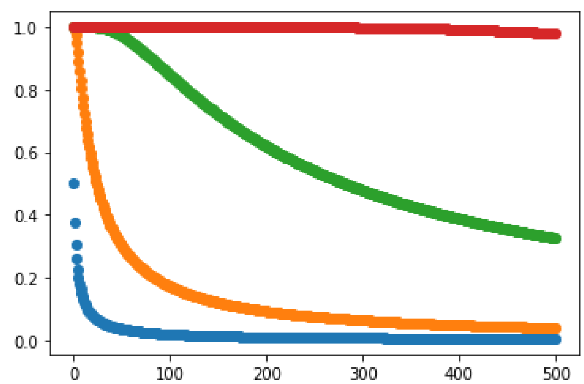
best__kvals.png



claude__kvals.png

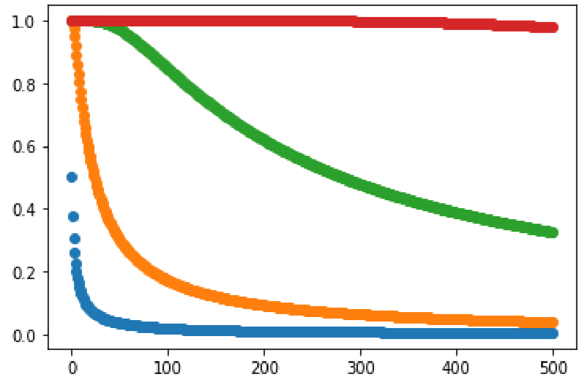


qwen__kvals.png



grok__IMG_ch3__kvals.png

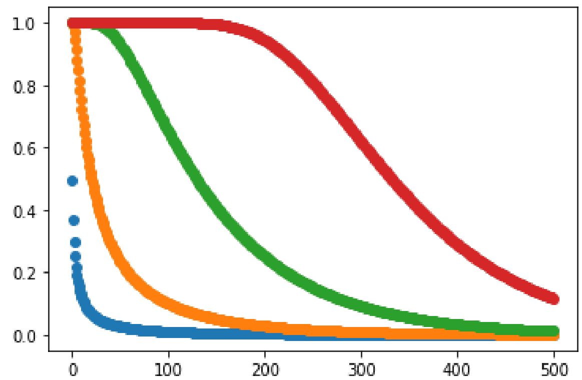
Figure 1.37: Candidates for group `kvals` (5 variants), part 1 of 2. Filenames are printed beneath the panels.



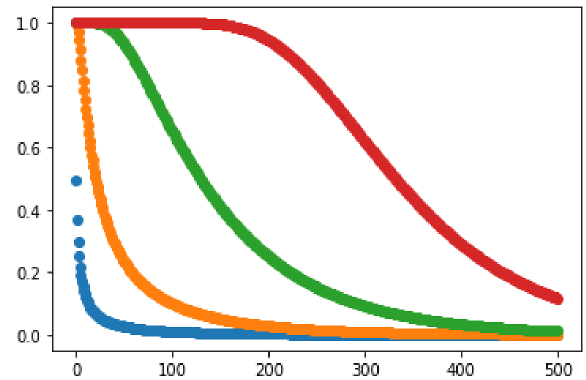
codex__IMG_ch3__kvals.png

Figure 1.38: Candidates for group kvals (5 variants), part 2 of 2. Filenames are printed beneath the panels.

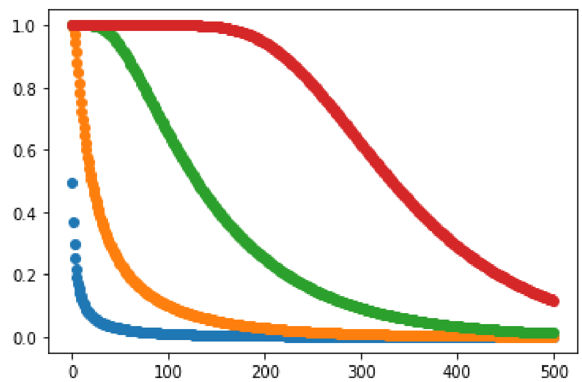
Group: kvals495



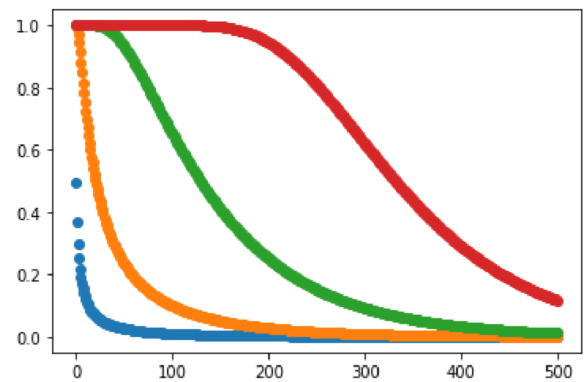
`best__kvals495.png`



`claude__kvals495.png`

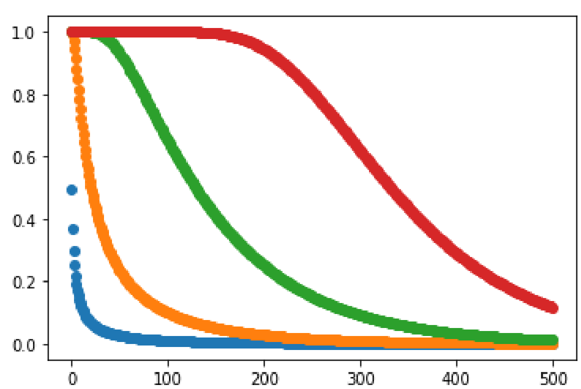


`qwen__kvals495.png`



`grok__IMG_ch3__kvals495.png`

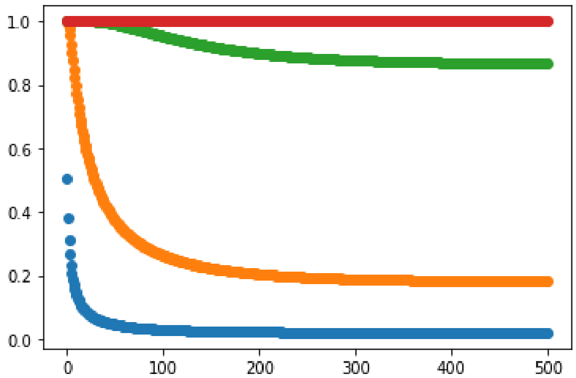
Figure 1.39: Candidates for group `kvals495` (5 variants), part 1 of 2. Filenames are printed beneath the panels.



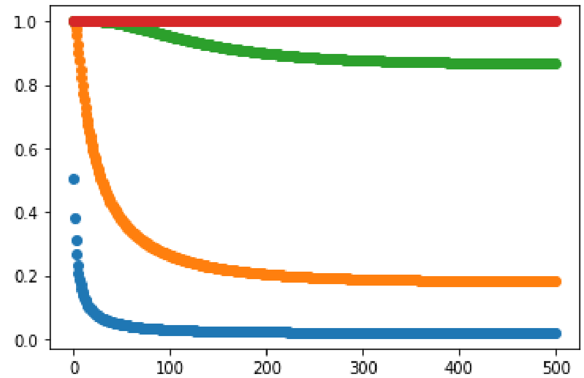
`codex__IMG_ch3__kvals495.png`

Figure 1.40: Candidates for group `kvals495` (5 variants), part 2 of 2. Filenames are printed beneath the panels.

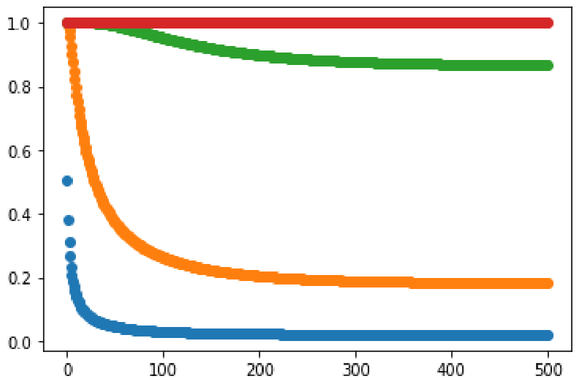
Group: kvals505



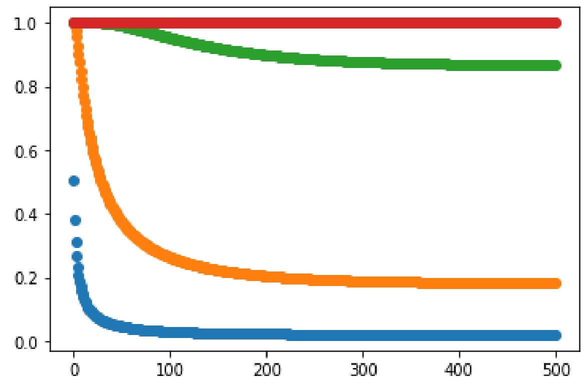
best__kvals505.png



claude__kvals505.png

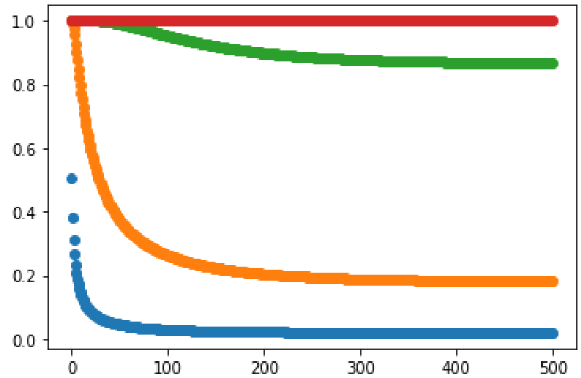


qwen__kvals505.png



grok__IMG_ch3__kvals505.png

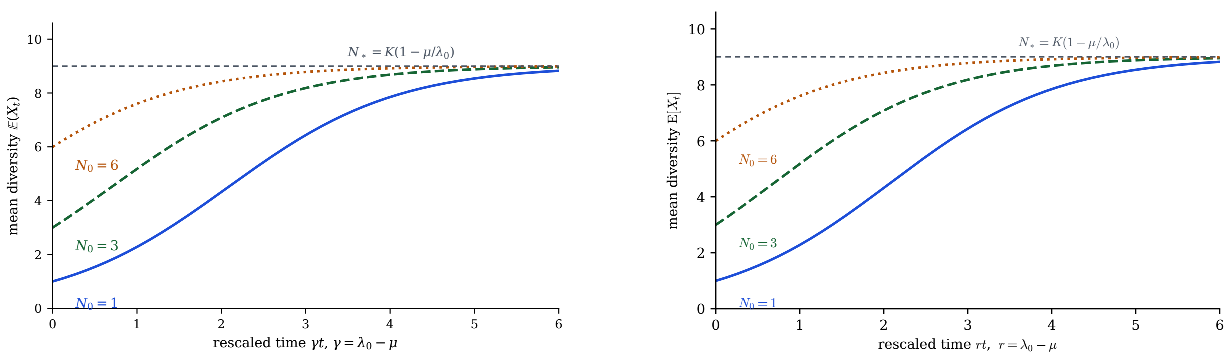
Figure 1.41: Candidates for group kvals505 (5 variants), part 1 of 2. Filenames are printed beneath the panels.



`codex__IMG_ch3__kvals505.png`

Figure 1.42: Candidates for group `kvals505` (5 variants), part 2 of 2. Filenames are printed beneath the panels.

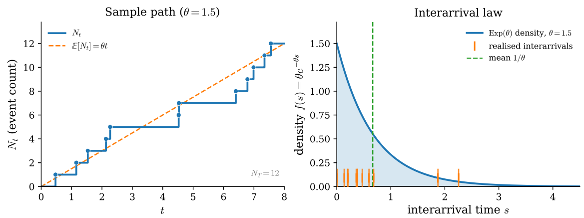
Group: `logspec_mean`



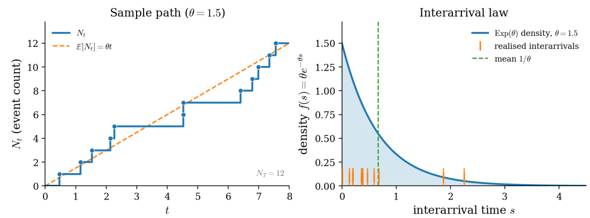
`best__logspec_mean.pdf` `qwen__logspec_mean.pdf`

Figure 1.43: Candidates for group `logspec_mean` (2 variants). Filenames are printed beneath the panels.

Group: `poisson_path`



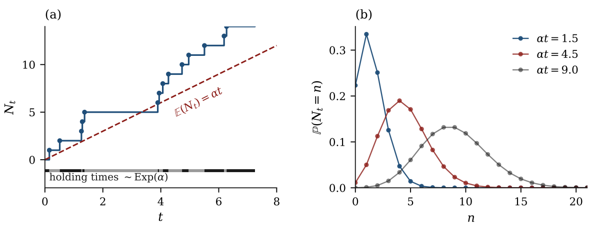
`grok__tikz_gen__poisson_path.pdf`



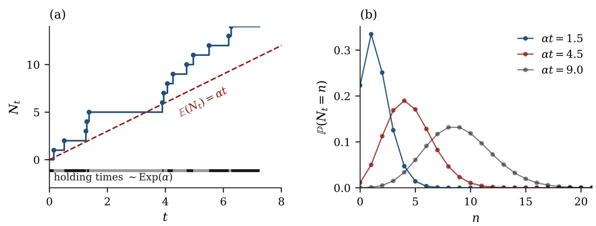
`grok__tikz_gen__poisson_path.png`

Figure 1.44: Candidates for group `poisson_path` (2 variants). Filenames are printed beneath the panels.

Group: `poisson_process`



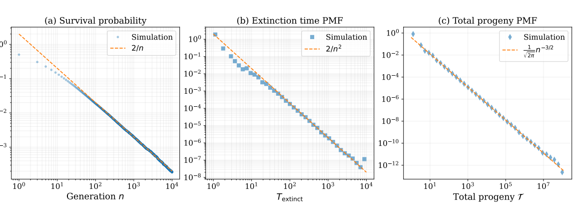
`best__poisson_process.pdf`



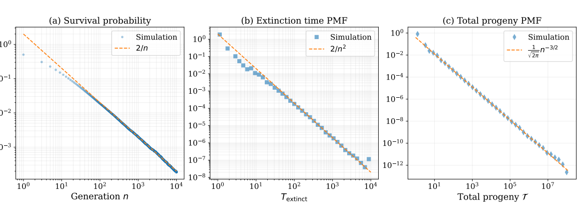
`claude__poisson_process.pdf`

Figure 1.45: Candidates for group `poisson_process` (2 variants). Filenames are printed beneath the panels.

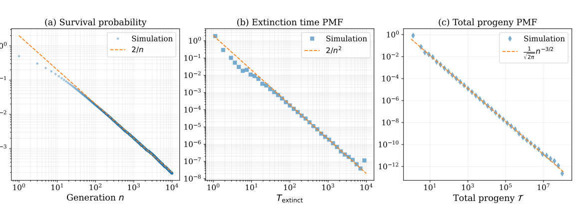
Group: `power_law_fixed`



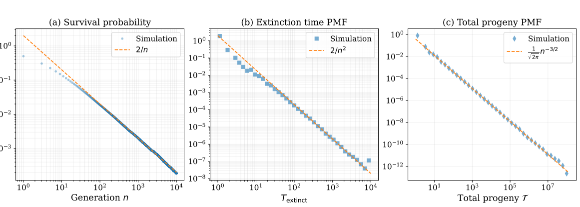
`best__power_law_fixed.pdf`



`claude__power_law_fixed.pdf`

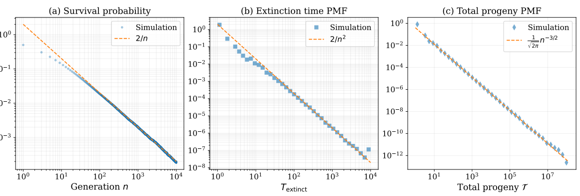


`qwen__power_law_fixed.pdf`



`grok__IMG_ch3__power_law_fixed.pdf`

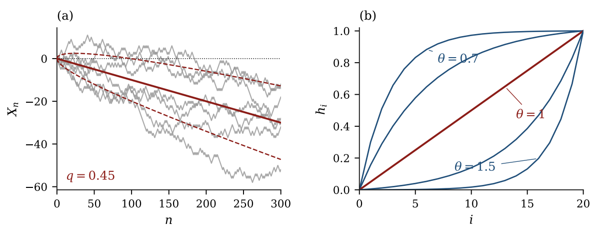
Figure 1.46: Candidates for group `power_law_fixed` (5 variants), part 1 of 2. Filenames are printed beneath the panels.



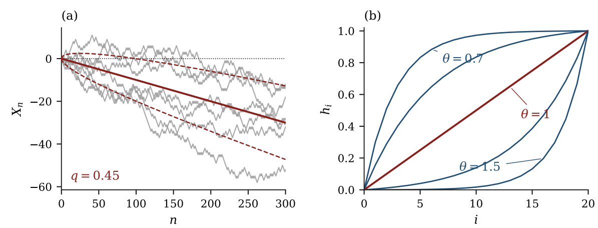
codex__IMG_ch3__power_law_fixed.pdf

Figure 1.47: Candidates for group power_law_fixed (5 variants), part 2 of 2. Filenames are printed beneath the panels.

Group: random_walk



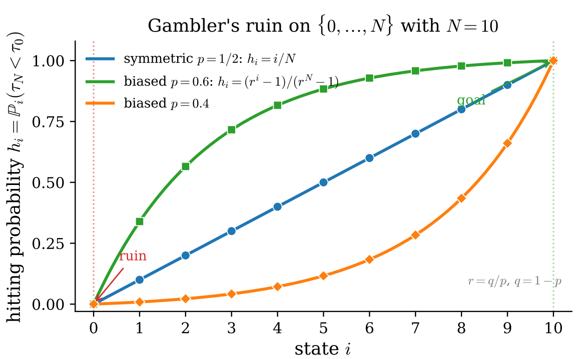
best__random_walk.pdf



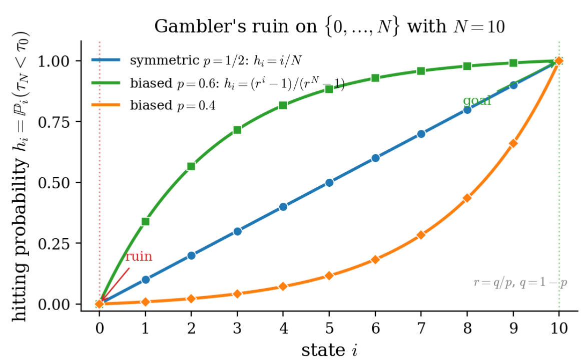
claude__random_walk.pdf

Figure 1.48: Candidates for group random_walk (2 variants). Filenames are printed beneath the panels.

Group: ruin_hitting



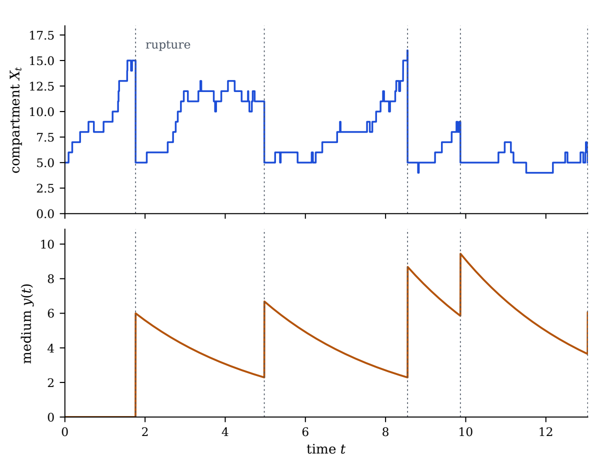
grok__tikz_gen__ruin_hitting.pdf



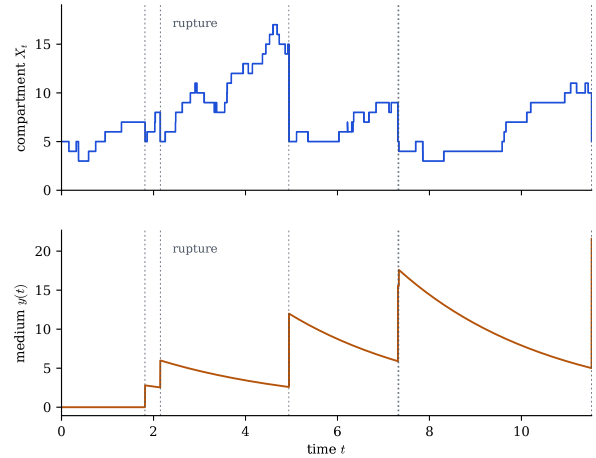
grok__tikz_gen__ruin_hitting.png

Figure 1.49: Candidates for group ruin_hitting (2 variants). Filenames are printed beneath the panels.

Group: rupture_sawtooth



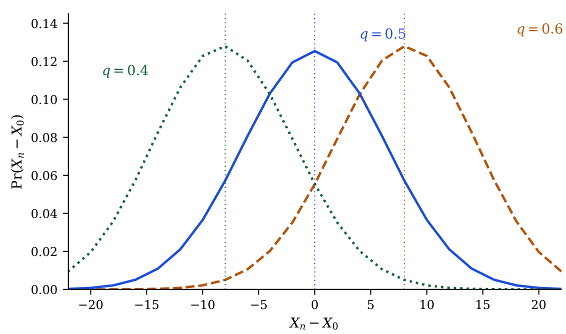
best__rupture_sawtooth.pdf



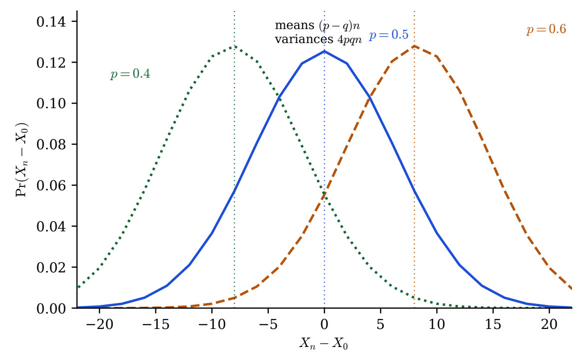
qwen__rupture_sawtooth.pdf

Figure 1.50: Candidates for group **rupture_sawtooth** (2 variants). Filenames are printed beneath the panels.

Group: rw_transition



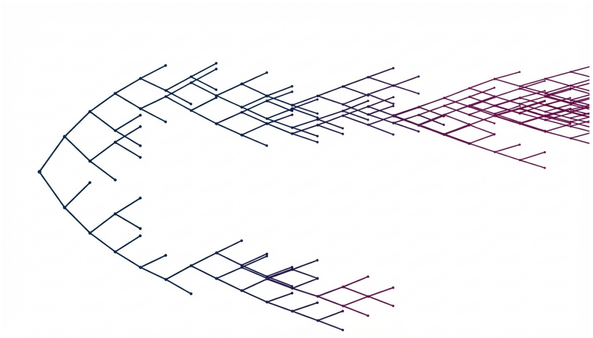
best__rw_transition.pdf



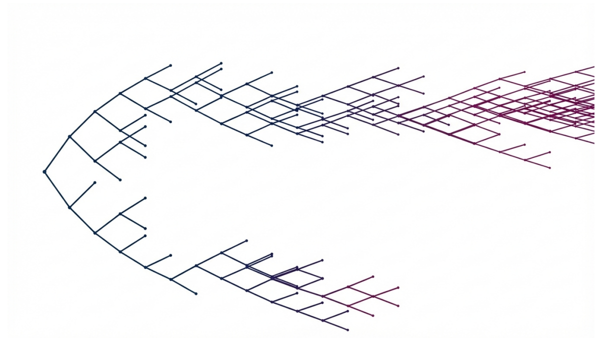
qwen__rw_transition.pdf

Figure 1.51: Candidates for group **rw_transition** (2 variants). Filenames are printed beneath the panels.

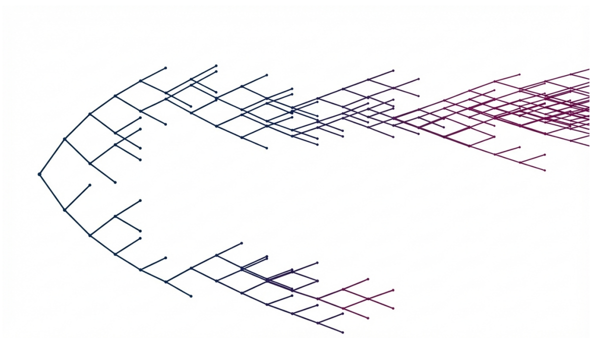
Group: simplegwvis



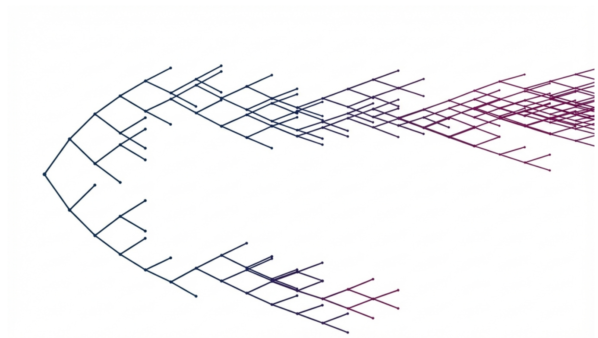
best__simpleGWvis.png



claude__simpleGWvis.png

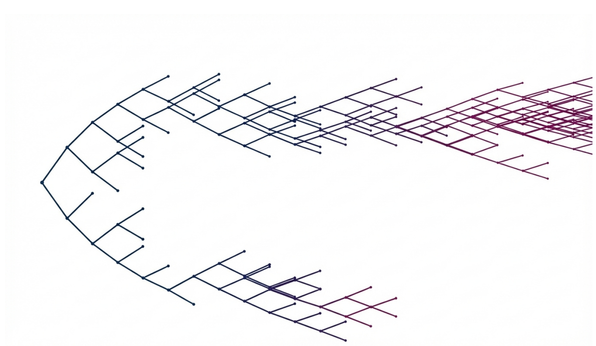


qwen__simpleGWvis.png



grok__IMG_ch3__simpleGWvis.png

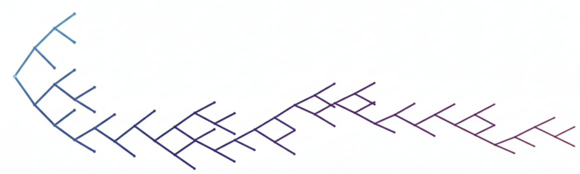
Figure 1.52: Candidates for group simplegwvis (5 variants), part 1 of 2. Filenames are printed beneath the panels.



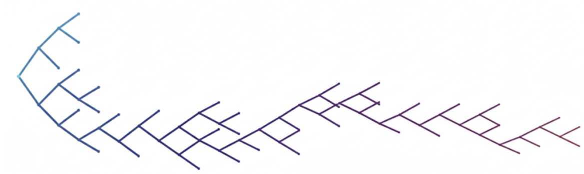
codex__IMG_ch3__simpleGWvis.png

Figure 1.53: Candidates for group simplegwvis (5 variants), part 2 of 2. Filenames are printed beneath the panels.

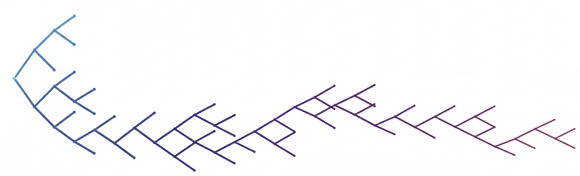
Group: subgwvis



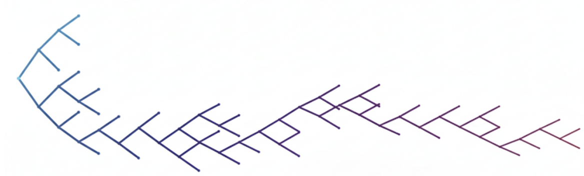
best__subGWvis.png



claude__subGWvis.png

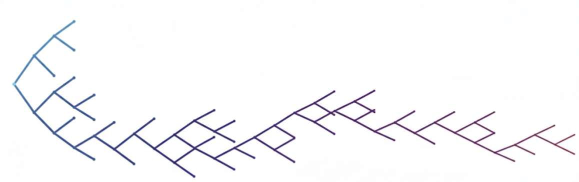


qwen__subGWvis.png



grok__IMG_ch3__subGWvis.png

Figure 1.54: Candidates for group subgwvis (5 variants), part 1 of 2. Filenames are printed beneath the panels.

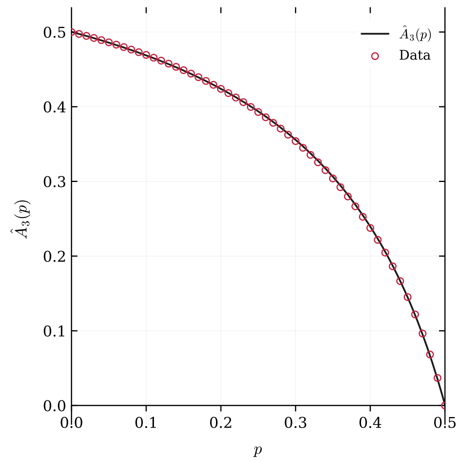


codex__IMG_ch3__subGWvis.png

Figure 1.55: Candidates for group subgwvis (5 variants), part 2 of 2. Filenames are printed beneath the panels.

Unique assets: a single source

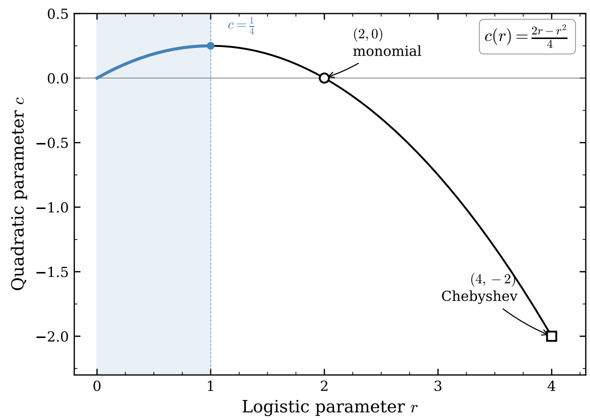
Group: a3_hat_plot



codex__IMG_ch3__A3_hat_plot.pdf

Figure 1.56: Candidates for group a3_hat_plot (1 variant). Filenames are printed beneath the panels.

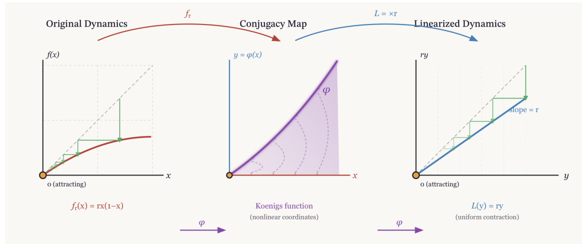
Group: figure1_parameter_conjugacy



codex__IMG_ch3__figure1_parameter_conjugacy.pdf

Figure 1.57: Candidates for group figure1_parameter_conjugacy (1 variant). Filenames are printed beneath the panels.

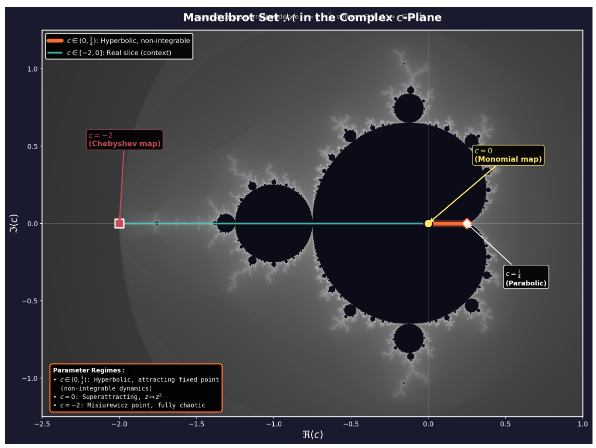
Group: figure2_koenigs_linearization



`codex__IMG_ch3__figure2_koenigs_linearization.png`

Figure 1.58: Candidates for group `figure2_koenigs_linearization` (1 variant). Filenames are printed beneath the panels.

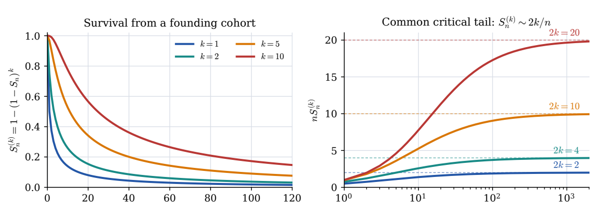
Group: `figure4_mandelbrot_context`



`codex__IMG_ch3__figure4_mandelbrot_context.png`

Figure 1.59: Candidates for group `figure4_mandelbrot_context` (1 variant). Filenames are printed beneath the panels.

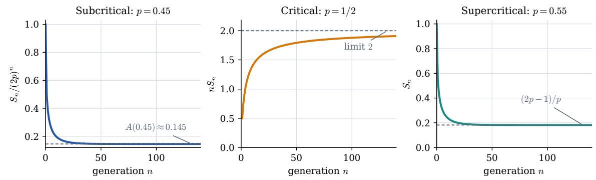
Group: `founder_cohort_survival`



`codex__generated__founder_cohort_survival.pdf`

Figure 1.60: Candidates for group `founder_cohort_survival` (1 variant). Filenames are printed beneath the panels.

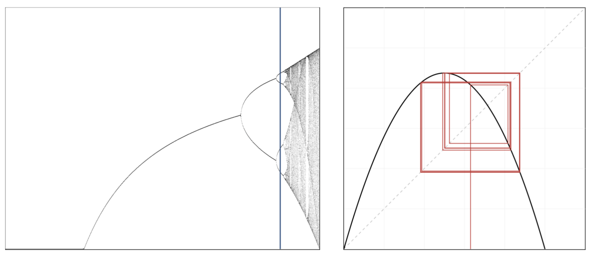
Group: gw_regime_diagnostics



codex__generated__gw_regime_diagnostics.pdf

Figure 1.61: Candidates for group gw_regime_diagnostics (1 variant). Filenames are printed beneath the panels.

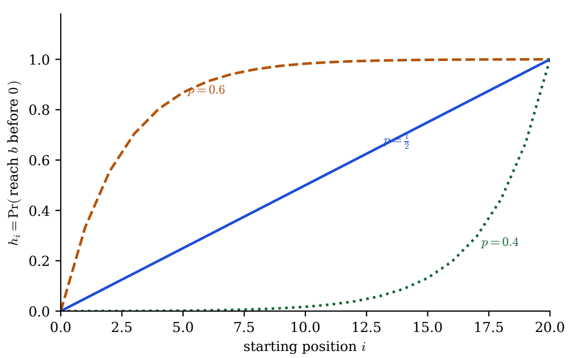
Group: period_double



codex__IMG_ch3__period_double.png

Figure 1.62: Candidates for group period_double (1 variant). Filenames are printed beneath the panels.

Group: ruin_prob



qwen__ruin_prob.pdf

Figure 1.63: Candidates for group ruin_prob (1 variant). Filenames are printed beneath the panels.

Candidates that failed the integrity check

None. Every staged candidate passed the header and `pdfinfo` checks, and every one of them is displayed above.

Bibliography

[1] James R. Norris. *Markov Chains*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, Cambridge, 1997.

[2] Samuel Karlin and Howard M. Taylor. *A First Course in Stochastic Processes*. Academic Press, New York, 2nd edition, 1975.

[3] Linda J. S. Allen. *An Introduction to Stochastic Processes with Applications to Biology*. CRC Press, Boca Raton, FL, 2nd edition, 2010.

[4] Theodore E. Harris. *The Theory of Branching Processes*. Springer, Berlin, 1963.

[5] Krishna B. Athreya and Peter E. Ney. *Branching Processes*. Springer, Berlin, 1972.

[6] Daniel T. Gillespie. Exact stochastic simulation of coupled chemical reactions. *The Journal of Physical Chemistry*, 81(25):2340–2361, 1977.

[7] Mark H. A. Davis. *Markov Models and Optimization*. Chapman & Hall, London, 1993.

[8] David G. Kendall. On the generalized "birth-and-death" process. *The Annals of Mathematical Statistics*, 19(1):1–15, 1948.

[9] Akiva M. Yaglom. Certain limit theorems of the theory of branching random processes. *Doklady Akademii Nauk SSSR (N.S.)*, 56:795–798, 1947.

[10] Sylvie Méléard and Denis Villemonais. Quasi-stationary distributions and population processes. *Probability Surveys*, 9:340–410, 2012.

[11] Pierre Collet, Servet Martínez, and Jaime San Martín. *Quasi-Stationary Distributions: Markov Chains, Diffusions and Dynamical Systems*. Springer, Berlin, 2013.

[12] Graham E. Budd and Richard P. Mann. History is written by the victors: The effect of the push of the past on the fossil record. *Evolution*, 72(11):2276–2291, 2018.

[13] Frank W. J. Olver, Daniel W. Lozier, Ronald F. Boisvert, and Charles W. Clark, editors. *NIST Handbook of Mathematical Functions*. Cambridge University Press, Cambridge, 2010.